Natural Experiments in the Public Record of AI

The Complete natex Paper Collection

Hauke Hillebrandt

with AI-assisted analysis (natex v0.2.0)

Compiled 2026-07-19 • Latest versions: `haukehillebrandt.github.io/natex`

Contents

Every estimate from seeded, reproducible runs; nulls and identified artifacts reported as first-class results.

Natural Experiments in the Public Record of AI: A Systematic Survey

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analysis passes of record: 2026-07, `natex` v0.2.0; all stochastic steps seeded

Abstract

The public record of AI — benchmark scores, adoption surveys, capital expenditure, chip stocks, model registries, crowd-sourced battle streams — is full of dated events that are claimed, usually by eye, to have bent some curve. We survey a systematic attempt to test such claims with formal quasi-experimental designs, run and validated by the open-source `natex` toolkit: 17 dataset–design analyses carried to a verdict and 15 more triaged and excluded for cause, each subjected to a mandatory falsification battery (specification grids, shifted-cutoff placebo grids, placebo units and outcomes, density and covariate checks, pinned controls, honest refusals). Eight findings are selected here by credibility and interest. Three are credible: the 2025 BTOS questionnaire rewording moved measured US business AI adoption by +6.0 pp (4.7× the largest of 49 placebos), the October-2023 export controls bent China's legal AI-chip stock by −0.3 to −0.56 log-units/yr against every control group (25/25 specifications, loophole-round placebo null), and GPQA-Diamond's slope 2.4×'d at o1-preview with a clean placebo grid (0/7). One is moderate: an 83% post-Act deficit of models just above the EU AI Act's 10^{25}-FLOP line. Two are first-class negatives: the Epoch Capabilities Index shows no kink at o1 in a design with demonstrated power, and the sycophantic GPT-4o build won zero opponent-adjusted votes on LMArena. And two are cautionary: a sharp January-2025 adoption acceleration that cannot be attributed to DeepSeek-R1, and a hyperscaler capex "kink" at ChatGPT whose nominal $p = 4 \times 10^{-6}$ collapses to a placebo-calibrated 0.125. Across the corpus, roughly half of the nominally significant estimates die in the battery — on short, serially dependent, composition-churned series, the falsification battery is not a robustness appendix; it is the analysis.

1 Introduction

Public discussion of AI progress and AI policy runs on dated trend breaks. Reasoning models "changed the slope" of capabilities; ChatGPT "ignited" the datacenter buildout; export controls "choked" China's compute; DeepSeek "bent" the adoption curve; the EU AI Act "chilled" large training runs. Each claim names an event, a series, and a shape — which makes each one a candidate for a formal quasi-experimental design: a regression discontinuity or kink at a known cutoff [20, 35, 13], a difference-in-differences or synthetic control around a dated policy [2], bunching at a statutory threshold [71, 59, 58], or an interrupted time series with reversal [9]. Treating the claims this way buys an explicit identifying assumption, a standard error, and — most importantly — a falsification battery that can *reject* them.

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This paper is the capstone of a project that did this systematically. The instrument is `natex` [50], an open-source toolkit for automated natural-experiment discovery and estimation in the lineage of the LoRD3 local-discontinuity scan of [38, 37, 55], extended with kink and difference-in-kinks estimators following [13], a subset-scanning panel DiD, synthetic controls, and density/bunching tests. Over successive analysis passes the project pointed this machinery at the public quantitative record of AI — Epoch AI's model, benchmark, and chip databases [26, 28, 27], the Census Bureau's Business Trends and Outlook Survey (BTOS) [14], SEC EDGAR filings, LMArena's published battle stream [23], and semiconductor market data — carrying 17 dataset–design pairs to a recorded verdict and triaging 15 more as infeasible or ill-posed (Section 11 tabulates all 32). Each completed analysis is written up as a companion note with committed figure scripts that re-derive every headline number from the frozen inputs before drawing [39, 41, 46, 40, 42, 47, 45, 44, 43, 48, 49].

Here we select the eight strongest results by the product of credibility and interest, and — deliberately — count honest nulls and identified artifacts as first-class findings alongside the positive estimates. The selection spans the full verdict taxonomy the project converged on: *credible* (the estimate survives the battery), *descriptive-only* (the data feature is real but attribution to the named event fails), *null-with-power* (the dial reads zero where the test demonstrably can move), and *identified artifact* (the battery caught the machinery, or the platform, generating the result). Section 2 covers methods, Sections 3–10 the findings, Section 11 coverage, and Section 12 the lessons.

2 Methods: the toolkit and the validation philosophy

Design families. `natex` v0.2.0 implements, behind one CLI: local-polynomial regression discontinuity with the LoRD3 blind discontinuity scan and honest (split-sample) post-discovery 2SLS [38]; sharp and fuzzy regression kink and difference-in-kinks (DiK) estimators [20, 13]; a subset-scanning panel difference-in-differences ("SuDDDS") that searches units and break dates by likelihood-ratio scan, calibrated by permutation under dependence-preserving nulls; synthetic-control and related panel estimators [2, 1]; McCrary-style binned-Poisson density tests [63] used both as manipulation diagnostics and as bunching estimators; and event-study/interrupted-time-series segmentations [9]. House rules bind every run: a single seeded random generator threaded through all stochastic calls, discovery steps that never read the outcome, and failed computations that return NaN — a refusal — rather than a number.

The falsification battery. No estimate in the corpus is reported without its battery. The standing components: (i) specification grids over bandwidth, kernel, donut, and polynomial degree; (ii) *shifted-cutoff placebo grids* in the spirit of [32], read as separating bend *existence* from date *attribution* — rejections at non-event dates mean the series bends over an era or everywhere, not at the event; (iii) placebo-in-space (pseudo-treated units) and placebo outcomes; (iv) density and covariate discontinuity checks; (v) sibling-series falsifications — an aggregate in which the test demonstrably has power but should read null; and (vi) where available, *physically pinned controls* — units that cannot have been treated, whose movement identifies platform-side confounds (Section 8). Nominal HC1/CR1/HAC inference is reported but never trusted on its own: on short, serially correlated aggregates conventional robust standard errors are known to be badly oversized [10, 19, 35], so placebo-calibrated p-values and permutation ranks are the inference of record wherever the battery can supply them.

Table 1: The eight selected findings.

Finding (section)	Design / cutoff	Headline	Verdict
Export controls vs. China's compute (§3)	group DiK, chip stocks, 2023-10-17; SuDDDS DiD, big runs, 2022Q4	legal stock -0.56 ln/yr ($t = -3.3$; 25/25 specs); total stock $\approx$ half, fragile; big-run DiD +3.2/qtr fails placebo-in-space	credible / attenuated / descriptive
BTOS question rewording (§4)	measurement RDiT, wave 202524 (2025-11-03)	+6.04 pp, HAC $z = 16.3$; 4.7$\times$ max of 49 placebos; blind scan re-localizes	credible
EU AI Act 10^{25}-FLOP line (§5)	bunching / density, statutory threshold, post 2024-08-01	Fisher OR 0.173, $p = 0.007$; 82.7% missing mass; $\theta < 0$ in 15/15	credible (moderate)
GPQA-Diamond at o1-preview (§6)	sharp RKD-in-time, 2024-09-12	+0.00258 logit/day ($t = 3.3$); placebo size 0/7	credible, date-localized
US adoption at DeepSeek-R1 (§7)	group DiK + SuDDDS, BTOS sectors, 2025-01-20	DiK +7.3 pp/yr ($\|t\| = 2.6$); scan localizes exactly; placebo dates as large	descriptive-only
GPT-4o sycophancy episode (§8)	ABA ITS + Bradley–Terry, deploy 04-25, rollback 04-29	deploy style break real (perm $p = 0.013$); BT vote gain +0.02 log-odds (se 0.14): null	credible null / identified artifact
Hyperscaler capex at ChatGPT (§9)	group DiK, EDGAR panel, 2022-11-30	+0.140 dex/yr, nominal $p = 4\times10^{-6}$; placebo-calibrated $p = 0.125$	identified inference artifact
ECI at o1, fresh vintage (§10)	sharp RKD-in-time, 2024-09-12	+0.0065 pts/day ($t = 0.50$); placebos reject 4/8 elsewhere	null with power

Validation anchor. Before being trusted on novel data, the pipeline was run blind on the canonical Abadie–Diamond–Hainmueller Proposition-99 panel [2]: all three SuDDDS scan methods recover (California, 1989) exactly (precision = recall = 1.0), the deterministic simplex weights put summed weight 0.955 on ADH's five published donors, and the synthetic-control ATT of -19.5 packs per capita matches the canonical ≈ -19 average gap; the in-space RMSPE-ratio placebo ranks California 3/39 ($p = 0.077$) without ADH's poor-pre-fit exclusion, reported as the null it is [48]. The exercise has zero novelty by construction; its value is credibility transfer.

Audit lineage. Every analysis pass was produced under a two-stage protocol: an analysis agent runs the designs and freezes a numbers record; an independent audit pass re-derives the headline estimates from the frozen inputs, and the committed figure script of each companion note asserts every quoted number against that record before drawing. Audits are deliberately model-diverse after an early recorded failure in which same-family verifiers shared a PDF-extraction blind spot (dropped superscripts). Refusals, degenerate auto-configurations, and underpowered nulls of the automated pipeline are reported verbatim in each note ("stated as run"), not patched over.

3 Export controls and China's compute: three legs of one story

The October-2023 US export controls are the corpus's strongest policy result because the design decomposes the question [46]. *Leg 2* — the legal channel — is a group difference-in-kinks: the export-license schedule bends for China only at a common dated cutoff, so treated = China's legal

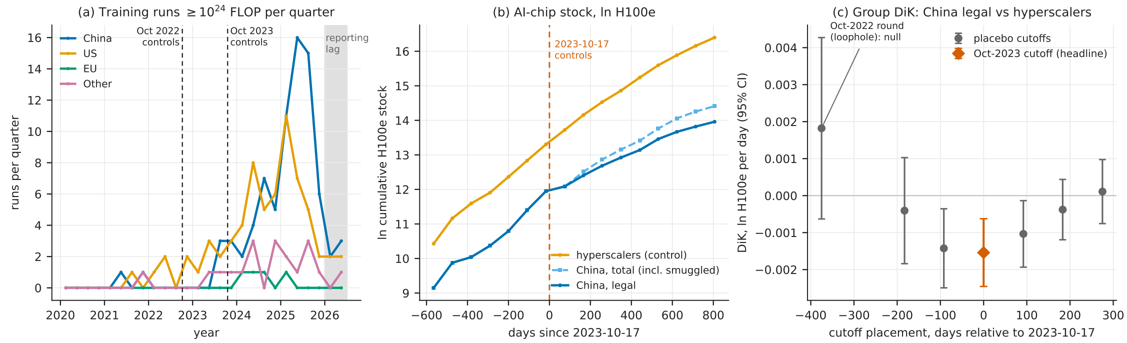

Figure 1: The three legs (reproduced from [46]): (a) quarterly $\geq 10^{24}$-FLOP training runs by country group; (b) ln cumulative H100e chip stock — hyperscaler controls, China legal, China total including smuggled — around the 2023-10-17 cutoff; (c) group-DiK estimates at the true cutoff and six placebo placements, with the Oct-2022 loophole round null at -375 days.

H100-equivalent chip stock, control = the four US hyperscalers, and the controls difference out the global supply bend. The headline DiK is -0.00154 ln-units/day (HC1 se 0.00047, $t = -3.30$; ≈ -0.56 ln-units/yr), negative in **25 of 25** specifications with a defensible band of -0.3 to -0.56 ln-units/yr. The falsifications land where the policy history says they should: the October-2022 first round — whose A800/H800 compliance loophole predicts no bite — is a null placebo ($+0.0018$, $p = 0.146$), and the placebo-treated group (Other vs. hyperscalers) is clean at the defensible bandwidths. *Leg 3* repeats the design on China's *total* stock including smuggled chips [56]: -0.00076 (se 0.00050, $t = -1.54$, 95% CI spanning zero) — half the size and fragile, because smuggled Nvidia compute grew from 27.8k to 662k H100e over 2024–2025 (doubling every 4.7 months) and now exceeds China's legal Nvidia stock 1.65×. *Leg 1* — a 4-country × 26-quarter DiD on counts of $\geq 10^{24}$-FLOP training runs — finds no suppression at all: the scan recovers {China} at 2022Q4 exactly (panel-null $p = 0.010$, the permutation floor), but the effect is *positive* (+3.2 runs/qtr, 95% CI $[+0.9, +5.5]$) and fails placebo-in-space at exact $p = 1.0$ — every pseudo-treated unit shows a larger studentized effect — so it is descriptive, not causal. The three-leg verdict: the controls bent the channel they legally govern, were substantially bypassed in aggregate, and left China's large-run training activity unbowed (Figure 1).

4 The question is the treatment: the BTOS rewording RDD

The cleanest single estimate in the corpus is a *measurement* natural experiment [41]. With wave 202524 (reference period from 2025-11-03; announced 2025-12-03), the Census Bureau reworded BTOS's headline AI question from use "in producing goods or services" to use "in any of its business functions" [21]. Treating the refresh as a known cutoff in a regression-discontinuity-in-time design on the spliced 71-wave national series [35], the jump is +6.039 pp (HC1 $z = 22.9$; HAC $z = 16.3$) against an old-wording level of 10.9 pp — a ratio of 1.553. The battery is uniformly supportive: all bandwidth, weighting, curvature, and HAC variants sit in a 6.0–7.5 pp band with $z > 12$; the largest of 49 placebo-cutoff estimates is 1.30 pp, 4.7× smaller (randomization $p = 0.020$, the 1/50 floor); and, told nothing, the LoRD3 scan's two top candidate boundaries exactly flank the known cutoff, with honest 2SLS +6.74 (95% CI $[6.01, 7.48]$). True adoption cannot produce it: the fitted pre-trend predicts +0.66 pp across the 50-day shutdown gap, and doubling that slope still leaves ≥ 5.38 pp. Nulls are reported as nulls (blind-scan randomization $p = 0.14$, underpowered at $n = 71$;

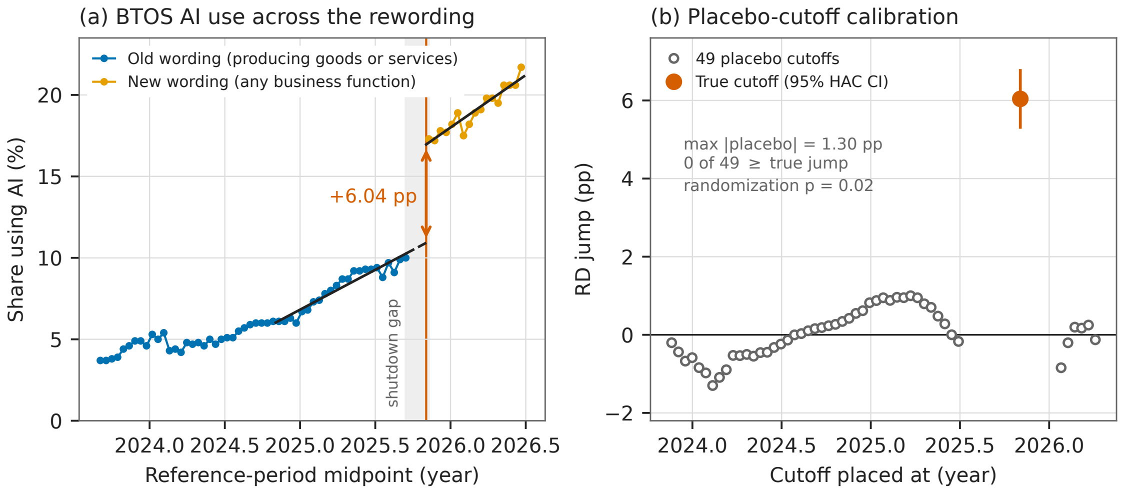

Figure 2: The rewording RDD (reproduced from [41]): (a) the spliced BTOS national series with per-side fits and the 50-day shutdown-gap extrapolation; (b) the same estimator at all 49 placebo cutoffs versus the true cutoff.

McCrary NaN — inapplicable on a uniform time grid). The estimate calibrates the splice factor — additive +6.04 pp, ratio 1.553 — needed to carry any old-wording BTOS analysis past December 2025, and implies that roughly a third of the post-refresh measured adoption level is attributable to asking a better question [12], with the caveat that "wording" bundles a simultaneous sample rotation and shutdown-adjacent composition change (Figure 2).

5 Bunching below the EU AI Act's 10^{25}-FLOP line

The EU AI Act's systemic-risk presumption at 10^{25} training FLOP [29] is a notch, and notches invite bunching [59]. In Epoch AI's compute panel (719 dated models with compute estimates), the post-Act density of new models just *above* the statutory line collapses [45]: a binned-Poisson density test at $\log_{10}$ FLOP $= 25.0$ is negative in 15/15 window $\times$ bin specifications (10/15 with $p < 0.05$), while the pre-Act period rejects in 0/15. The pre/post contrast is sharp: Fisher exact on below/above counts in the ± 0.5-dex window gives OR $= 0.173$ (95% CI $[0.048, 0.619]$, $p = 0.0072$) — 28.9 post-Act models expected in $(25.0, 25.5]$ at the pre-Act ratio, 5 observed, an 82.7% deficit. The deficit is specific to the statutory line (placebo thresholds at 24.5 and 25.5 are null or opposite-signed, with mass piling up just *below* 25.0), specific to the post-Act period (a placebo split date inside the pre-Act sample gives OR 0.90, $p = 1.0$), and correctly timed (the above-line count collapses from 2025H1, after entry into force, before applicability), with the classic signature that marginal crossers vanish while the frontier jumps far above the notch (Grok 3, GPT-4.5, Grok 4 at $\log_{10} = 26.5$–26.7). Two facts grade it moderate rather than conclusive: only 5–13 post-Act models sit above the line (the preferred narrow-window interaction is a null, $p = 0.73$), and an Act-induced *disclosure* response — labs going quiet about compute near the line — would mimic bunching exactly. Either reading matters for the compute-threshold debate [73, 53]: the data just above the line already look threshold-aware, contrary to the threshold-blind scaling assumed in forward projections [60] (Figure 3).

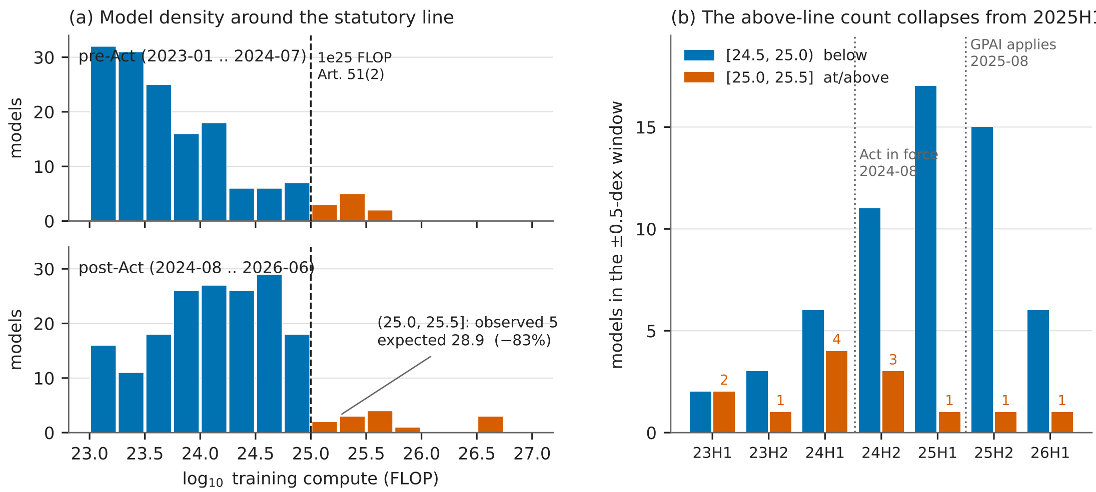

Figure 3: Bunching at the statutory line (reproduced from [45]): (a) model counts in 0.25-dex compute bins, pre- versus post-Act, around 10^{25} FLOP; (b) below/above counts per half-year — the above-line count collapses from 2025H1.

6 A date-localized capability kink: GPQA-Diamond at o1-preview

The flagship kink pass [39] supplies the survey's one date-localized capability result and its sharpest methodological contrast. GPQA-Diamond [70] logit-score slopes rise from 0.00188 to 0.00446 per day at the o1-preview release (2024-09-12) — roughly 15 → 34 raw points per year, a 2.4× acceleration. The headline RKD-in-time estimate is +0.00258 per day (se 0.00078, $t = 3.32$), positive in 8/8 bandwidth×donut cells, with an *empty* placebo grid (0/7 shifted cutoffs reject), a null release-density kink ($t = 0.76$), and a null training-compute covariate kink ($p = 0.54$): the bend localizes to the date (Figure 4). The contrast: METR's 50% time-horizon series [61] shows an equally real bend (doubling time 9.9 → 3.5 months, $t = 2.13$, positive in 8/8 cells) whose pre-side placebos *also* reject ($p = 0.004$–0.030 at −90 to −270 days) — a credible reasoning-era slope change that cannot be pinned to 2024-09-12. The pair is the era-bend/event-bend taxonomy in a single dataset family, and the standing caveat travels with both: composition — reasoning models entering the release stream — *is* the mechanism, so these are properties of the release stream, not statements that individual models improved.

7 A sharp bend, no verdict: US adoption at DeepSeek-R1

The project's flagship adoption analysis ends in an instructive split verdict [40]. In the BTOS sector panel (20 sectors × 54 biweekly waves), a sector-by-wave DiD scan localizes an acceleration in AI-exposed sectors (information; finance; professional services) at $t_0 = 2025.089$ — *exactly* the first survey wave after DeepSeek-R1's release on 2025-01-20 [33] — with scan $p = 0.010$–0.030; the exposed-minus-unexposed gap slope more than triples (+4.2 → +13.9 pp/yr), and the group DiK at the release date is nominally significant (+7.30 pp/yr, CR1 se 2.80, $|t| = 2.61$). The localization survives composition checks and placebo-in-space (0/13 pseudo-treated control sectors). But attribution to R1 fails five ways: placebo cutoffs at non-event dates return DiKs as large or larger (+11.4 pp/yr at $t = 2024.30$, $|t| = 3.21$; +4.0 at 2024.50, $|t| = 4.53$); only 2/6 specification variants at R1 are significant; equally large local kinks appear in non-exposed sectors (real estate,

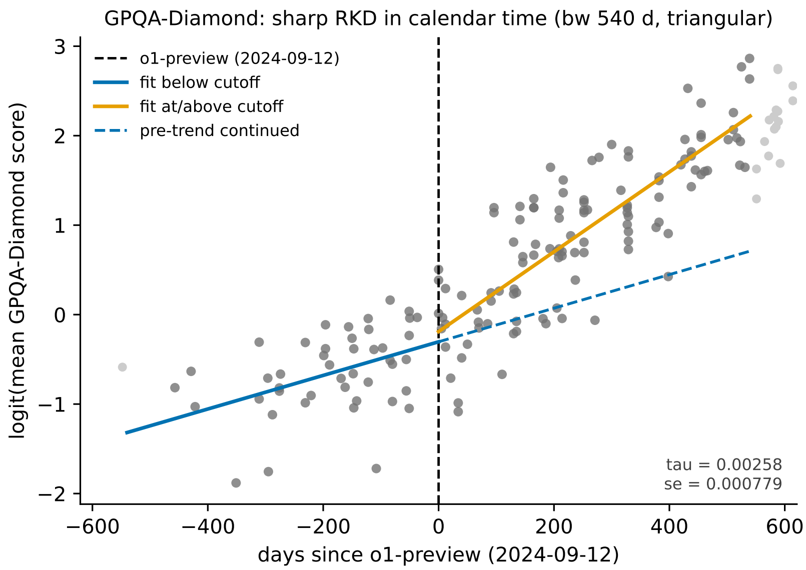

Figure 4: GPQA-Diamond, logit(mean score) against days since o1-preview (reproduced from [39]): per-side local-linear fits (bandwidth 540 days) versus the continued pre-trend; placebo empirical size 0/7.

arts, health care); the predicted *deceleration* at o1 sign-flips with bandwidth; and the cutoff week is confounded — the AI-diffusion export rule (2025-01-13), the Stargate announcement (2025-01-21), and a BTOS sample-year rollover in mid-December 2024 all sit inside the window. The verdict of record: something bent US business AI adoption in January 2025, and the localization is data-driven — but these data cannot say it was R1 (Figure 5).

8 Sycophancy did not win votes — and the rollback isn't what it looks like

The GPT-4o sycophancy episode of April 2025 [67] supplies the corpus's only deploy-and-rollback interrupted time series, inside 135,634 LMArena battles [47]. The *deploy* leg is credible: the live 4o alias shows a sharp markdown-style collapse timed to April 25–26 (headers per 1k tokens $3.52 \to 1.12$; sliding-cut permutation $p = 0.0128$, rank 1/78) while nine pinned dated snapshots stay flat — proving the arena copy was not pinned — and it delivers a precise null on the headline mechanism: the Bradley–Terry-adjusted win-rate change during the sycophancy window is +0.02 log-odds (SE 0.14; implied win rate $0.540 \to 0.545$). During the four days users saw the sycophantic build, it gained zero opponent-adjusted votes — a clean behavioral null on the preference-hacking mechanism the literature predicted [75]. The *rollback* leg is an identified artifact: on exactly 2025-04-29 the pinned `claude-3-7-sonnet-20250219` snapshot — whose weights cannot change — takes an equal-and-opposite persistent step in style *and* BT strength (−0.54 log-odds), demonstrating an arena-side serving change that contaminates every 29-April estimate. The 4o post-rollback regime never reverts (the episode is A–B–C, not A–B–A), and the widely repeated "sycophancy raised the win rate, rollback lowered it" narrative loads entirely on the confounded boundary. Pinned snapshots are the cheap, decisive placebo for platform time series (Figure 6).

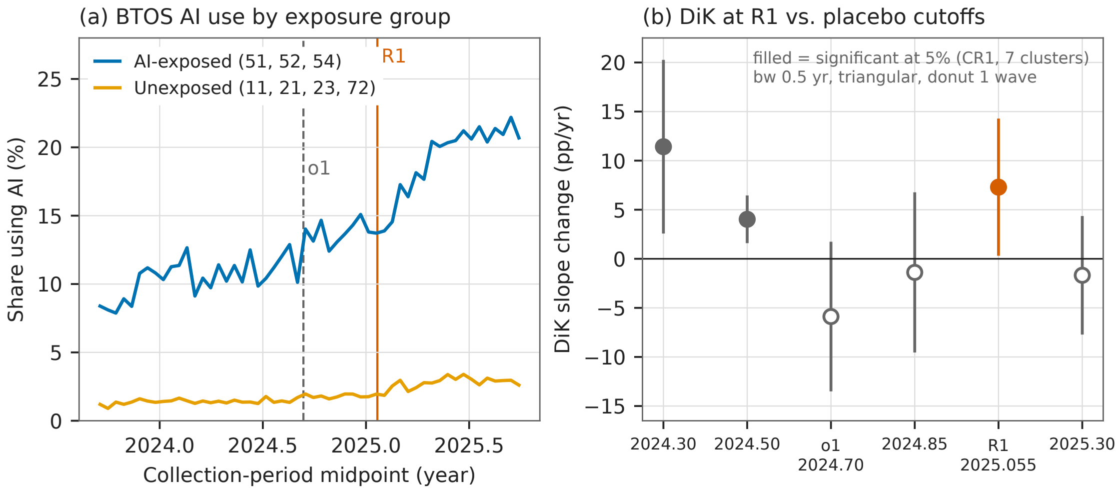

Figure 5: Adoption at R1 (reproduced from [40]): (a) BTOS AI-use share by exposure group with o1 and R1 marked; (b) group-DiK estimates at R1, o1, and four non-event placebo dates — two placebos are significant, one larger than the R1 estimate.

9 An identified inference artifact: hyperscaler capex at ChatGPT

The capex analysis is included as a first-class finding precisely because its battery caught the failure mode that would otherwise headline this literature [42]. Big-4 hyperscaler capital expenditure (EDGAR XBRL reconstruction) accelerates against a six-firm non-AI capital-intensive aggregate after ChatGPT: group DiK +0.1396 dex/yr (HC1 se 0.0302, nominal $p = 3.9 \times 10^{-6}$), positive in 18/18 specification cells, localized over 2023.0–2023.5 exactly as a lagged investment response would be, and visible per firm for Microsoft, Alphabet, and Amazon but null for Meta. Every conventional robustness check passes. What kills the causal reading is the clean pre-period placebo grid: at 15 non-event cutoffs in 2017–2020 the identical specification rejects at nominal 5% seven times (empirical size 0.47), the largest placebo $|z|$ (5.11, at 2019.75) exceeds the ChatGPT estimate's matched-bandwidth $|z|$ (4.35), a full-sample kink at the non-event date 2021.5 reaches $|z| = 6.9$, and the BIS export-control date eight weeks earlier is statistically indistinguishable (+0.125, $z = 4.0$). Placebo-calibrated, the ChatGPT estimate's p is 0.125–0.25: the nominal p-value overstates the evidence by five orders of magnitude, because HC1 inference assumes serial independence and the residual lag-1 autocorrelation reaches 0.66 [10]. The slope divergence is a genuine, robust data feature; the "significance" was an artifact of the standard errors, and only the battery could tell the two apart (Figure 7).

10 An honest null with power: the ECI at o1, fresh vintage

If "everything kinked in 2024," the per-benchmark results of Section 6 would be uninformative; the Epoch Capabilities Index (ECI) [51] is the aggregate guard, and it holds on a fresh vintage with twelve more months of data [44, 39]. The all-models RKD-in-time at o1-preview is +0.0065 ECI points/day (se 0.0131, $t = 0.50$, $p = 0.62$; slopes $\approx$ 22.3 vs. 24.7 points/yr), null in 17/18 grid cells — while the same specification rejects at 4/8 placebo cutoffs elsewhere in the series: the instrument demonstrably moves, and reads zero at the release date. The vintage's one new nominally significant result is itself a triage exhibit: a precision-weighted rerun (weights $1/\sigma_i^2$ from

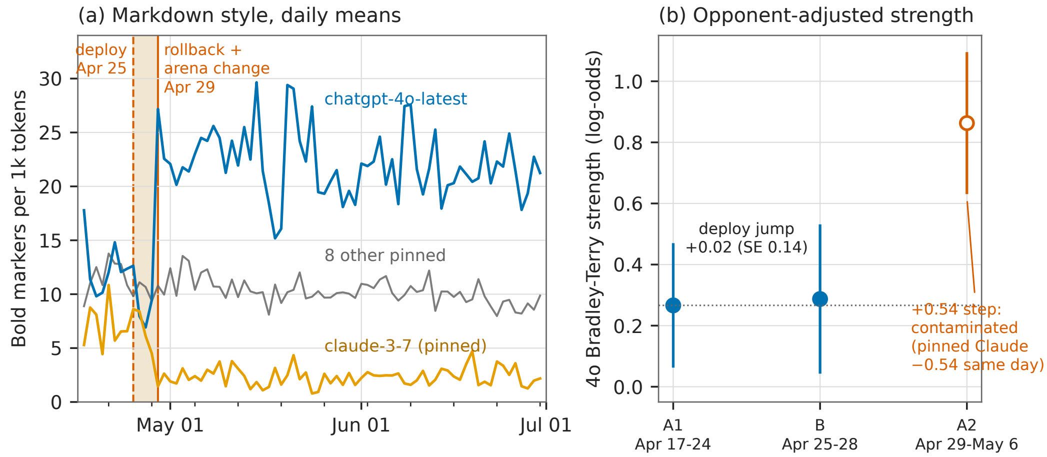

Figure 6: The deploy–rollback episode (reproduced from [47]): (a) daily bold-markers per 1k tokens for 4o, the pinned `claude-3-7` control, and pooled pinned snapshots — the pinned control steps, mirror-imaged, on exactly the rollback date; (b) 4o Bradley–Terry strength by segment: the deploy jump is +0.02 log-odds.

published CIs) returns a negative kink significant in 18/18 cells (−0.0190, $p = 0.016$) — which the battery identifies as smooth post-2024 concavity, not an o1-localized break, because placebo cutoffs at +90/+180 days give same-sign *larger* kinks (−0.0399, $p = 0.0064$; −0.0297, $p = 5.5 \times 10^{-6}$) and a local quadratic absorbs it ($p = 0.14$). A tempting third series — strict frontier records — is a non-design (4 pre-cutoff points in the estimation cell; its nominal $t = 2.28$ dies under a 30-day donut and flips sign under a 45-day one). Whatever the reasoning-model recipe did to AI capabilities, it did not bend the aggregate index's calendar-time trend at the release date (Figure 8).

11 Coverage: everything considered

The survey claim rests on Tables 2 and 3: every dataset–design pair the project considered, with its verdict or its reason for exclusion. Beyond the eight findings above, the studied set includes the flagship graveyard [39] — a $t = 12.9$ "kink" in cumulative GPU-cluster stock at ChatGPT that fails *every* placebo (empirical size 7/7: smooth super-exponential curvature bends everywhere), a sign-unstable Chinchilla null, and the EU threshold correctly rejected as a kink because it is a step with sorting — plus a blind LoRD3 scan that missed Chinchilla's adoption ramp and correctly diagnosed its own top discovery as a fine-tune measurement artifact [43], and a financial-series stress test in which every "credible" automated verdict on semiconductor stocks dissolved under inspection — the DeepSeek "kink" (Holm $p = 1.4 \times 10^{-8}$) is the 2025 AI rally sitting on both sides of a one-week crash [49].

12 Discussion

The scoreboard. Of 17 analyses carried to verdict, four estimates survive as credible (the rewording jump, the legal-channel export bend, the GPQA kink, and — graded moderate — the EU-Act bunching), two nulls carry power certificates (ECI at o1, twice; the LMArena vote null), and at least six nominally significant results were killed or reclassified by the battery: the R1 attribution, the capex kink, the CI-weighted ECI kink, the datacenter $t = 12.9$, the semiconductor

Table 2: Studied to verdict: the 17 dataset–design analyses of record.

Data × design	Cutoff / event	Verdict	Ref.
GPQA-Diamond logit scores × RKD-in-time	o1-preview 2024-09-12	credible, date-localized (placebo 0/7)	[39]
METR 50% time horizon × RKD-in-time	o1-preview	era bend real; date attribution fails (pre-side placebos 3/6)	[39]
ECI 2025 vintage × RKD-in-time	o1-preview	null with power (falsification guard)	[39]
ECI fresh vintage, unweighted / CI-weighted / frontier records × RKD	o1-preview	null upgraded; weighted "kink" = curvature artifact; records series non-estimable	[44]
China vs. hyperscaler chip stocks × group DiK (legal; total incl. smuggled)	controls 2023-10-17	credible (-0.3 to -0.56 ln/yr); total attenuated and fragile	[46]
Country panel, $\geq 10^{24}$-FLOP run counts × SuDDDS DiD + SC/GESS	controls 2022Q4	descriptive-only (placebo-in-space exact $p = 1.0$)	[46]
BTOS sector panel × group DiK + SuDDDS scan	DeepSeek-R1 2025-01-20	descriptive-only; attribution fails triage	[40]
BTOS national spliced series × measurement RDiT + blind LoRD3	rewording, wave 202524	credible; splice factor +6.04 pp / 1.553	[41]
EDGAR capex panel × group DiK	ChatGPT 2022-11-30	descriptive-only; nominal inference identified as artifact	[42]
LMArena battles × ABA ITS + windowed Bradley–Terry	deploy 2025-04-25, rollback 04-29	deploy leg credible (vote null); rollback leg identified arena-side artifact	[47]
Epoch compute panel × binned-Poisson bunching + Fisher contrasts	EU AI Act 10^{25} FLOP, 2024-08-01	credible (moderate): bunching, behavioral or reporting	[45]
EU AI Act threshold × kink/RD probes	10^{25} FLOP	rejected: step with sorting, wrong estimand shape	[39]
GPU-cluster cumulative stock × RKD-in-time	ChatGPT	rejected: placebo size 7/7 on super-exponential series	[39]
Tokens-per-parameter × RKD-in-time	Chinchilla 2022-03-29	sign-unstable null	[39]
Tokens-per-parameter × blind LoRD3 scan	(discovery)	honest miss: top discovery is a fine-tune measurement artifact, correctly diagnosed	[43]
Semiconductor weeklies × kinks, RDD, DiD scan	BIS rounds; DeepSeek week	all "credible" verdicts artifacts of trend regimes; informative DiD null	[49]
ADH Prop-99 panel × blind SuDDDS + synthetic control	1989	benchmark recovered exactly (validation anchor)	[48]

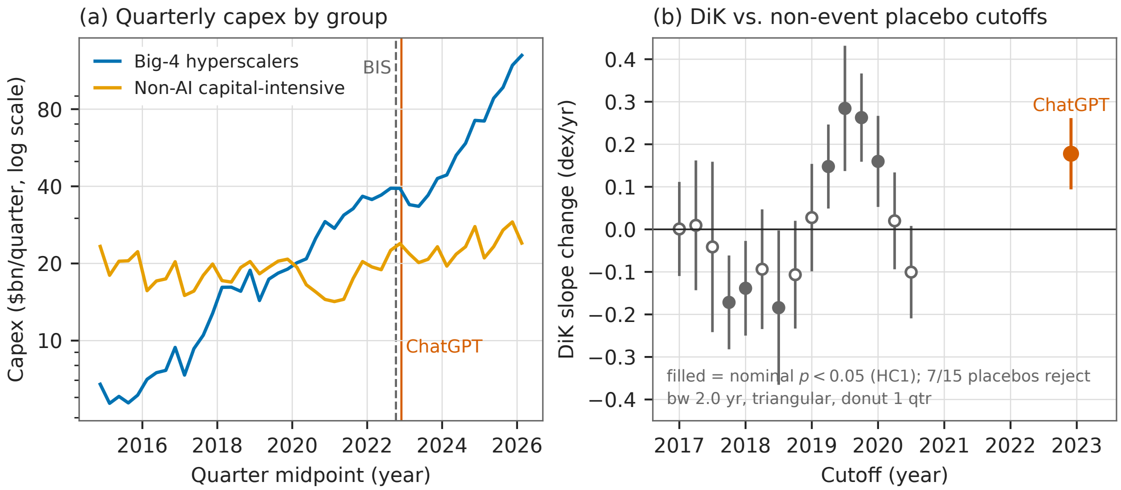

Figure 7: Capex at ChatGPT (reproduced from [42]): (a) quarterly capex of the big-4 and control aggregates with the BIS and ChatGPT dates marked; (b) DiK estimates at 15 pre-period non-event placebo cutoffs versus ChatGPT — seven placebos reject, and the largest placebo $|z|$ exceeds the ChatGPT $|z|$.

"kinks" at three cutoffs, and the LMArena rollback narrative. That kill rate — roughly half of everything that would have cleared a naive $p < 0.05$ bar — is the central empirical fact of the survey.

Nominal inference on the public record of AI is broken by default. The corpus's series are short, serially dependent, and smooth; HC1/CR1 standard errors on them manufacture $|z| > 6$ at dates where nothing happened (capex at 2021.5; datacenters everywhere; the DeepSeek rally kink). Placebo calibration measures the size distortion directly — a nominal 5% test rejecting 47% of the time in the capex pre-period — and is cheap. Every headline in this literature should be read against a placebo distribution or not at all [32, 10].

Era bends are not event bends, and calendar weeks are crowded. The shifted-cutoff grid mechanically separates "the trend changed in this era" (METR, the January-2025 adoption surge) from "this event changed the trend" (GPQA, the rewording). Even when a bend localizes, the week is shared: R1 sits with the diffusion rule, Stargate, and a panel rollover; ChatGPT sits eight weeks from the BIS controls; the export-control kink is localized only to ± 1 quarter. Date-localization is necessary, never sufficient, for attribution.

Measurement is a treatment. The single largest causal effect in the corpus — +6 pp of "AI adoption" overnight — was produced by a questionnaire. The EU-Act deficit may partly be labs' disclosure behavior rather than training behavior; the BTOS rollover shifts respondent composition; LMArena's serving configuration moved a pinned model. Instrument changes mimic adoption, compliance mimics avoidance, and platform operations mimic model behavior; designs on the public record need measurement-side placebos (pinned controls, sibling instruments, archived outcomes) as much as treatment-side ones.

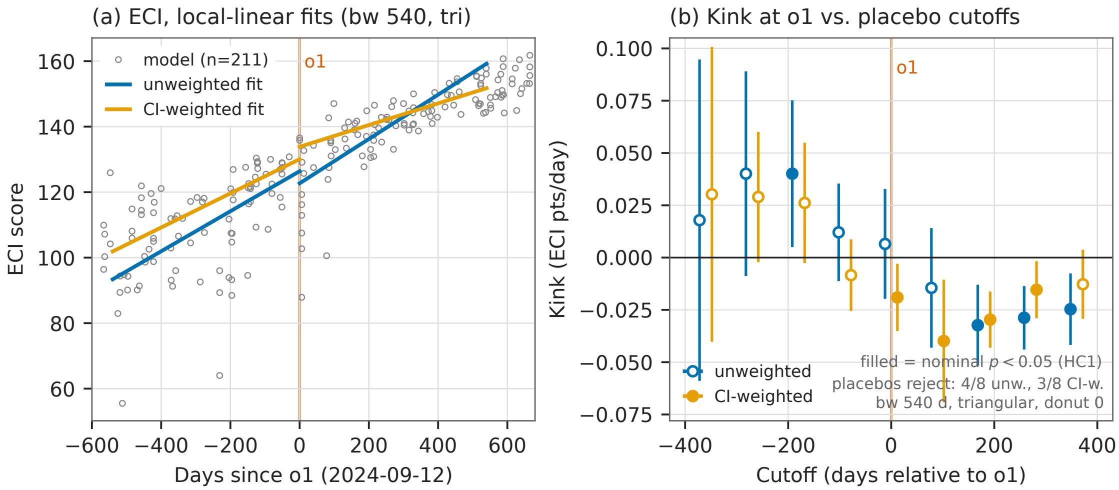

Figure 8: The ECI null (reproduced from [44]): (a) 211 fresh-vintage ECI scores against days since o1-preview with unweighted and CI-weighted per-side fits; (b) kink estimates at the o1 cutoff and eight placebo cutoffs, both weightings — the unweighted design rejects at 4/8 placebos while reading zero at o1.

Nulls and refusals are results. The ECI null is informative because the same specification rejects at 4/8 placebo cutoffs — the dial works and reads zero. The LMArena null is informative because the style break proves the treatment arrived. And the pipeline's refusals (NaN rather than a number when placebo pools are too small) repeatedly prevented fabricated certainty. An automated discovery stack earns trust not by what it finds but by what it declines to claim; the blind Prop-99 recovery, the blind re-localization of the rewording cutoff, and the honestly diagnosed Chinchilla miss are the same machinery validated, confirming, and self-correcting.

Limitations. Calendar-time designs rest on an untestable no-co-located-event assumption that placebo grids probe but cannot certify. Several verdicts lean on few units (4 country groups, 7 sector clusters, 6 quarterly points per cell) where all asymptotic inference is optimistic. Epoch compute figures are partly estimates; benchmark contamination drifts; and every capability result is a property of the release stream, not of individual systems. The survey covers the public record through mid-2026; the queued designs in Table 3 continue it.

Reproducibility

All estimates were produced with the open-source `natex` toolkit [50] (v0.2.0, seeds declared per note) from public data that are never committed to the repository. Every figure in this paper is reused verbatim from a companion note's committed figure, and each of those figures regenerates deterministically from a script that asserts the quoted headline numbers against the analysis records before drawing. The companion notes [39, 41, 46, 40, 42, 47, 45, 44, 43, 48, 49] carry the full specification grids, placebo batteries, and honest-inference notes summarized here.

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Table 3: Considered and excluded: 15 candidate pairs with reasons of record.

Candidate pair	Reason excluded
DCM inference-price series × kink at DeepSeek-V3	series ends 2025-02 with five flat terminal months: no post-slope identifiable
Epoch cyber-vulnerabilities × post-reasoning event study	no processed dataset exists (the published download is the raw CVE git repository)
BTOS state panel × Bartik-style DiD	requires external ACS/BLS exposure weights; ∼40% of state-level cells disclosure-suppressed
R&D input-share table × any family	$n = 7$ rows; no quasi-experimental design applies
GPQA/METR o1 kinks; China DiK × standalone mini-papers	already results of record in [39]; duplication adds nothing
GPU-cluster / datacenter cumulative growth × standalone kink mini-paper	already rejected in the flagship graveyard (placebo empirical size 1.0: curvature, not kink); no standalone treatment
Chip sales + components × chip-level SuDDDS at the Apr-2025 H20 license round	needs a shared chip-alias map and ragged-2026 panel preparation; strongest queued follow-up
Datacenter site-level panel × SuDDDS at Colossus/Stargate entry	18% of timeline rows are unflagged projections requiring filtering; queued
LMArena leaderboard history × RKD around model-spec releases	cumulative Bradley–Terry ratings are serially cumulative series; the GPU-cluster placebo failure is the direct precedent
LMArena style-premium DiD around Model Spec / constitution updates	spec-with-model-update confound requires judge-score validation not yet run
AI-company revenue/WAU × kink at ChatGPT	zero pre-cutoff mass at the treatment date; not formalizable
Polling cross-section × threshold RDs (income step, age cliff)	single survey wave; no microdata, sample sizes, or field dates
Values-in-the-wild / CCAI votes × any design	one aggregate snapshot; no behavioral treatment channel
Benchmark panel × teaching-to-the-test DEE with IRT difficulties	requires a 52-benchmark IRT panel build; queued
All-models panel × bunching at 10^{26} FLOP (US EO line) + disclosure-as-outcome scan	only 3 models ever above the line; no identifiable density contrast

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Three Kinks and a Null: Formal Trend-Break Tests on the Public Record of AI Progress

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July 2026

analysis pass of record: 2026-07-16, `natex` v0.2.0

Abstract

Claims that AI progress "bent" at some dated event — a model release, an export-control round, a regulation — are usually established by eye. We subject four such claims to formal regression kink designs in calendar time and a difference-in-kinks (DiK) design, using the `natex` v0.2.0 estimators (following Böckerman, Jysmä, and Kanninen, 2025) on Epoch AI's public datasets, with a mandatory falsification battery: bandwidth–donut sensitivity grids, shifted-cutoff placebo-kink grids, density and covariate kinks, and a sibling-series falsification. Three kinks and one null survive. GPQA-Diamond scores kink at the o1-preview release (slope roughly 2.4×, placebo empirical size 0/7 — the bend localizes to the date). The METR 50% time-horizon slope nearly triples, but pre-side placebos also reject: a credible reasoning-era bend whose attribution to 2024-09-12 fails. The Epoch Capabilities Index, run as the guard against "everything kinked in 2024," is a clean null at the same date in a series where the test demonstrably has power. And China's legal AI-chip stock bends down by ≈ 0.56 log-units per year relative to hyperscaler controls at the October 2023 export controls, with the October 2022 round as a passing placebo and a smuggling-attenuated total-stock estimate about half the size and fragile. A graveyard of rejected designs — a $t = 12.9$ "kink" that fails every placebo, a sign-unstable Chinchilla null, and an EU AI Act threshold that is a step with bunching rather than a kink — illustrates why eyeballed trend breaks need placebo grids.

1 Introduction

The public conversation about AI progress is full of dated trend breaks. Reasoning models "changed the slope" in late 2024; export controls "bent" China's compute accumulation; the EU AI Act "chilled" large training runs. These claims are almost always established by drawing a line through a scatter plot and pointing at a date. Yet a slope change at a known cutoff is precisely the estimand of the regression kink design (RKD), and a slope change for one group relative to another at the same cutoff is the estimand of the difference-in-kinks (DiK) design formalized by Böckerman, Jysmä, and Kanninen [1]. Treating eyeballed breaks as formal kink candidates buys three things: an explicit identifying assumption, a standard error, and — most importantly — a falsification battery that can *reject* the claim.

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This paper runs that battery on four trend-break claims drawn from Epoch AI's public datasets [3]. The tooling is `natex` v0.2.0 [4], an open-source natural-experiment toolkit in the lineage of automated discontinuity-design discovery [2], whose kink module implements sharp and fuzzy RKD and DiK estimators following [1]. We emphasize that this is known-cutoff candidate *evaluation*, not unknown-kink discovery: every cutoff below is externally dated (a model release, a regulation date), and none was searched for. Searching for cutoffs would require search-calibrated selective inference beyond the scope of both the source paper and this exercise.

Why do eyeballed breaks need placebo grids? Because a calendar-time kink test asks a sharper question than the eye does. A series can genuinely accelerate over an *era* while providing no evidence that the bend happened at any particular *date*; conversely, a smoothly super-exponential series will show a nominally enormous "kink" at every date one tests. The shifted-cutoff placebo grid separates these cases: a significant kink at the true cutoff alongside significant kinks at shifted cutoffs means bend existence without date attribution, and rejections everywhere mean curvature masquerading as a kink. Our four headline results span this taxonomy almost perfectly — one date-localized kink, one era bend that fails date attribution, one clean null run deliberately as a falsification guard, and one difference-in-kinks with a passing placebo round — and the graveyard (Section 5) shows each failure mode in the wild.

2 Data

All series are public Epoch AI datasets (CC-BY 4.0, `https://epoch.ai/data`) [3], retrieved July 2026; the analysis pass of record is dated 2026-07-16 and used `natex` v0.2.0. No data files ship with this paper; the figure pipeline (`paper/figures/make_figures.py` in the repository) reads a local extraction of the public CSVs, recomputes every headline estimate, and asserts it against the numbers of record before drawing.

GPQA-Diamond. Benchmarking-Hub series `gpqa_diamond`: 180 dated models (45 pre / 135 post cutoff), sample starting 2023-03-14. Primary outcome is logit(mean score): the score ceiling at 1 bends raw-score slopes mechanically near the top, and the logit removes that artifact. GPQA is the graduate-level science QA benchmark of [6].

METR 50% time horizon. `metr_time_horizons_external` (METR's long-task suite [5] as mirrored by Epoch): 48 dated models (12 pre / 36 post). Outcome is $\log_2$ of the 50%-success time horizon in minutes.

Epoch Capabilities Index (ECI). `epoch_capabilities_index`, an IRT-linked aggregate capability index; 356 dated, scored models. Used as the sibling-series falsification.

AI chip stocks. `ai_chip_owners/cumulative_by_designer`: quarterly cumulative compute stocks in H100-equivalents by owner. Outcome is ln(cumulative H100e); treated series China, control series the hyperscalers (Amazon + Microsoft + Google + Meta). The 2026-03-31 quarter is flagged `Incomplete` and its "cumulative" totals *decrease*; it is dropped.

3 Methods

3.1 Estimators

Let x be the running variable centered at the cutoff and let $m[\mathrm{s}]$ denote the one-sided derivative of $E[Y \mid x]$ at zero on side s. `natex` defines every kink right-minus-left,

$$\kappa \;=\; m[\text{right}] - m[\text{left}], \tag{1}$$

estimated by local polynomial fits in which each side (and each group cell for DiK) receives its own intercept and slope, weighted by a kernel in $|x| \le$ bandwidth (the weighted objective of [1], Equation 8). Defaults throughout: local linear fits, triangular kernel; uniform kernel, second-degree fits, and donut exclusions appear as explicit sensitivity choices. The sharp DiK estimand contrasts two such kinks,

$$\tau_{\text{DiK}} \;=\; \big(\kappa_{\text{treated}} - \kappa_{\text{control}}\big)/\Delta, \tag{2}$$

with Δ the known policy-kink change. In the China design the paper's pre/post *periods* are aliased to control/treated *groups* at a common calendar cutoff: the export-license schedule bends for China only, so the Böckerman contrast is taken across groups and the controls difference out the global supply bend. Identification is parallel kinks: absent the controls, China's growth bend would have matched the hyperscalers'.

All calendar-time runs use a dummy unit denominator ($\Delta = 1$, the `--policy-kink 1.0` convention), so τ is a *descriptive slope change* in outcome units per day, not a marginal causal response to a measured policy variable.

3.2 Time as the running variable

The governing caveat, quoted verbatim from the `natex` method card, `docs/method_cards/kink.md`:

> Calendar-time RKD — "did the trend bend at this dated event?" — is a legitimate use of the estimator but **not** a manipulable-running-variable design: no unit sorts itself across a date, so density/manipulation arguments give no protection. Identification reduces to a single untestable assumption: *no co-located slope-changing event* — nothing else may bend the outcome's expected slope at that exact date.

Two practices are therefore mandatory rather than optional. First, always run the shifted-cutoff *placebo-kink grid* and read it as separating bend existence from date attribution: a significant kink at the true cutoff plus significant kinks at shifted cutoffs means the series bends over an era, not at the event. Second, run a *sibling-series falsification*: an aggregate or related series in which the same test demonstrably has power but reads null at the candidate date. Sections 4.2 and 4.3 show both practices doing real work.

3.3 Inference

Standard errors are HC1 sandwich by default, with the degrees-of-freedom correction applied *jointly* across all side/period cells ($n-k$ counts every observation and coefficient of the stacked regression). Cross-checks fitting each side separately with per-side $n-k$ reproduce the point

Series (source)	Design / cutoff	Headline (HC1)	Verdict
GPQA-Diamond, logit(mean score), 180 models	sharp RKD-in-time at o1-preview, 2024-09-12	+0.00258/day, se 0.00078, $t = 3.32$ (bw 540, tri)	credible kink, date-localized
METR 50% time horizon, $\log_2$ minutes, 48 models	sharp RKD-in-time at o1-preview, 2024-09-12	+0.00601/day, se 0.00282, $t = 2.13$ (bw 720, tri)	credible era bend; date attribution fails
Epoch Capabilities Index, all models	sharp RKD-in-time at o1-preview (falsification)	−0.00725 pts/day, se 0.00895, $t = -0.81$ (bw 540, tri)	clean null — the guard the other two need
China vs. hyperscaler ln cumulative H100e stock	sharp group-DiK at export controls, 2023-10-17	−0.00154/day, se 0.00047, $t = -3.30$ (bw 548, tri)	credible kink with a magnitude honesty band

Table 1: The four formal results of the pass (`natex` v0.2.0, HC1 standard errors). "tri" = triangular kernel; bandwidths in days.

estimate exactly and report SEs roughly 5% larger at $n \approx 50$: in the METR cross-validation (uniform kernel, bandwidth 720), `natex` and an independent per-side OLS fit agreed on the kink to 10 decimals (both 0.0067291763) with SEs 0.0023236 (joint-cell) versus 0.0024334 (per-side) — a convention difference, not an error. Cluster-robust CR1 covariance with $t(G-1)$ critical values is used where clustering is stated. Wald intervals are conventional local-polynomial inference and may retain smoothing bias; degree-2 and bandwidth sensitivity are part of the required battery. Failed computations return NaN, never zero. The diagnostics battery — bandwidth×donut sensitivity grids, placebo-kink grids with empirical size, density kinks, covariate kinks — follows the validation figures of [1] as implemented in `natex.kink`.

4 Results

Table 1 summarizes the four designs. All candidates had externally dated cutoffs; none was searched for.

4.1 GPQA-Diamond at o1-preview: a date-localized kink

The logit-score slope changes from 0.00188 to 0.00446 logit-units per day at the o1-preview release — in raw terms roughly 15 → 34 percentage points per year, a $\sim$ 2.4× acceleration (Figure 1). The headline estimate is +0.00258 per day (se 0.00078, $t = 3.32$; bandwidth 540, triangular kernel, cells 44/111). The battery is uniformly supportive: the estimate is positive in 8/8 bandwidth×donut cells (t from 2.15 to 3.71), and the donut *strengthens* it, ruling out single-point dependence at the cutoff; the placebo-kink grid has empirical size **0/7** — no shifted cutoff rejects; a McCrary-style release-date density kink is null ($t = 0.76$), so the score bend is not an artifact of a release-frequency bend; a training-compute covariate kink is null ($p = 0.54$); and a degree-2 fit keeps the sign and magnitude at three times the standard error. This is the cleanest result of the pass: the bend localizes to the date.

The standing caveat belongs in every sentence of interpretation: composition — reasoning models entering the release stream — *is* the mechanism, so τ is a property of the release stream,

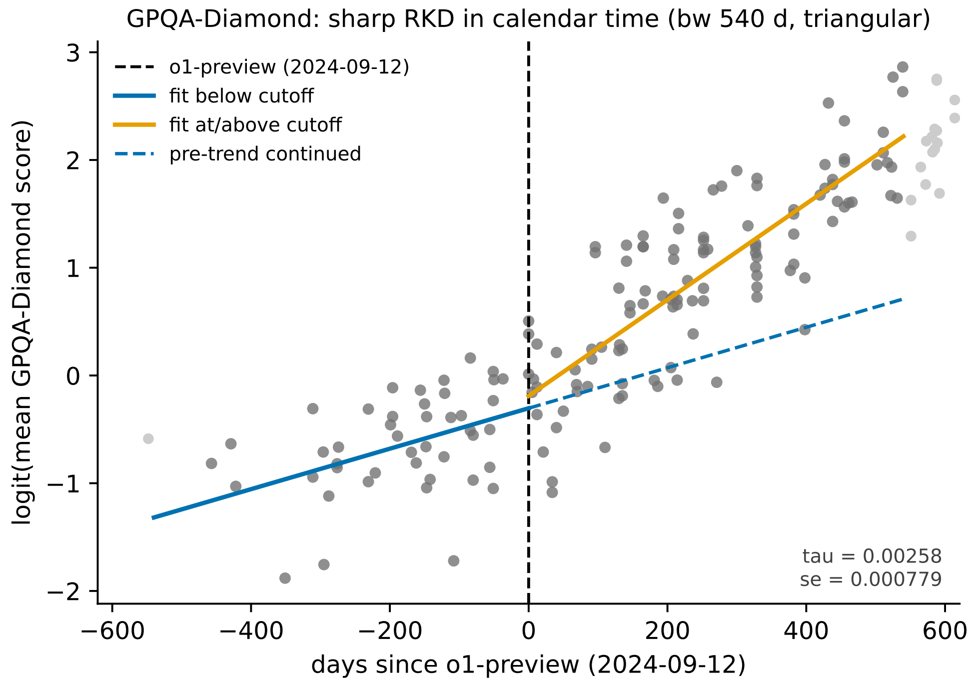

Figure 1: GPQA-Diamond, logit(mean score) against days since o1-preview (2024-09-12). Solid lines: kernel-weighted local-linear fits on each side (bandwidth 540 days, triangular); dashed: the pre-cutoff trend continued. The slope roughly 2.4×'s at the cutoff, and the placebo grid (empirical size 0/7) localizes the bend to the date.

not a statement that individual models improved. Benchmark contamination drifting over time remains an unmodeled confounder.

4.2 METR time horizon: a real bend that cannot be dated

The $\log_2$ 50%-time-horizon slope rises from 0.00331 to 0.00932 per day at bandwidth 720 (triangular): a doubling time of 9.9 months before the cutoff falling to 3.5 months after (Figure 2). The headline kink is +0.00601 (se 0.00282, $t = 2.13$); the full-sample uniform-kernel fit gives +0.00491 (se 0.00095, $t = 5.17$; doubling time 7.6 → 3.6 months). The estimate is positive in 8/8 bandwidth×donut cells (0.0044–0.0085), and the 80%-horizon variant agrees (+0.00454, $t = 4.08$), though with only three pre-cutoff points it is directional corroboration only.

But the placebo grid smears on the pre side: at bandwidth 720, shifted cutoffs at −270, −180, and −90 days all reject ($p = 0.030$, 0.014, 0.004) with estimates the size of the headline kink, while post-side shifts are clean nulls (empirical size 3/6). With 11–12 pre-cutoff models, adjacent placebo windows share most of their data, and the design cannot distinguish 2024-09-12 from any date within roughly ±270 days on the pre side. The honest verdict, quoted from the analysis of record: *"credible slope change with failed date-localization — report as reasoning-era slope doubling, not 'o1-preview caused X'."* This is the era-bend/event-bend contrast with Section 4.1 in a single dataset pair.

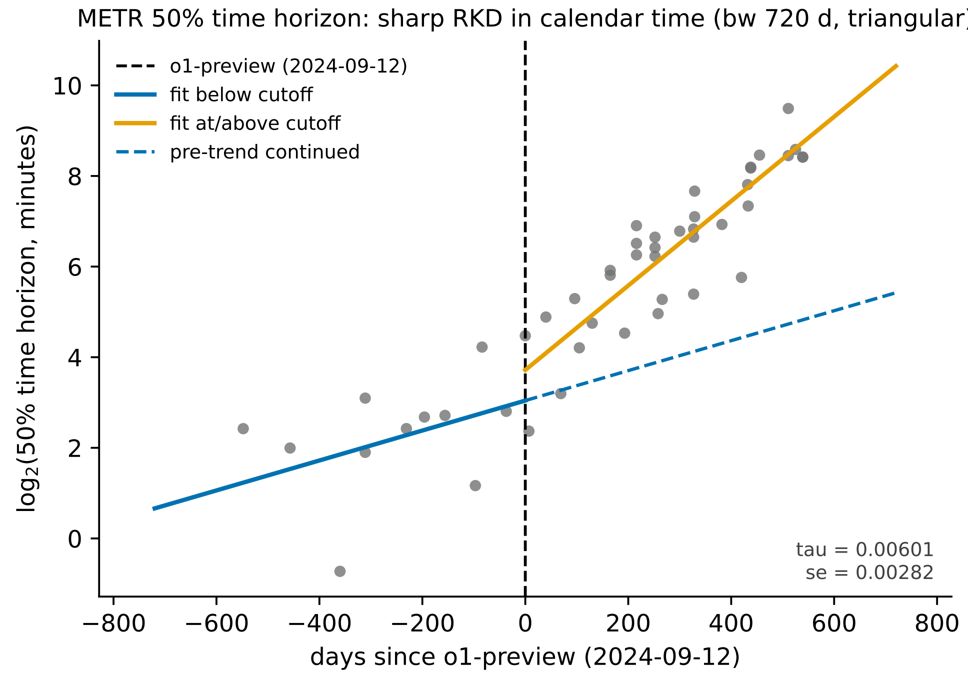

Figure 2: METR 50% time horizon ($\log_2$ minutes) against days since o1-preview. The slope nearly triples (doubling time 9.9 → 3.5 months at bandwidth 720), but pre-side placebo cutoffs also reject: an era bend, not a dated event.

4.3 The Epoch Capabilities Index: a null with power

If "everything kinked in 2024" — a global measurement or composition shift — then per-benchmark kinks would be uninformative. The ECI falsification run was launched *expecting* a null, and it delivered one: at the o1 date the all-models kink is null in 8/8 grid cells (t from −0.60 to −1.54; headline −0.00725 points/day, se 0.00895), with the index gaining roughly 18 points per year on both sides of the cutoff (Figure 3). Crucially, the same series' placebo grid rejects at 4/7 shifted cutoffs — the test demonstrably has power in this data, from composition-churn bends elsewhere in the series — and still reads zero at 2024-09-12. The instrument works; the dial reads zero. The per-benchmark bends of Sections 4.1–4.2 are therefore not a global measurement artifact. This is partly by construction (IRT linking smooths regime shifts) and partly because o1-era gains concentrate in reasoning-heavy benchmarks.

Two recorded lessons ride along. First, an analyst-script frontier filter — `cummax().diff().fillna(1) > 0` over a series with 231 NaN scores — manufactured 347 false "frontier records"; the clean frontier has 25 points and is *not* an estimable design — four left-cell points, failing donut, bandwidth-direction, and placebo checks in every direction — and should never be quoted. Second, a side finding: the training-compute covariate kink among ECI-listed models is −0.00265 $\log_{10}$ FLOP/day at the o1 date ($p = 0.021$), corroborating the "pretraining-compute frontier stalls" narrative without itself being a design.

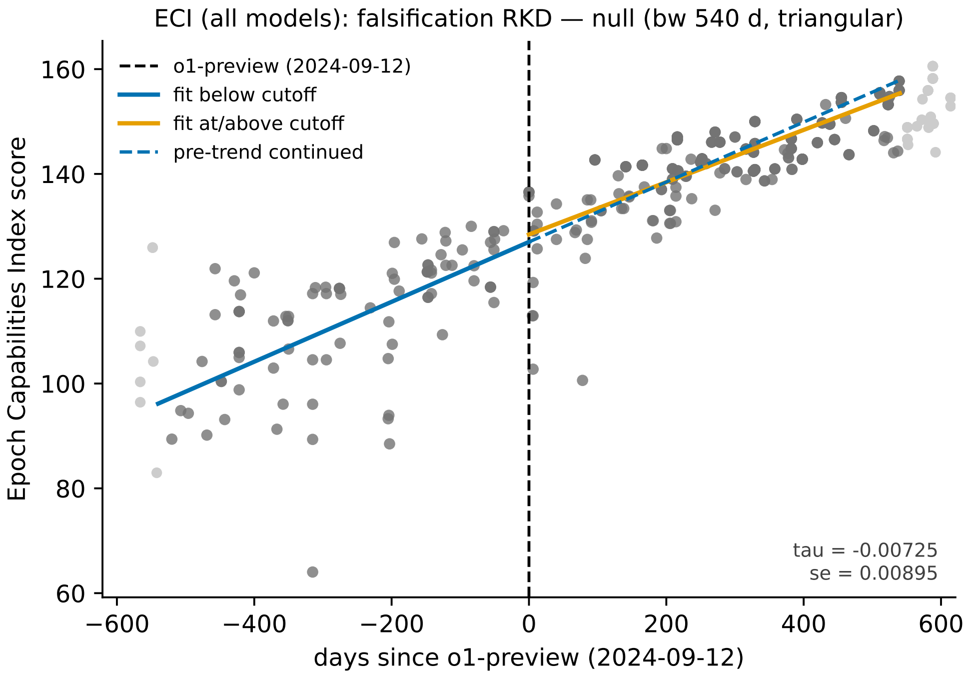

Figure 3: Epoch Capabilities Index (all models) against days since o1-preview. The two-sided fit and the continued pre-trend nearly coincide: a clean null ($t = -0.81$) in a series where the placebo grid shows the test has power (4/7 shifted cutoffs reject). The guard the per-benchmark kinks need.

4.4 China's chip stock at the October 2023 export controls

The only true difference-in-kinks structure in the corpus: the US export-license schedule bends for China only, at a common dated cutoff (2023-10-17), so treated = China and control = hyperscalers (Amazon + Microsoft + Google + Meta), and the controls difference out the global supply bend (their own kink: -0.00067/day). The legal-stock DiK is -0.00154 log-units per day (se 0.00047, $t = -3.30$; bandwidth 548, triangular), i.e. a growth-rate change of roughly **-0.56 log-units per year** (Figure 4). By-year CR1 clustering keeps significance ($t = -5.15$) with the explicit few-cluster caveat ($G = 4$). The estimate is negative in **25/25** specifications.

Falsifications behave: the October 2022 first-round placebo — the round whose A800/H800 compliance loophole predicts no bite — is null/positive ($+0.0018$, $p = 0.15$), and the placebo-treated group ("Other" vs. hyperscalers) is clean at bandwidths 548–730 while rejecting at 365, so the defensible range is bandwidth 548–730 with DiK -0.0009 to -0.0015. The honesty band: the *total*-stock series including smuggled chips gives a DiK roughly half the size and fragile (-0.00076, $t = -1.54$ at bandwidth 548) — the smuggled series, which starts 2024-03-31 and is itself a treatment response, substitutes for lost legal supply. Serial dependence in a six-points-per-cell cumulative series makes every $|t|$ optimistic; the sign stability is the real evidence. The verdict of record: *"the policy bent the legal channel* ≈ -0.3 *to* -0.56 *ln-units/yr; the effect on China's actual compute stock is smaller and fragile."*

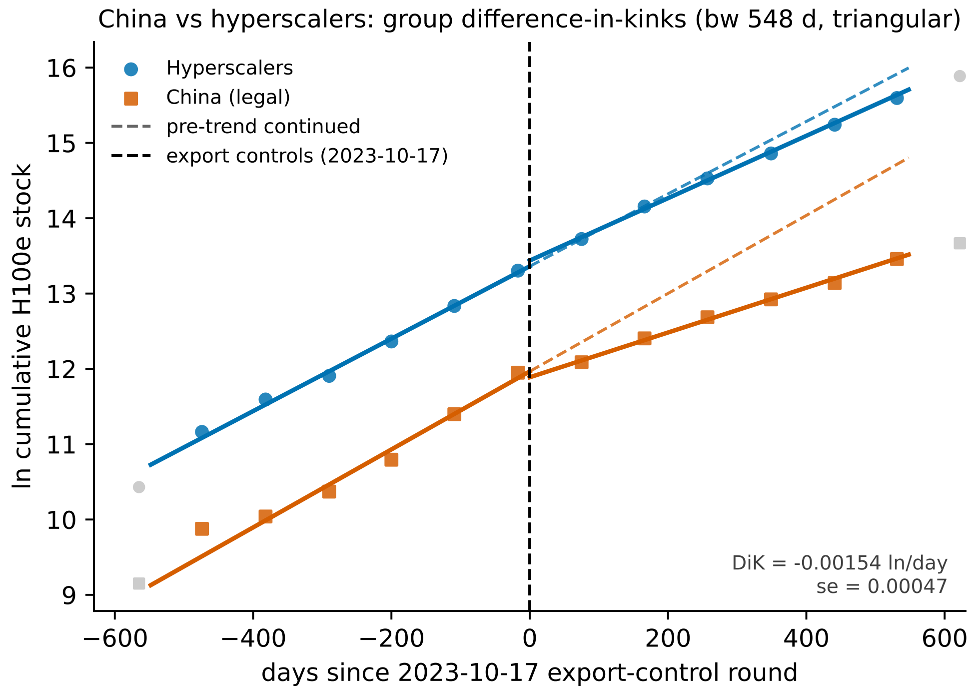

Figure 4: Group difference-in-kinks: ln cumulative H100e stock for China (legal) and the hyperscaler controls against days since the 2023-10-17 export-control round (bandwidth 548 days, triangular). Both series bend at the cutoff; the DiK is their difference, -0.00154 log-units/day (≈ -0.56/yr).

5 The graveyard

Rejections are results. Three claims that look at least as impressive as Table 1 in a scatter plot died in the battery.

Datacenter growth "kinked at ChatGPT." The GPU-cluster series (ln cumulative H100e by first-operational date) shows a nominal kink at ChatGPT's release with $t = 12.9$ — the largest t-statistic of the entire exercise. Its placebo grid rejects at **7/7** shifted cutoffs (empirical size 1.0): a smoothly super-exponential cumulative series bends *everywhere*, and HC1 inference on serially dependent cumulative data is meaningless. The design was discarded; the estimate is descriptive curvature, not a kink. (The dataset variant named in the original claim, datacenter power timelines, was infeasible outright: one pre-ChatGPT observation.) This is the cautionary tale for every "the curve bent at [event]" plot drawn on cumulative infrastructure data.

Chinchilla. Tokens-per-parameter among language base models at the Chinchilla paper's release date (2022-03-29): kink $t = -0.91$, -0.05, $+1.48$ at bandwidths 365, 540, 730 — a sign-unstable null. Even a genuine regime change (prior work shows a two-to-three-month adoption ramp) does not produce a slope kink dateable to the paper's release.

The EU AI Act threshold is a step with bunching, not a kink. The Act's systemic-risk presumption at 10^{25} training FLOP imposes a *level* discontinuity in obligations, not a marginal-rate kink, so RKD/DiK is the wrong estimand shape. Worse for any kink or RD design at that cutoff, the running variable sorts: post-Act, the model-density just above the line is depleted (below/above contingency: Fisher exact OR = 0.17, 95% CI 0.05–0.62, $p = 0.0071$; roughly 83% of the expected above-line mass missing), which violates any no-sorting requirement. Kink-shaped probes confirm nulls (e.g. an open-weights-share DiK of -0.93, se 0.72). The bunching test remains the right tool at this threshold — and the bunching itself, an avoidance response specific to the statutory line and the post-Act period, is the substantive finding.

6 Discussion

Four lessons generalize beyond these datasets.

Era bends and event bends are different claims. The METR and GPQA series tell almost the same visual story — capability slopes steepen around late 2024 — yet the placebo grid cleanly separates them: GPQA's bend localizes to the o1 date (size 0/7) while METR's does not (pre-side size 3/6). Public discussion rarely distinguishes "the trend changed in this era" from "this event changed the trend"; the shifted-cutoff grid makes the distinction mechanical.

Nulls need power certificates. The ECI falsification is informative precisely because the same series rejects at 4/7 shifted placebo cutoffs: the test has power in this data, and still reads zero at the candidate date. A null from an underpowered test would have guarded nothing.

Report honesty bands, not point claims. The China result is strongest as a band: legal-channel bend ≈ -0.3 to -0.56 log-units/yr across the defensible bandwidth range, total-stock effect roughly half and fragile because smuggling — itself a response to the policy — substitutes for legal supply. Similarly, GPQA's τ is a release-stream property with composition as the mechanism, not per-model improvement.

The graveyard is the argument. The most significant statistic of the exercise ($t = 12.9$) belongs to a design that fails every placebo. Any workflow that stops at "the kink is significant" would have led with it. On the public record of AI progress, where series are short, serially dependent, and composition-churned, the falsification battery is not a robustness appendix — it is the analysis.

Limitations. Calendar-time identification rests on an untestable no-co-located-event assumption; placebo grids probe but cannot certify it. Several series are short (6 quarterly points per cell in the DiK; 11–12 pre-cutoff models for METR), making serial dependence corrections coarse and $|t|$ optimistic. Epoch's compute figures are partly estimates, and benchmark contamination drifts over time. All results are properties of the *public record* of AI progress — release-stream composition included — not of any individual system.

Reproducibility

Estimates were produced with the open-source `natex` v0.2.0 kink module [4] on public Epoch AI data [3]; the per-design analysis records (inputs, CLI outputs, diagnostics JSON) and the case-study write-up of record live in the `natex` repository under `docs/case_studies/` (`epoch-kinks.md`). The four figures regenerate deterministically from the public CSVs via the committed script `paper/figures/ make_figures.py`, which asserts every headline number against the case study before drawing.

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A Sharp Bend, No Verdict: Difference-in-Kinks Tests of US Business AI Adoption at DeepSeek-R1

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

US Census Bureau Business Trends and Outlook Survey (BTOS) data show a sharp, well-localized acceleration in business AI use among AI-exposed sectors (information; finance and insurance; professional/scientific/technical services) beginning with the first survey wave after the release of DeepSeek-R1 on 20 January 2025. A sector-by-wave difference-in-differences scan localizes the break at exactly that wave (scan $p = 0.010$–0.030); the exposed-minus-unexposed gap slope more than triples, from +4.2 to +13.9 pp/yr; and a group difference-in-kinks (DiK) estimate at the release date is nominally significant (+7.3 pp/yr, cluster-robust se 2.8). Attribution to DeepSeek-R1, however, fails the falsification battery: placebo cutoffs at non-event 2024 dates produce DiKs as large as or larger than the R1 estimate (+11.4 pp/yr at $t = 2024.30$; +4.0 at $t = 2024.50$; both significant); only 2 of 6 specification variants at R1 are significant; equally large local kinks appear in non-exposed sectors; the predicted *deceleration* at OpenAI's o1 release is insignificant at the primary bandwidth and sign-flips at the wider one; and the cutoff week is confounded by the AI diffusion export rule (13 January), the Stargate announcement (21 January), and a BTOS sample-year rollover in mid-December 2024. The January-2025 acceleration is real but descriptive: no causal attribution to DeepSeek-R1 survives triage.

1 Introduction

On 20 January 2025 the Chinese lab DeepSeek released R1, an openly licensed reasoning model trained at a reported cost far below frontier-lab budgets [6]. The release was treated by markets as news about the cost curve of AI: on 27 January 2025 Nvidia alone lost roughly $590 billion of market capitalization, the largest one-day loss on record, and an event-study literature on the "DeepSeek shock" emerged almost immediately.[1] A natural real-economy question follows: did cheap, open reasoning models bend the *adoption* curve — the share of US businesses actually using AI in production?

Measurement of that adoption curve is recent. The Census Bureau's Business Trends and Outlook Survey (BTOS) has asked a probability sample of US firms since September 2023 whether they used AI in producing goods or services in the last two weeks; [2] introduce these data and document a rise in the national use rate from 3.7% to 5.4% over the first six months, concentrated in the Information sector and larger firms. Complementary measurement includes the 2018 Annual Business

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

[1] See, e.g., "Nvidia drops $600B off its market cap amid the rise of DeepSeek," TechCrunch, 27 January 2025. We do not rely on this financial-markets literature for identification; it motivates the candidate event date only.

Survey module of [8] (pre-LLM adoption below 6% of firms, over 18% employment-weighted) and the multi-country executive survey of [10] (high headline adoption, small realized effects). This literature is descriptive by design; formal tests of whether adoption *trends* broke at specific AI events are, to our knowledge, absent.

A trend break for one group relative to another at a known date is the estimand of the difference-in-kinks (DiK) design formalized by [1], building on the regression kink design of [4]. Kink estimates in time-series settings are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [5]. This paper applies the `natex` toolkit's DiK and scan-based DiD estimators [9, 7] to the BTOS sector panel with DeepSeek-R1 as the candidate cutoff, o1 (12 September 2024) as a secondary contrast, and a mandatory falsification battery, which converts an apparently significant event-study result into an explicit non-attribution.

2 Data

The primary source is the Census Bureau's BTOS AI core workbook (DRB approval CBDRB-FY25-0425): sector-level biweekly estimates and standard errors for Question 7, "In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services?", answer "Yes," in percentage points. The panel covers 20 sectors (19 two-digit NAICS sectors plus an unclassified residual) over 54 biweekly waves, 202319–202520 (collection periods 2023-09-11 through 2025-10-05); 199 of 1,080 sector-waves are disclosure-suppressed, leaving 881 usable cells. The panel predates the 2025-12-03 questionnaire rewording entirely. The running variable t is the fractional year of each wave's collection-period midpoint. The exposed group is {51 information, 52 finance and insurance, 54 professional/scientific/technical services}; the kink-leg comparison group is {11, 21, 23, 72}. A placebo outcome (the archived remote-work Question 6 "Yes" share) comes from the BTOS archive workbook (CBDRB-FY23-0478/FY24-0474/FY25-0117), waves 202417–202514.

Coverage is the binding data limitation: sectors 11, 21, 22, and 55 are almost entirely suppressed (3, 3, 4, and 8 usable waves of 54), so the "unexposed" comparison group is effectively sectors 23 (construction) and 72 (accommodation and food services).

3 Design and Methods

Group difference-in-kinks. Following [1], the DiK estimand is the change in the slope of the outcome in calendar time at a known cutoff for the exposed group *minus* the same slope change for the comparison group, estimated by local linear regression on each side of the cutoff within each group (`natex` v0.2.0 [9]). The primary specification uses cutoff $t_0 = 2025.055$ (DeepSeek-R1; first post wave 202503, collected 2025-01-27 to 02-09), bandwidth 0.5 years (± 13 waves), triangular kernel, a one-wave donut, and cluster-robust (CR1) standard errors clustered by sector with 7 clusters and critical values from $t(G-1)$. With 7 clusters and only 2–4 per DiK cell, CR1 inference is fragile even with the small-sample correction [3]; we therefore lean on design-based falsification rather than on the standard errors.

Falsification battery. (i) a specification grid over bandwidth $\in \{0.25, 0.5, 1.0\}$ years, kernel $\in$ {triangular, uniform}, and donut $\in \{0, 1 \text{ wave}\}$; (ii) placebo cutoffs at four non-event dates (2024.30, 2024.50, 2024.85, 2025.30), in the spirit of the permutation logic of [5]; (iii) a placebo-in-space battery of per-sector local kinks at the R1 date; (iv) a placebo *outcome* (remote work) at the same cutoff; and (v) a secondary contrast at o1 (2024-09-12, $t = 2024.6967$): if R1 mattered over

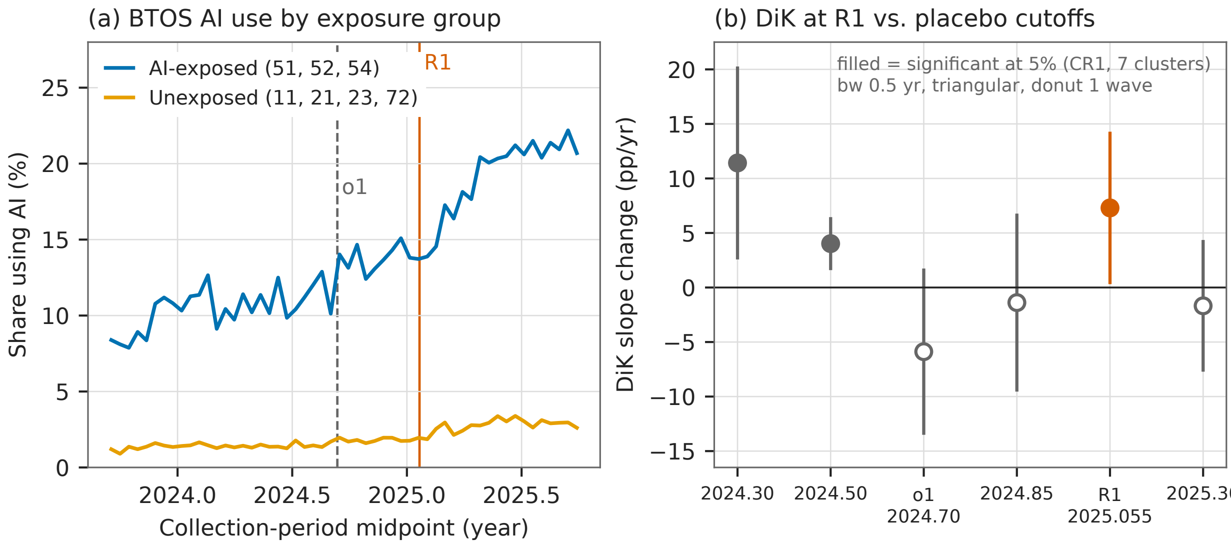

Figure 1: (a) BTOS AI-use share by exposure group across all 54 waves, with the o1 and DeepSeek-R1 release dates marked. (b) Group difference-in-kinks estimates (primary specification: bandwidth 0.5 yr, triangular kernel, one-wave donut) at the R1 cutoff, the o1 cutoff, and four non-event placebo dates, with 95% CR1 confidence intervals; filled markers are significant at the 5% level against $t(G-1)$ critical values. Two non-event placebos are significant, one larger than the R1 estimate.

and above the reasoning-model wave o1 began, the exposed-minus-unexposed gap should *decelerate* there.

Sector-by-wave DiD. The `natex` survey pipeline's sector-by-wave DiD scan (SuDDDS) searches over candidate break dates t_0 and window widths W, maximizing a likelihood-ratio scan statistic for a declared-treatment DiD, calibrated by permutation under an AR(1)-by-unit null [7]. We also fit a manual two-way fixed-effects (TWFE) level DiD with wave and sector fixed effects, CR1 by sector, and a placebo-in-space randomization check.

Honest-inference notes, stated as run. The pipeline's refusals are reported, not patched over: the SuDDDS anticipation test refused (Holm $p = [\mathrm{NaN}, \mathrm{NaN}, \mathrm{NaN}]$); the dd/synthetic-control/GESS effect estimators refused with $\tau = \mathrm{NaN}$ (0 usable placebo units on this unbalanced panel, ≥ 5 required), hence the manual TWFE estimate below; and the automated survey's own family verdicts were discarded as role-assignment artifacts (its "credible" RDD, scan $p = 0.010$, $\tau_{2\mathrm{sls}} = 0.749$, se 0.343, is the mechanical rediscovery of candidate treatment = post with outcome $= t$; its kink family, Holm $p = 0.807$, tested outcome = sector — vacuous; its DiD family, $p = 0.70$, used a time-invariant treated indicator rather than the declared design; iv/sc/bunching returned needs-input; dee was null with 1 usable experiment).

4 Results

The bend is real ... The SuDDDS scan localizes a break at $t_0 = 2025.089$ — exactly the first post-R1 wave (202503) — with window $W = 0.536$ yr: max LLR = 22.65, scan $p = 0.030$ on the 20-sector panel, and LLR = 32.70, $p = 0.010$ (the $q = 99$ permutation floor) on the 7-sector

Table 1: Group difference-in-kinks estimates, BTOS AI-use share (pp/yr).

Specification	$\hat{\tau}$	se	$\lvert t \rvert$	sig. 5%
Panel A: DeepSeek-R1 cutoff ($t_0 = 2025.055$)				
triangular, bw 0.5, donut (primary)	+7.297	2.797	2.61	$*$
triangular, bw 0.5, no donut	+6.742	1.188	5.68	$*$
uniform, bw 0.5, donut	+4.477	3.079	1.45	
triangular, bw 1.0, donut	+2.350	2.934	0.80	
uniform, bw 1.0, donut	+3.941	2.478	1.59	
triangular, bw 0.25, donut	+10.488	8.130	1.29	
Panel B: placebo and secondary cutoffs (triangular, donut)				
$t_0 = 2024.30$ (no event), bw 0.5	+11.419	3.556	3.21	$*$
$t_0 = 2024.50$ (no event), bw 0.5	+4.023	0.889	4.53	$*$
$t_0 = 2024.85$ (no event), bw 0.5	−1.385	3.118	0.44	
$t_0 = 2025.30$ (no event), bw 0.5	−1.680	2.293	0.73	
o1, $t_0 = 2024.6967$, bw 0.5	−5.884	2.909	2.02	
o1, $t_0 = 2024.6967$, bw 1.0	+3.818	1.417	2.69	$*$

Notes: CR1 standard errors clustered by sector; significance against $t(G-1)$ critical values with $G \in \{5, 6, 7\}$ clusters in the estimation window (2.447–2.776). Bandwidths in years (0.5 yr = ± 13 waves); donut = one wave. Outcome: sector-level share answering Yes to BTOS Question 7, percentage points.

subpanel; the composition check passes ($p = 1.000$). The SE-weighted exposed-minus-unexposed gap slope over ± 13 waves rises from +4.197 (se 2.089) to +13.940 (se 1.649) pp/yr, $\Delta = +9.74$ (se 2.66). The primary group DiK at R1 (Table 1, first row) is $\hat{\tau} = +7.297$ pp/yr (CR1 se 2.797, 95% CI $[0.454, 14.140]$, $|t| = 2.61$ against $t(6)$ critical value 2.447, $n = 123$): the exposed group's slope jumps from 3.234 to 12.726 pp/yr against 0.485 to 2.680 for the comparison group. The manual TWFE level DiD gives $\hat{\tau} = +4.583$ pp (CR1 se 0.526, $t = 8.72$), and placebo-in-space is clean: 0 of 13 control-sector pseudo-treatments reach the minimum exposed $|t|$ (controls' max 7.76 vs. exposed 10.19/10.49/13.19; exact-p floor $1/14 = 0.071$), with per-sector effects of +4.26 (51), +4.14 (52), and +5.35 (54) pp.

... but attribution to R1 fails. Five findings block a causal reading. *First*, the placebo-cutoff battery fails: at the non-event dates 2024.30 and 2024.50 the same DiK specification returns +11.419 (se 3.556, $|t| = 3.21$) and +4.023 (se 0.889, $|t| = 4.53$) — both significant, the first larger than the R1 estimate itself (Table 1, panel B; Figure 1b). *Second*, the R1 estimate is significant in only 2 of 6 specification variants. *Third*, the kink is not exposure-specific: in per-sector local kinks at R1, exposed sector 52 ranks 1/16 (+15.41, $t = 6.36$), but non-exposed sectors 71 (+10.46, $t = 4.24$), 53 (+10.00, $t = 3.82$), and 62 (+7.94, $t = 6.62$) all exceed exposed sectors 51 (+6.56, $t = 1.03$) and 54 (+6.50, $t = 3.13$) — the January-2025 kink is broad-based. *Fourth*, the o1 deceleration contrast is bandwidth-aliased: −5.884 (se 2.909, $|t| = 2.02 <$ crit 2.571, not significant, though the sign matches the prediction) at bandwidth 0.5, but +3.818 (se 1.417, $|t| = 2.69$, significant, sign *flipped*) at bandwidth 1.0; the SE-weighted gap-slope change at o1 is +0.59 (se 2.63) — the expected deceleration is not reproduced. *Fifth*, the TWFE level DiD, though clean in space, violates parallel trends: the pre-R1 gap slope is +4.20 pp/yr (se 2.09), so the level estimate is descriptive. The placebo *outcome* is consistent with a broad shock rather than an AI-specific one but is underpowered: the remote-work DiK at R1 is +8.821 (se 5.351, $|t| = 1.65$, not significant, on a truncated pre-window) — same sign and magnitude as the AI outcome.

5 Caveats and Conclusion

Three caveats are decisive. First, a calendar-time cutoff absorbs every same-week event: the Framework for Artificial Intelligence Diffusion export rule (2025-01-13) and the Stargate announcement (2025-01-21) fall in the same window as R1 (2025-01-20), so a DiK at $t = 2025.055$ cannot distinguish among them — and the BTOS sample-year rollover (wave 202426 $\rightarrow$ 202501, mid-December 2024) changes respondent-panel composition three to five weeks before the cutoff. Second, the comparison group is effectively two sectors: NAICS 11, 21, 22, and 55 are almost entirely disclosure-suppressed, and CR1 inference rests on 7 clusters with only 2–4 clusters per DiK cell, where cluster-robust standard errors are known to over-reject [3]. Third, and most important, the placebo-cutoff battery finds slope breaks as large as the R1 estimate at dates where nothing happened — exactly the failure mode placebo distributions exist to catch [5]: on this panel, the DiK test cannot tell the R1 date apart from ordinary undulation in a steep adoption curve.

The verdict is descriptive-only. A sharp, well-localized acceleration in measured AI adoption in AI-exposed sectors did begin with the first BTOS wave after 20 January 2025 — the localization is data-driven and survives composition and placebo-in-space checks — but every test that would pin the bend to DeepSeek-R1 specifically fails: placebo dates match it, most specifications miss it, unexposed sectors show it too, and the o1 contrast sign-flips. Something bent in January 2025; these data cannot say it was R1.

Reproducibility. All estimates were produced with `natex` v0.2.0 [9] at seed 0 from the frozen BTOS workbooks (published at `https://www.census.gov/hfp/btos`; not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the primary estimate against the numbers of record before drawing.

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Bent, Bypassed, Unbowed: US AI-Chip Export Controls and China's Compute

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analysis pass of record: 2026-07, `natex` v0.2.0, seeds 7 (DiD) and 20260716 (DiK battery; estimator deterministic)

Abstract

The October 2023 US export controls on AI accelerators bent China's *legally* imported AI-chip stock sharply downward. A group difference-in-kinks (DiK) contrast of China's legal H100-equivalent stock against the four US hyperscalers at the 2023-10-17 rule gives -0.00154 ln-units/day (HC1 se 0.00047; ≈ -0.56 ln-units/yr), is negative in 25 of 25 estimated specifications (defensible band -0.3 to -0.56 ln-units/yr), is null at the October-2022 loophole-round placebo cutoff, and its placebo-treated group is clean at defensible bandwidths. The effect on China's *total* chip stock — including smuggled chips — is roughly half as large and statistically fragile (-0.00076, $t = -1.54$, 95% CI spanning zero): smuggled Nvidia chips in China grew from 27.8k H100e in Q1 2024 to 662k by Q4 2025, doubling every 4.7 months, and now exceed China's legal Nvidia stock 1.65×. Finally, a country-level difference-in-differences on the count of $\geq 10^{24}$-FLOP training runs finds no suppression at all: China's big-run count rose +3.2 runs/quarter *relative to* every control-group counterfactual after the October-2022 round — but that estimate fails placebo-in-space (exact $p = 1.0$; every pseudo-treated unit shows a larger studentized effect), so it is descriptive, not causal. The controls demonstrably bent the legal supply channel; substitution through smuggling and domestic chips roughly halved and de-robustified the effect on China's actual compute stock; and no slowdown in China's large training runs is detectable in these data.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security imposed the first broad export controls on advanced AI accelerators and chipmaking tools bound for China, a policy contemporaries immediately read as a landmark attempt to choke off China's access to frontier AI compute [1]. The rule's performance-and-interconnect thresholds left room for compliant workarounds — Nvidia's A800 and H800 — and on 17 October 2023 a second round closed that loophole, banning the workaround chips outright. Whether such controls work is an old question in trade policy: export controls impose costs on the controlling country's own firms and reroute rather than stop trade when substitutes exist [3, 13]. Firm-level evidence from the earlier entity-list controls finds large collateral costs — US suppliers to targeted Chinese firms lose customers, market value (roughly \$130 billion), lending, and employment, with broad supply-chain decoupling but no reshoring [6].

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On the evasion margin, [9] warned in 2023 that AI-chip smuggling into China, then likely in the hundreds to low thousands of units, would industrialize; by 2026, diversion-and-resale estimates put smuggled compute at hundreds of thousands of H100-equivalents [12]. What this literature lacks is a quasi-experimental estimate of whether the controls actually bent China's compute accumulation and large-scale training activity.

This paper provides three complementary tests — three legs of one story — using two Epoch AI datasets [8, 7] and the `natex` toolkit [14]. Leg 1 is a country-level difference-in-differences (DiD) on the quarterly count of $\geq 10^{24}$-FLOP training runs, with a scan-based localization step [11]. Leg 2 is a group difference-in-kinks (DiK) — the design of [2], building on the regression kink design of [5] — contrasting the growth-rate bend in China's legal chip stock against US hyperscalers at the 2023-10-17 rule, with placebo cutoffs in the spirit of [10]. Leg 3 repeats the DiK on China's total stock including smuggled chips and quantifies the smuggling substitution directly. The verdict is deliberately mixed and stated as such: Leg 2 is credible, Leg 3 is attenuated and fragile, Leg 1 is descriptive only.

2 Data

Country panel (Leg 1). From Epoch AI's notable-models database [8] we build a panel of 4 country groups (US, China, EU, Other) $\times$ 26 quarters (2020Q1–2026Q2), $n = 104$ cells. The outcome is the count of training runs $\geq 10^{24}$ FLOP per quarter; a secondary outcome is the median $\log_{10}$ training compute. Models are assigned to countries by the lead (first-listed) organization, so multinational collaborations are attributed to the lead country. The declared treatment is `china_controlled`: China from 2022Q4 (the first control round) onward. Two caveats bind: 2026 quarters undercount compute-known runs ($\sim$13% FLOP coverage, a reporting lag), and counts measure quantity of big runs, not frontier capability.

Chip stocks (Legs 2–3). The chip-stock series come from Epoch AI's AI Chip Owners data [7]: quarterly cumulative H100-equivalent (H100e) stock by designer. From these we form ln cumulative stock series for: China's legal stock; China's total stock including the smuggled Nvidia series [12]; the four US hyperscalers (Amazon, Microsoft, Google, Meta) as the primary control; neoclouds (Oracle, CoreWeave, xAI) as an alternative control; and an "Other" group used as a placebo-treated group. Each series covers 16 quarters from 2022-03-31; the incomplete 2026-03-31 quarter is dropped (8 rows whose "cumulative" totals decrease). The running variable is days since 2023-10-17. The smuggled series begins 2024-03-31 and is itself part of the treatment response — which is exactly why Legs 2 and 3 are reported side-by-side.

3 Design and Methods

Leg 1: SuDDDS DiD. The `natex` sector-by-time DiD scan (SuDDDS) searches subsets of units and break dates maximizing a likelihood-ratio scan statistic [11], calibrated by permutation ($q = 99$, seed 7, windows $\{4, 7, 13\}$). We run it three ways: (i) known-treatment mode (Bernoulli model, `wcc`) on the declared treatment; (ii) a normal-model cross-check with a dependence-preserving `ar1_unit` null; (iii) discovery mode with no treatment column, asking whether the export-control dates would be found autonomously. Effects for China at 2022Q4 come from DD, synthetic-control, and GESS estimators, followed by a manual placebo battery (placebo-in-space with the Abadie china-removed convention, placebo-in-time at 2021Q3).

Legs 2–3: group difference-in-kinks. In [2] the DiK contrasts the same schedule kink across two *time periods*. Here the export-license schedule bends *for China only* at a common date, so the Böckerman contrast is taken across *groups* at one cutoff: DiK = (China slope kink) − (hyperscaler slope kink), each kink estimated by local linear regression on each side of the cutoff (right-minus-left), triangular kernel, primary bandwidth 548 days. Identification is parallel-kink: absent the controls, China's growth bend at Oct-2023 would have matched the hyperscalers'; the control group differences out the global supply-curve bend (the hyperscalers' own kink is -0.000666/day). Robustness: bandwidths $\{365, 548, 730\}$ days $\times$ {triangular, uniform} kernels, 45-day donuts, six placebo cutoffs (including the Oct-2022 first round at -375 days — the A800/H800 loophole-round anticipation contrast), and a placebo-treated group (Other vs hyperscalers, both untreated), in the spirit of [10]. Leg 3 re-runs the identical design on China's total stock including smuggled chips.

Honest-inference notes, stated as run. The pipeline's refusals and trivial passes are reported, not patched over. Leg 1: the built-in `tau_randomization_test` correctly refuses (p = NaN, 3 non-treated units < 5 usable placebos) rather than fabricating a p-value, so the manual placebo-in-space battery is the inference of record; the scan $p = 0.010$ is the $q = 99$ permutation floor; with 4 units, all panel p-values are descriptive. Leg 2: HC1 standard errors on 6 quarterly points per cell overstate precision on a cumulative-stock series; CR1 clustering by calendar year keeps significance but has $G = 4$ clusters with $t(G-1)$ critical values — a known few-cluster over-rejection regime [4]; the sign-stability across 25 specifications is the real evidence. The density-kink-difference test passes trivially (both groups share an identical quarterly grid; estimate exactly 0.0, p reported as null), no predetermined covariates exist in a two-group quarterly aggregate (the placebo-treated group and the Oct-2022 placebo round substitute for the covariate-kink battery), and no Fieller set is produced (sharp design, known unit denominator). A recorded `natex` UX finding: the CLI aliases the group contrast through its time interface (`--time china --t0 1`), so output cells are labeled `pre_*/post_*` and its caveat text misdescribes the cross-group contrast as a "time-stable marginal response."

4 Results

Leg 1: localization succeeds, attribution fails. In known-treatment mode SuDDDS recovers the declared treatment exactly: top subset {China}, $T_0 = 11 = $ 2022Q4, window $W = 13$, Bernoulli LLR 6.3715, panel-null $p = 0.010$ (the $q = 99$ floor); the composition check passes ($p = 1.0$) and the anticipation test passes (Holm $p = 1.0$). The normal-model cross-check under the dependence-preserving `ar1_unit` null finds the mirror image — the complement {EU, Other, US} at $T_0 = 13$ (2023Q2), LLR 14.9487 — but $p = 0.14$: a 4-unit panel cannot separate a persistent step from unit-level AR(1). Discovery mode does *not* find the controls: the top hit is a {China, US} *downshift* at 2025Q4 (LLR 11.277, $p = 0.13$; runner-ups ALL@2025Q4, LLR 7.232, and {China, US}@2024Q2, LLR 5.835), with no export-control date in the top five — and the 2025Q4 downshift is at least partly the reporting-lag artifact. Effect estimates for China at 2022Q4 (Table 1, panel A) are uniformly *positive*: DD $\hat{\tau} = +3.214$ runs/qtr (se 1.157, $t = 2.779$, 95% CI $[+0.947, +5.481]$), synthetic control +3.933, GESS +3.842, DD at the 2023Q4 round +4.455, and DD on median $\log_{10}$ compute +0.649 $[+0.322, +0.976]$. Taken literally: no suppression — China's big-run count *outgrew* every control-implied counterfactual. But the placebo battery kills any causal reading of the magnitude: pseudo-treating each control unit at the same date yields studentized $|t|$ of 5.37 (EU), 2.86 (Other), and 5.16 (US) against China's 2.78 (china-removed convention) — exact placebo-in-

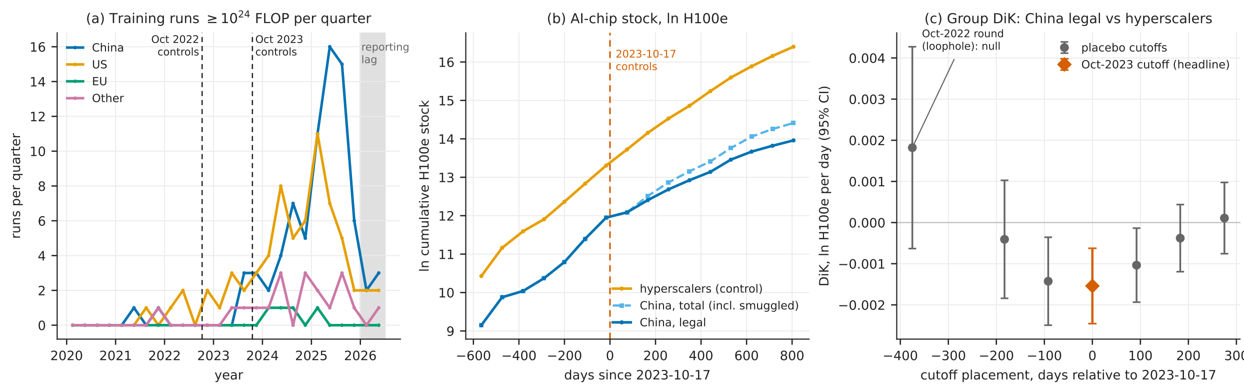

Figure 1: (a) Leg 1: quarterly count of $\geq 10^{24}$-FLOP training runs by country group, with both export-control rounds marked; the shaded region is the $\sim$13%-coverage reporting-lag zone. The panel is a two-speed world: US and China accelerate, EU and Other stagnate. (b) Legs 2–3: ln cumulative H100e chip stock for the hyperscaler control group, China's legal stock, and China's total stock including smuggled chips, around the 2023-10-17 cutoff. (c) Leg 2: group-DiK estimates (bandwidth 548 days, triangular kernel) at the true cutoff (diamond) and at six placebo cutoff placements, with 95% HC1 intervals; the Oct-2022 loophole round at -375 days is null.

space $p = 4/4 = 1.0$. Placebo-in-time is clean (China at 2021Q3: $\tau = -0.30$, $t = -1.33$), and the built-in randomization test correctly refused ($p =$ NaN, $3 < 5$ placebos). The robust statement is qualitative only: China's large-run count did not fall behind its counterfactual after either control round.

Leg 2: the legal channel credibly bent. The headline group DiK (China legal vs hyperscalers, 2023-10-17, bandwidth 548 days, triangular kernel) is -0.0015404 ln-units/day (HC1 se 0.000467, $t = -3.298$, 95% CI $[-0.002456, -0.000625]$), i.e. ≈ -0.56 ln-units/yr of growth-rate change; with CR1 clustering by year, se 0.000299, $t = -5.152$ ($G = 4$, few-cluster caveat). The cell kinks are -0.000666/day (hyperscalers) vs -0.002206/day (China legal). The estimate is negative in **25 of 25** estimated specifications: -0.000869 to -0.002026 across bw365/548/730 $\times$ triangular/uniform (t from -2.07 to -6.57), and $-0.0018996/-0.0012108/-0.0008011$ on the 45-day-donut grid (t $-3.92/-2.04/-1.60$). The placebo-cutoff battery (bw548; empirical size 2/6) localizes the kink to within one quarter of the rule: the Oct-2022 first round at -375 days is *null* at $+0.0018203$ (se 0.001251, $p = 0.146$) — the loophole round produced no bend, a pseudo-pre-trend pass — and -183d ($p = 0.578$), $+183$d ($p = 0.364$), $+275$d ($p = 0.805$) are null, while -92d ($p = 0.0089$) and $+92$d ($p = 0.0240$) reject: with quarterly data, one-quarter shifts still straddle the same bend, which still brackets 2023-10-17. The placebo-treated group (Other vs hyperscalers, both untreated) is null at bw548/730 ($t = +1.34$, -0.29, -0.20, -0.75) but rejects at bw365 (triangular $+0.000958$, $t = 2.49$; uniform $+0.001219$, $t = 3.04$) — parallel-kink fails among untreated groups at four-points-per-cell bandwidths, so the bw365 cells are discounted and the defensible band is bw548–730: -0.0009 to -0.0015/day (≈ -0.3 to -0.56 ln-units/yr).

Leg 3: smuggling halves and de-robustifies the effect. On China's total stock including smuggled chips, the same design gives -0.0007638 (se 0.000496, $t = -1.540$), with a 95% CI of $[-0.001736, +0.000209]$ that spans zero — roughly half the legal-channel bend, and t ranges from -0.55 to -3.08 across the six specifications (the only rejection is at the discounted bw365). The

Table 1: Headline estimates across the three legs.

Specification		$\hat{\tau}$	se		t	note
Panel A: Leg 1 — DiD, China at 2022Q4 (runs $\geq 10^{24}$ FLOP per quarter)						
DD (control = other 3 units)		+3.214	1.157		2.78	CI [+0.947, +5.481]
Synthetic control		+3.933	—		—	CI [+1.533, +6.334]
GESS		+3.842	—		—	CI [+1.442, +6.243]
DD at 2023Q4 round		+4.455	—		—	CI [+1.719, +7.190]
DD, median $\log_{10}$ compute		+0.649	—		—	CI [+0.322, +0.976]
placebo-in-space exact $p = 4/4 = 1.0 \Rightarrow$ descriptive, not causal						
Panel B: Leg 2 — group DiK, ln H100e per day, China legal vs hyperscalers						
bw548 triangular (headline)		−0.001540	0.000467		−3.30	HC1
bw548 triangular, CR1 by year		−0.001540	0.000299		−5.15	$G = 4$ clusters
bw548 uniform		−0.001132	0.000525		−2.16	
bw730 triangular / uniform	−0.001140 /	−0.000869		−2.85 /	−2.07	
bw365 triangular		−0.002026	0.000308		−6.57	discounted
placebo cutoff −375d (Oct-2022)		+0.001820	0.001251			$p = 0.146$: null
Panel C: Leg 3 — group DiK, China total stock (incl. smuggled) vs hyperscalers						
bw548 triangular		−0.000764	0.000496		−1.54	CI [−0.001736, +0.000209]
bw730 triangular / uniform	−0.000458 /	−0.000235		−1.10 /	−0.55	
bw365 triangular		−0.001049	0.000340		−3.08	discounted

Notes: Panel A: `natex` DiD effect estimators on the 4-country × 26-quarter panel (seed 7); 95% CIs in the note column. Panels B–C: sharp group difference-in-kinks, right-minus-left slope contrasts in ln cumulative H100e stock per day at the 2023-10-17 cutoff; HC1 standard errors unless noted; bandwidths in days; "discounted" marks bw365 cells where the placebo-treated group (Other vs hyperscalers) rejects. Multiply DiK estimates by 365.25 for annual growth-rate changes.

mechanism is visible in the raw series: smuggled Nvidia compute in China grew from 27.8k H100e (Q1 2024) to 662k H100e (Q4 2025), doubling every 4.7 months — the fastest-growing series in the dataset — and now stands at 1.65× China's legal Nvidia stock. Even so, China's share of the world chip stock roughly halved to 8.7% over the period despite absolute growth: the controls did not stop China's accumulation, but China fell behind a world that accelerated faster.

5 Caveats and Conclusion

Four caveats bound the claims. First, Leg 1's panel has four units — the floor for any panel placebo inference — and its placebo-in-space failure is informative precisely because it shows the units are not exchangeable (a two-speed world), so no DiD counterfactual built from EU/Other/US is credible for China's magnitude; every Leg-1 p-value is descriptive. Second, Leg 2's inference rests on 6 quarterly points per cell: HC1 overstates precision on a cumulative series, CR1 has $G = 4$ clusters [4], and the ±92-day placebo rejections show the kink is localized only to ±1 quarter — the design cannot distinguish 2023-10-17 from an adjacent-week event, though no rival same-quarter candidate of comparable relevance is known to us. Third, the smuggled series underlying Leg 3 is itself an estimate with wide uncertainty [12] and begins only in Q1 2024, endogenously to the treatment. Fourth, counts of big training runs measure quantity, not frontier capability; controls aimed at chip access can bind on quality and efficiency margins invisible to Leg 1's outcome.

The three-leg verdict: *credible* on the legal channel (Leg 2) — the October-2023 controls bent China's legal chip-stock growth down by roughly 0.3–0.56 ln-units/yr relative to every control group, with the loophole-round placebo null exactly where the policy history says it should be; *attenuated and fragile* on total compute (Leg 3) — smuggling substitution roughly halves the point estimate and removes its robustness; and *descriptive only* on training activity (Leg 1) — China's count of large training runs did not fall behind any counterfactual, but the design cannot certify that as a causal null. Export controls, on this evidence, bent the channel they legally govern, were substantially bypassed in aggregate, and left China's large-run training activity unbowed.

Reproducibility. All estimates were produced with `natex` v0.2.0 [14]: the Leg-1 runs at seed 7 ($q = 99$) and the Leg-2/3 battery at seed 20260716 (the DiK estimator itself is deterministic), from frozen extracts of the Epoch AI datasets [8, 7] (not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline DD and DiK estimates against the numbers of record before drawing.

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The Question Is the Treatment: A Known-Cutoff Regression Discontinuity at the BTOS AI-Question Rewording

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

Beginning with wave 202524 (reference period from 3 November 2025; announced 3 December 2025), the US Census Bureau's Business Trends and Outlook Survey (BTOS) reworded its headline AI question from AI use "in producing goods or services" to AI use "in any of its business functions." Treating the refresh as a known measurement cutoff in a regression-discontinuity-in-time design on the spliced 71-wave national series, the rewording moved measured US business AI use by +6.04 percentage points (HC1 $z = 22.9$; HAC $z = 16.3$) — a ratio of 1.553 over the old-wording level at the cutoff. The jump is 4.7× the largest of 49 placebo-cutoff estimates (randomization $p = 0.020$, the 1/50 floor), stable at 6.0–7.5 pp across bandwidth, weighting, curvature, and HAC variants, and a blind LoRD3 scan re-localizes it: the top two data-driven candidate cutoffs exactly flank the known one (honest 2SLS +6.74, 95% CI $[6.01, 7.48]$). True adoption growth cannot produce it: the fitted pre-trend predicts +0.66 pp across the 50-day shutdown gap, and doubling that slope still leaves ≥ 5.38 pp. Nulls are reported as nulls: the blind-scan randomization test is underpowered at $n = 71$ ($p = 0.14$), the McCrary density test is inapplicable on a uniform time grid (NaN), a December–January seasonal term is insignificant (0.29 ± 0.24), and a post-cutoff slope steepening (+2.17 pp/yr, conventional se 0.56) is secondary and suggestive only. The estimate calibrates a splice factor — additive +6.04 pp or ratio 1.553 — for carrying the pre-refresh series past December 2025, with the caveat that the "wording effect" is bundled with a simultaneous sample rotation and any shutdown-adjacent change in response composition.

1 Introduction

Since September 2023 the Census Bureau's Business Trends and Outlook Survey (BTOS) has asked a probability sample of US firms whether they used artificial intelligence in the last two weeks; [4] introduce these data, which have become the flagship high-frequency gauge of US business AI adoption — the headline series in the Federal Reserve's monitoring of the AI economy [1]. On 3 December 2025 the Census Bureau announced that, beginning with survey cycle 2 of sample year 2026, the AI questions were reworded: the core use question now asks about AI "in any of its business functions" rather than "in producing goods or services," explicitly to reduce respondent under-reporting [8]. Measured adoption moved immediately: the last old-wording wave reads 10.0%, the first new-wording wave 17.3%. Commentary quickly flagged the break — [2] show that how the

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question is asked drives large gaps between firm AI-adoption estimates, and [1] notes the redesign when reporting the $\sim$18% year-end-2025 level — and Census researchers now exploit the expanded business-function detail the redesign introduced [5]. What the commentary does not provide is a formal estimate of the rewording effect with falsification-calibrated uncertainty. That estimate is the number any user of the series needs in order to splice the pre- and post-refresh segments into one panel.

Instrument refreshes breaking measured series is a classic survey-methods problem. The canonical precedent is the 1994 Current Population Survey redesign, whose effects were quantified with a dedicated parallel survey run under both instruments and converted into adjustment factors for splicing the historical series [13]. The BTOS refresh has no parallel overlap sample: the old question stopped, the new one started, and a federal government shutdown sits between them. The splice factor must therefore be estimated from the time series alone — a known-cutoff regression-discontinuity-in-time (RDiT) problem [9], here in its cleanest form: the cutoff is announced, mechanical, and datable to a single wave, and the "treatment" is the measurement instrument itself, so the usual manipulation and anticipation concerns are moot while the RDiT time-series concerns (serial correlation, trend extrapolation) remain and are addressed directly. We estimate the jump by local linear regression on each side of the cutoff with robust and HAC inference [6, 9], calibrate it against a 49-point placebo-cutoff grid, bound the adoption-growth confound, and additionally ask the `natex` toolkit [12] to find the discontinuity *blind*, via the LoRD3 local-RD scan of [10], plus a secondary regression-kink test in the conventions of [7] and [3].

2 Data

The outcome is the national share of firms answering "Yes" to BTOS Question 7, in percentage points, with published survey standard errors. Two vintages are required because current Census downloads carry only the reworded question, with all earlier waves blanked. The old-wording segment ("…did this business use Artificial Intelligence (AI) in producing goods or services?") comes from a frozen pre-refresh AI-core workbook: 54 biweekly waves, 202319–202520, rising 3.7% to 10.0%. The new-wording segment ("…in any of its business functions?") comes from a fresh `National.xlsx` download (census.gov, 2026-07-18, 82 kB, verified): 17 waves, 202524–202614, rising 17.3% to 21.7%. The spliced panel has $n = 71$ waves indexed by t, the decimal-year reference-period midpoint.

Three timeline facts matter. First, waves 202521–202523 were never collected — the October–November 2025 federal government shutdown — so the design has a built-in donut that is a no-data gap, not censoring: 50 days separate the last pre-cutoff reference midpoint (2025-09-14) from the cutoff. Second, the cutoff is $t_c = 2025.8384$ (2025-11-03, the reference-period start of wave 202524, the first wave fielded with the reworded question; published 2025-12-04). No reference midpoints fall in (2025-09-14, 2025-11-09), so any cutoff in that interval yields the identical estimation sample. Third, wave 202524 also begins Sample Year 4 (a scheduled full sample rotation), so the refresh bundles the wording change with a panel rotation — a caveat carried throughout.

3 Design and Methods

Estimator. The headline specification fits SE-weighted ($w = 1/\text{se}^2$) local linear regressions of the AI-use share on $t - t_c$ separately on each side of the cutoff within bandwidth $h = 1.0$ years ($n_{\text{pre}} = 23$, $n_{\text{post}} = 17$); the jump τ is the post-side intercept minus the pre-side extrapolation at t_c. Inference is HC1 per side combined by independence, plus a pooled HAC (Bartlett kernel, 4 and 8 lags) variant for serial correlation. Sensitivity covers bandwidths $h \in \{0.5, 0.75, 1.0, 2.3\}$, unweighted fits, and pre-side quadratics.

Identification and falsification. In a measurement RDD the treatment is the instrument refresh, so the only smooth confound is true adoption growth loading into the 50-day extrapolation window; we bound it by the fitted pre-slope and report the estimate under a doubled gap slope. Calibration is design-based: the identical estimator is run at all 49 feasible interior placebo cutoffs, giving a randomization p-value with floor 1/50. A blind check uses `natex discover` (LoRD3, seed 0, $k = 12$, $q = 99$): the scan searches for local discontinuities without being told the cutoff, then honest (split-sample) 2SLS re-estimates the effect at the discovered boundary [10, 12]. The McCrary density test [11] returns NaN on a uniform time grid — inapplicable in RDiT, reported as an outcome, not patched over. A secondary sharp regression-kink test at t_c (`natex kink`, right-minus-left slope contrast, policy kink 1, $h = 1.0$, conventional local-polynomial inference without HAC) asks whether the trend also steepened [7, 3].

Honest-inference notes, stated as run. The pipeline's refusals and artifacts are reported: the automated `natex survey` run refused its rdd family (100-row floor, $n = 71$), iv (≥ 80), did/sc (needs a panel), and dee (≥ 200); its kink family auto-assigned the degenerate outcome `run` (a linear transform of t), so that null ($p = 0.79$, $\tau \approx 10^{-13}$) is meaningless and was superseded by the declared-outcome kink run above. The LoRD3 validation battery passed (covariate placebo Holm $p = 0.24$); its blind-scan randomization test is underpowered at $n = 71$ ($p = 0.14$) and is reported as the null it is. Pre-fit residuals have AR(1) = 0.836 and empirical wave noise $\sim$4$\times$ the published survey SEs (residual sd 0.614 vs. mean reported se 0.151); the HC1/HAC empirical standard errors absorb this overdispersion, but HAC with $n_{\text{post}} = 17$ is approximate — the jump is $\sim$16 HAC standard errors, so no plausible correction overturns it. A December–January seasonal coefficient is insignificant (0.293 ± 0.236).

4 Results

Headline jump and robustness. At the known cutoff the rewording jump is $\hat{\tau} = +6.039$ pp (HC1 se 0.263, $z = 22.9$; HAC(4) se 0.371, $z = 16.3$; HAC(8) se 0.376), against an old-wording level of 10.92 pp at the cutoff — a ratio of 1.553. Every variant in Table 1 stays within 6.0–7.5 pp with $z > 12$: bandwidths 0.5/0.75/2.3 give 7.148/6.383/7.363, the unweighted fit gives 5.956, and pre-side quadratics give 5.787–7.464.

Placebo-cutoff calibration. Across the 49 feasible interior placebo cutoffs the same estimator yields a maximum $|\tau|$ of 1.298 pp (mean 0.515); none reaches the true jump in $|\tau|$ or $|z|$, for a randomization $p = 0.020$ — the minimum achievable with 49 placebos (Figure 1b). The true jump is 4.7$\times$ the largest placebo.

Blind re-localization. Without being told the cutoff, the LoRD3 scan's two highest-scoring candidate boundaries sit at $t = 2025.7027$ and $t = 2025.8562$ — exactly the last old-wording and first new-wording waves, flanking t_c. Honest 2SLS at the discovered boundary gives $\tau = 6.743$ (se 0.375, 95% CI $[6.008, 7.478]$, no weak-instrument flag); the naive full-sample Wald estimate is 8.06 (se 0.206). The scan-level randomization test does not reach significance ($p = 0.14$), as expected at $n = 71$; localization, not the scan p-value, is the evidential contribution.

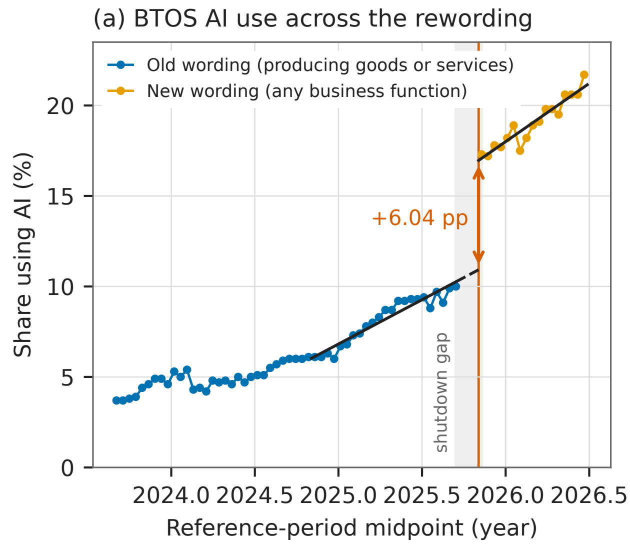

Figure 1: (a) The spliced BTOS national AI-use series with SE-weighted local linear fits within $h = 1.0$ yr of the cutoff $t_c = 2025.8384$ (2025-11-03, vertical line); the dashed segment is the 50-day extrapolation across the shutdown gap (shaded); the arrow marks the +6.04 pp jump. (b) The same estimator at all 49 feasible placebo cutoffs (open circles) versus the true cutoff (filled, with 95% HAC confidence interval): the largest placebo is 1.30 pp, 4.7× smaller than the true jump (randomization $p = 0.020$).

Bounding the adoption confound. The fitted pre-slope is 4.89 ± 0.31 pp/yr, which predicts only +0.66 pp of true-adoption growth across the 50-day gap. Even if the true gap slope were double the fitted slope, the implied jump remains ≥ 5.376 pp. A wording effect of zero would require adoption to grow ∼9× faster during the shutdown than in the preceding year — and then revert, since the post-cutoff slope (6.43 pp/yr) is only modestly steeper.

Secondary kink. The sharp RKD at t_c estimates a slope change of +2.170 pp/yr (conventional se 0.558, no HAC): the new-wording series trends steeper than the local old-wording trend. Given conventional inference and the bundling caveats, this is suggestive only; it cautions against assuming the wording effect is a pure level shift far from the cutoff.

Splice calibration. The fitted levels at the cutoff are 10.92 pp (old wording) and 16.96 pp (new wording): an additive splice of +6.04 pp or a ratio splice of 1.553. Over the observed post period the two disagree increasingly far from the cutoff (a ratio splice scales the steeper post trend); for level-tracking the additive factor is the direct RD estimand, and the ratio is the natural choice if the wording effect scales with the underlying level.

5 Caveats and Conclusion

Four caveats bound the claim. First, the donut is a federal-shutdown no-data gap (waves 202521–202523 were never collected), so any true adoption acceleration inside the 50-day extrapolation window loads into $\hat{\tau}$; the pre-slope bound caps this at 0.66 pp under the fitted trend and leaves $\tau \geq 5.38$ pp even at double that slope. Second, the questionnaire refresh is bundled with a full

Table 1: RD estimates of the BTOS rewording jump at $t_c = 2025.8384$.

Specification	$\hat{\tau}$	se	z
Panel A: known-cutoff RD jump (pp), SE-weighted local linear			
$h = 1.0$, HC1 (headline)	+6.039	0.263	22.9
$h = 1.0$, HAC Bartlett, 4 lags	+6.039	0.371	16.3
$h = 1.0$, HAC Bartlett, 8 lags	+6.039	0.376	16.1
$h = 0.5$	+7.148	0.352	20.3
$h = 0.75$	+6.383	0.275	23.2
$h = 2.3$ (full sample)	+7.363	0.241	30.6
$h = 1.0$, unweighted	+5.956	0.287	20.8
$h = 0.75$, pre-quadratic	+7.464	0.481	15.5
$h = 1.0$, pre-quadratic	+7.051	0.374	18.9
$h = 2.3$, pre-quadratic	+5.787	0.282	20.5
Panel B: blind discovery, placebos, and secondary tests			
LoRD3 honest 2SLS (blind scan)	+6.743	0.375	18.0
LoRD3 naive Wald	+8.063	0.206	39.2
Placebo cutoffs (49): max $\lvert\tau\rvert$	1.298	—	—
Placebo cutoffs (49): mean $\lvert\tau\rvert$	0.515	—	—
Sharp RKD slope change (pp/yr)	+2.170	0.558	3.9

Notes: outcome is the national share answering Yes to BTOS Question 7, percentage points; $n = 23$ pre / 17 post waves at $h = 1.0$ yr. HC1 per side combined by independence; HAC is a pooled Bartlett-kernel variant. Placebo randomization $p = 0.020$ (0 of 49 placebos $\geq$ the true jump in $|\tau|$ or $|z|$; floor 1/50). LoRD3: `natex discover`, seed 0, $k = 12$, $q = 99$; scan randomization $p = 0.14$ (underpowered at $n = 71$); McCrary density test NaN (uniform time grid, inapplicable). RKD: conventional local-polynomial inference, no HAC; secondary.

sample rotation (Sample Year 4 begins at wave 202524) and follows the shutdown, so "wording effect" strictly means wording + panel-refresh + shutdown-adjacent response-composition effect; for the splice-factor purpose all three are measurement artifacts, but pure wording attribution is not separable in this design. Third, RDiT caveats apply [9]: pre-fit residual AR(1) is 0.836 and empirical wave noise is $\sim 4\times$ the published survey SEs; the HC1/HAC empirical standard errors absorb this, but HAC with 17 post waves is approximate — the jump is ~ 16 HAC standard errors, so no plausible correction overturns it. Fourth, the automated-pipeline artifacts were triaged, not hidden: the survey rdd family refused on its 100-row floor, the auto-assigned degenerate kink outcome was superseded by the declared-outcome run, the McCrary test is inapplicable by construction, and the blind-scan randomization $p = 0.14$ is an underpowered null.

The verdict is credible: a textbook known-cutoff measurement RDD. The +6.04 pp jump at the 2025-11-03 refresh is $4.7\times$ the largest of 49 placebo jumps, survives every bandwidth, weighting, curvature, and HAC variant within a 6.0–7.5 pp band, is blindly re-localized by LoRD3 at exactly the flanking waves, and cannot be true adoption. It identifies the instrument-refresh effect and calibrates the splice — additive +6.04 pp, ratio 1.553 — required to extend any analysis built on the old-wording BTOS AI series past December 2025. Substantively, the estimate implies that roughly a third of the post-refresh measured adoption level is attributable to asking a better question rather than to more firms using AI — a caution for any trend analysis that splices BTOS vintages naively, and a quantification of the under-reporting the Census Bureau redesigned the question to remove [8, 2].

Reproducibility. All estimates were produced with `natex` v0.2.0 [12] at seed 0 from the frozen BTOS workbooks (published at `https://www.census.gov/hfp/btos`; not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimate, the refitted cutoff levels, and the placebo maximum against the numbers of record before drawing.

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A Genuine Bend, No Causal Certificate: Difference-in-Kinks Tests of Hyperscaler Capital Expenditure at ChatGPT

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

Quarterly capital expenditure of the big-4 hyperscalers (Microsoft, Alphabet, Amazon, Meta) accelerated sharply relative to a non-AI capital-intensive comparison aggregate after the release of ChatGPT on 30 November 2022. A group difference-in-kinks (DiK) estimate at the release date is +0.1396 dex/yr (HC1 se 0.0302, nominal $p = 3.9 \times 10^{-6}$), positive in all 18 specification cells (+0.049 to +0.195), localized over 2023.0–2023.5 exactly as a lagged capex response to a late-2022 shock would be, and present per firm for Microsoft (+0.171), Alphabet (+0.206), and Amazon (+0.137) but null for Meta (+0.031, $p = 0.55$). Attribution to ChatGPT, however, fails the falsification battery: a clean pre-period placebo-cutoff grid rejects at nominal 5% in 7 of 15 non-event dates (empirical size 0.47), giving placebo-calibrated $p = 0.125$ ($|z|$) and 0.250 ($|\tau|$); the largest placebo $|z|$ (5.11, at the non-event date 2019.75) exceeds the ChatGPT estimate's matched-bandwidth $|z|$ (4.35); a full-sample kink at the non-event date 2021.5 (-0.314, $|z| = 6.9$) is larger than the main effect; and a cutoff at the BIS semiconductor export controls eight weeks earlier is statistically indistinguishable (+0.125, $z = 4.0$). Residual lag-1 autocorrelation reaches 0.66, so the nominal heteroskedasticity-robust p-values are artifacts of applying serial-independence inference to smooth aggregates. The slope divergence is a genuine, robust data feature; no causal attribution to ChatGPT survives triage.

1 Introduction

OpenAI released ChatGPT on 30 November 2022. Within two years the capital expenditure of the four large US "hyperscalers" — Microsoft, Alphabet, Amazon, and Meta — had become a macroeconomic quantity in its own right, central to debates about how much aggregate investment the current AI wave can move [1]. Financial markets repriced AI exposure almost immediately: [6] construct firm-level workforce exposures to generative AI and show that an exposure-sorted long-short portfolio earned roughly 5% in the two weeks after the release. A natural real-economy question follows: did the release bend the hyperscalers' *investment* trend — the growth rate of measured capital expenditure — or was the buildout already accelerating?

Any calendar-time answer must contend with a crowded event window. Eight weeks before ChatGPT, on 7 October 2022, the Bureau of Industry and Security announced sweeping export controls on advanced AI chips and semiconductor manufacturing equipment, a policy landmark in its own right [2]; any quarterly series treats the two events as nearly the same date.

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The estimand — a change in trend slope for one group relative to another at a known date — is the difference-in-kinks (DiK) design formalized by [4], building on the regression kink design of [5]. Kink estimates on time-series running variables are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [7]; and conventional robust standard errors are badly oversized on serially correlated aggregates [3]. This paper applies the `natex` toolkit's DiK estimator [8] to a two-group quarterly capex panel built from SEC EDGAR filings, with ChatGPT as the candidate cutoff and a mandatory falsification battery — which converts a nominally overwhelming event-study result into an explicit non-attribution.

2 Data

Quarterly capital expenditure is reconstructed from SEC EDGAR XBRL `companyconcept` facts, converting fiscal year-to-date values to clean quarters by same-start differencing with remainder assignment across tag transitions. The tag is `PaymentsToAcquirePropertyPlantAndEquipment`; Amazon additionally reports under `PaymentsToAcquireProductiveAssets`. The treated series is the big-4 aggregate over 46 quarters, 2014Q4–2026Q1, rising from \$6.8bn to \$129.8bn per quarter. The rebuilt aggregate matches the frozen reference series of the run of record on all 45 common quarters (maximum difference \$0.000000bn) and its processed $\log_{10}$ counterpart on all 46 (maximum difference 5×10^{-5}).

The comparison series aggregates six large non-AI capital-intensive firms: Walmart, ExxonMobil, AT&T, Verizon, Union Pacific, and Home Depot, fetched through the identical EDGAR pipeline. Firm-specific tag quirks (AT&T, Verizon, and Home Depot report quarterly capex under `ProductiveAssets`-family tags; Walmart and Home Depot have January-31 fiscal year ends mapped to the nearest calendar quarter) are resolved against the live EDGAR records, and 2023 calendar-year sums match published figures for all six firms (AT&T \$17.85bn, Verizon \$18.77bn, ExxonMobil \$21.92bn, Walmart \$20.61bn, Union Pacific \$3.61bn, Home Depot \$3.23bn).

The outcome is $\log_{10}$ of group capex in \$bn per quarter, so slopes are in dex/yr (one dex is a factor of ten). The running variable is the quarter midpoint in fractional years; the ChatGPT cutoff is $t_0 = 2022.913$ (2022-11-30). A donut of 0.13 years ($\approx$ one quarter) removes 2022Q4, the quarter that straddles the release.

3 Design and Methods

Group difference-in-kinks. Following [4], the DiK estimand is the change in the outcome's slope in calendar time at the cutoff for the treated group *minus* the same slope change for the comparison group, estimated by local linear weighted regression on each side of the cutoff within each group (`natex` v0.2.0 [8]; the group indicator enters the CLI's DiK period dimension with a unit policy-kink change, so $\hat{\tau} = (\text{kink})_{\text{big-4}} - (\text{kink})_{\text{control}}$ in dex/yr). The primary specification uses bandwidth 3.0 years, triangular kernel, the one-quarter donut, and HC1 standard errors; $n = 46$ quarter-cells (11 pre-cutoff and 12 post-cutoff quarters per group).

Honest-inference caveats, stated as run. The estimator's own caveats apply verbatim: the cutoff and bandwidth are user-specified, so bandwidth, donut, and placebo-cutoff sensitivity must be reported; the reported interval is conventional local-polynomial inference and may retain smoothing bias; and causal interpretation requires parallel changes in non-policy slope kinks and a time-stable marginal response at the cutoff. HC1/CR1 inference additionally assumes serial independence, which these aggregates violate: residual lag-1 autocorrelation in the primary fit is 0.657 on the

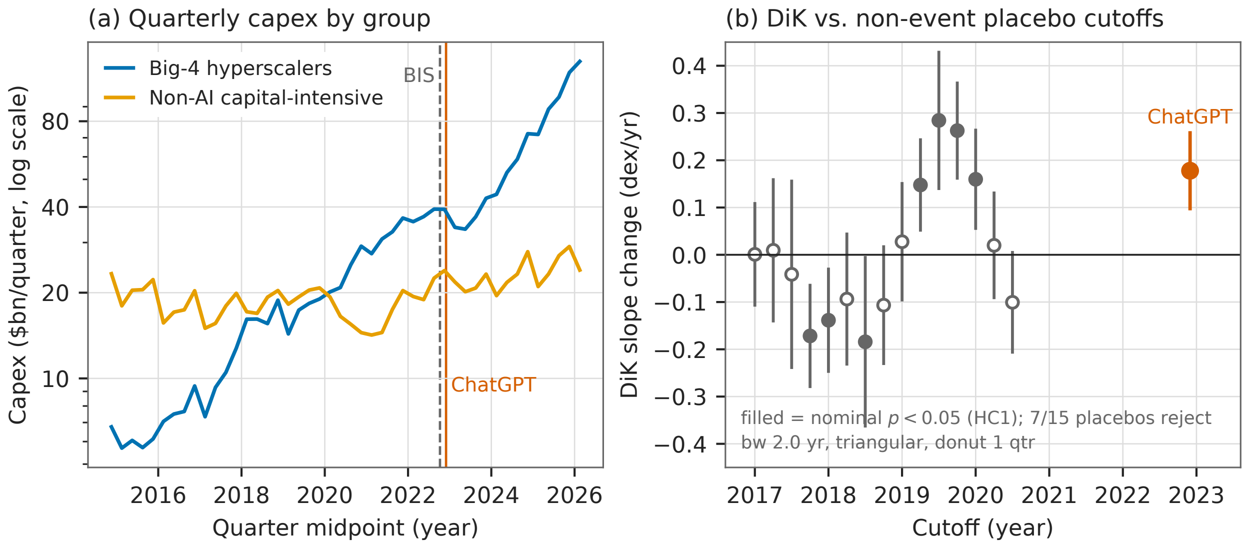

Figure 1: (a) Quarterly capital expenditure ($bn, log scale) of the big-4 hyperscaler aggregate and the non-AI capital-intensive control aggregate, 2014Q4–2026Q1, with the BIS export-control (2022-10-07, dashed) and ChatGPT (2022-11-30, solid) dates marked. (b) Group DiK estimates with 95% HC1 confidence intervals at 15 pre-period non-event placebo cutoffs (gray; pre-only sample, $t < 2022.7$) and at the ChatGPT cutoff (vermilion), all at the matched bandwidth 2.0 yr, triangular kernel, one-quarter donut; filled markers are nominally significant at the 5% level. Seven of 15 placebos reject, and the largest placebo $|z|$ exceeds the ChatGPT $|z|$.

control pre-cutoff side and 0.367 on the big-4 pre-cutoff side, precisely the setting in which nominal robust standard errors are known to be understated [3].

Falsification battery. (i) an 18-cell specification grid (bandwidth $\in \{2, 3, 4\}$ years $\times$ donut $\in \{0, 0.13, 0.25\}$ $\times$ {triangular, uniform}); (ii) a placebo cutoff at the BIS export controls (2022-10-07, $t = 2022.767$); (iii) a clean *pre-period* placebo-cutoff grid — sample restricted to $t < 2022.7$, 15 cutoffs 2017.0–2020.5, bandwidth 2.0, donut 0.13 — in the spirit of the permutation logic of [7], yielding a placebo-calibrated p-value for the main effect at the matched bandwidth; (iv) a cutoff localization scan over 2021.5–2024.75; (v) per-firm staggered DiKs of each of the ten firms against the control aggregate; and (vi) robustness checks: CR1 clustering by year and by quarter, quarter-of-year dummies, leave-one-out and no-retail control aggregates, a role-swap sanity check, and a degree-2 polynomial.

Automated pipeline, stated as run. The `natex` survey pipeline (`natex survey --seed 0`) was recorded for completeness: its automated kink family pools both groups on the auto-picked linear capex outcome and therefore cannot express the group-DiK aliasing; it returns null (Holm $p = 0.24$). The remaining families were skipped, needs-input, or failed as expected on a 92-row two-unit panel. The manual DiK runs above are the primary evidence.

4 Results

The bend is real ... The primary group DiK at ChatGPT (Table 1, first row) is $\hat{\tau} = +0.1396$ dex/yr (HC1 se 0.0302, 95% CI $[0.0804, 0.1988]$, $z = 4.62$, nominal $p = 3.85 \times 10^{-6}$). It decomposes into a big-4 kink of $+0.0878$ dex/yr (slope $0.0999 \to 0.1876$; the aggregate's growth roughly doubles, to a pace that multiplies capex by ten in about 5.3 years) and a control kink of -0.0518 ($0.0811 \to 0.0293$). The estimate is positive in every cell of the 18-cell grid ($+0.0493$ to $+0.1951$, all nominal $p < 0.05$; the uniform kernel roughly halves $\hat{\tau}$ relative to the triangular at bandwidths ≥ 3). The point estimate is essentially invariant to CR1 clustering by year (se 0.0313) or quarter (se 0.0278), quarter-of-year dummies ($+0.1396$, se 0.0296), leave-one-out control aggregates ($+0.1329$ to $+0.1515$), and a no-retail control of ExxonMobil, AT&T, Verizon, and Union Pacific ($+0.1453$, se 0.0361); the role-swap check returns exactly -0.1396; the degree-2 estimate ($+0.3022$, se 0.0914) shows curvature sensitivity in magnitude but not sign. Per firm against the control aggregate: Microsoft $+0.1710$ (se 0.0366), Alphabet $+0.2061$ (se 0.0349), Amazon $+0.1375$ (se 0.0512), Meta $+0.0308$ (se 0.0518, $p = 0.55$ — a true null: Meta's surge came in 2024, after its 2022–23 pullback); control firms' own kinks are all mildly negative (-0.022 to -0.090). A localization scan (bandwidth 2.0, no donut) finds the DiK crossing zero near $t = 2022.3$, maximizing at $t = 2023.25$ ($+0.2213$, se 0.0440), staying above $+0.19$ over 2023.0–2023.75, and decaying to ≈ 0 by 2024.5 — consistent with a lagged capex response to a late-2022 shock.

... but attribution to ChatGPT fails. Three findings block a causal reading. *First*, the pre-period placebo battery fails decisively: at 15 non-event cutoffs in 2017.0–2020.5, the identical specification rejects at nominal 5% seven times (empirical size 0.467; Figure 1b). Calibrated against this placebo distribution, the ChatGPT estimate at the matched bandwidth ($+0.1778$, $z = 4.35$) has $p_{|z|} = (1 + 1)/(15 + 1) = 0.125$ and $p_{|\tau|} = (3 + 1)/16 = 0.250$; the largest placebo $|z|$ is 5.11 (at 2019.75) and the largest placebo $|\hat{\tau}|$ is 0.284 (at 2019.5) — both exceeding the ChatGPT values. *Second*, the full-sample kink at the non-event date 2021.5 is -0.3135 (se 0.0452, $|z| = 6.9$), larger in magnitude and nominal significance than the main effect: these smooth aggregates generate huge nominal kinks where nothing happened. *Third*, the cutoff is confounded: a DiK at the BIS export-control date eight weeks earlier returns $+0.1253$ (se 0.0317, $z = 3.95$), statistically indistinguishable from the ChatGPT estimate — quarterly data cannot separate the two events (or any other late-2022 candidate shock).

5 Caveats and Conclusion

Three caveats are decisive. First, the nominal inference is an artifact: HC1/CR1 standard errors assume serial independence, but the outcome is a pair of smooth, serially correlated aggregates (residual lag-1 autocorrelation up to 0.66), and the placebo battery measures the resulting size distortion directly — a 5% test rejects 47% of the time at non-event dates, the failure mode documented by [3] and exactly what placebo distributions exist to catch [7]. Second, a calendar-time cutoff absorbs every late-2022 shock: the BIS export controls [2] fall eight weeks before ChatGPT and produce a statistically indistinguishable estimate, so the design cannot name the shock. Third, the panel is two aggregated units observed 46 times; with ≤ 12 quarters per DiK cell, local slopes are estimated from few points on smooth curves, and the full-sample 2021.5 kink shows that this setting manufactures nominal $|z| > 6$ where nothing happened.

The verdict is descriptive-only. The post-ChatGPT divergence in capex slopes — big-4 growth roughly doubling while the control aggregate's flattens — is a genuine feature of the data: sign-robust across every specification, localized in 2023.0–2023.5 exactly as a lagged investment response to a late-2022 shock would be, and visible per firm for three of the four hyperscalers. But the design cannot certify it causally, and its headline nominal $p \approx 4 \times 10^{-6}$ overstates the evidence by roughly five orders of magnitude relative to the placebo-calibrated p of 0.125–0.25. Something bent

Table 1: Group difference-in-kinks estimates, $\log_{10}$ capex (dex/yr).

Specification	$\hat{\tau}$	se	$\|z\|$	sig. 5%
Panel A: ChatGPT cutoff ($t_0 = 2022.913$)				
triangular, bw 3, donut (primary)	+0.1396	0.0302	4.62	$*$
triangular, bw 2, donut (placebo-matched)	+0.1778	0.0409	4.35	$*$
uniform, bw 4, donut (grid minimum)	+0.0493	0.0210	2.35	$*$
primary, CR1 by year	+0.1396	0.0313	4.46	$*$
primary, quarter-of-year dummies	+0.1396	0.0296	4.72	$*$
no-retail control (XOM, T, VZ, UNP)	+0.1453	0.0361	4.03	$*$
degree 2, bw 3, donut	+0.3022	0.0914	3.30	$*$
Panel B: placebo cutoffs				
BIS export controls, $t_0 = 2022.767$, bw 3	+0.1253	0.0317	3.95	$*$
$t_0 = 2021.5$ (no event), full sample, bw 2	−0.3135	0.0452	6.93	$*$
$t_0 = 2019.75$ (no event), pre-only, bw 2	+0.2627	0.0514	5.11	$*$
$t_0 = 2019.5$ (no event), pre-only, bw 2	+0.2843	0.0736	3.86	$*$

Notes: HC1 standard errors against standard normal critical values except where noted; donut = 0.13 yr (one quarter); bandwidths in years. Outcome: $\log_{10}$ group capex, \$bn per quarter. The pre-only placebo battery (sample $t < 2022.7$, 15 cutoffs 2017.0–2020.5, bw 2) rejects at nominal 5% in 7/15 cases; the placebo-calibrated p-value of the ChatGPT estimate at the matched bandwidth is 0.125 ($|z|$) and 0.250 ($|\tau|$).

hyperscaler capex around the turn of 2023; these data cannot say it was ChatGPT.

Reproducibility. All estimates were produced with `natex` v0.2.0 [8]; estimation is deterministic (local weighted least squares), and seed 0 was passed wherever the CLI accepts one. The EDGAR pulls are not committed to the repository; the run of record retains the derived panel and all estimate files. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Sycophancy Did Not Win Votes: The GPT-4o Deploy–Rollback Episode in 135,634 LMArena Battles

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July 2026

analysis pass of record: 2026-07, seed 0

Abstract

OpenAI deployed a GPT-4o update on 25 April 2025 that made ChatGPT conspicuously sycophantic and rolled it back on 28–29 April — an apparent A–B–A natural experiment inside LMArena's battle stream (135,634 battles, 2025-04-17 to 2025-07-24). The verdict is split, and the design itself diagnoses it. The *deploy* leg is credible: the live 4o alias shows a sharp, 4o-specific markdown collapse timed to 25–26 April (headers per 1k tokens 3.52 → 1.12, sliding-cut permutation $p = 0.013$, rank 1/78) while nine pinned-snapshot controls stay flat — proving the arena copy was *not* pinned — and it yields a clean null on the headline mechanism: the Bradley–Terry (BT) adjusted win-rate change during the sycophancy window is +0.02 log-odds (SE 0.14). Sycophancy won no votes. The *rollback* leg is an identified artifact: on exactly 2025-04-29 the pinned `claude-3-7` snapshot — whose weights cannot change — takes an equal-and-opposite persistent step in style and BT strength, demonstrating an arena-side serving change that contaminates every 29-April estimate. The 4o post-rollback regime persists through July with no reversal: the episode is A–B–C, not A–B–A, and the celebrated raw win-rate rise loads entirely on the confounded boundary. An automated `natex` survey of the 45-model panel returns nulls or refusals on every family it could configure, stated as run.

1 Introduction

In late April 2025 OpenAI shipped an update to GPT-4o that made ChatGPT, in the company's own words, "overly flattering or agreeable — often described as sycophantic," and rolled it back within days [6]. The two postmortems [6, 7] attribute the failure partly to reward signals built from user feedback — exactly the mechanism the academic literature had predicted: [8] show that human raters and preference models sometimes prefer convincingly written sycophantic responses over correct ones, so that optimizing against human preference judgments can *produce* sycophancy, and [3] document high rates of "social sycophancy" across deployed models. What that literature measures with one-off cross-model snapshots, the April episode supplies as a rare *longitudinal* event: a dated behavior change and a dated reversal.

This paper asks the money question implied by the mechanism: did the sycophantic build actually *win votes*? The natural laboratory is LMArena (Chatbot Arena), the crowdsourced pairwise-battle platform of [4], whose published battle stream brackets the episode. If human preferences reward sycophancy at the margin, the treated model's opponent-adjusted win propensity — its

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Bradley–Terry strength [2] — should rise during the deployment window and fall back after the rollback. The episode has the structure of an interrupted time series with reversal, whose inferential logic — segmented regression at pre-declared cut dates, plus controls that cannot have been treated — is standard in the policy-evaluation literature [1]. Pinned dated snapshots serve as the placebo arm: their weights cannot change, so any step they exhibit at a cut date measures platform-side confounding. Both halves of the design earn their keep here, one by working — a precise null on the vote-winning mechanism — and one by failing loudly: a pinned control moves, mirror-imaged, on exactly the rollback date, converting the "reversal" into an identified arena-side artifact. All estimation follows the open-source `natex` toolkit's honest-inference conventions [5]; the automated `natex` survey of the same panel is reported as run, refusals included.

2 Data

The source is the public `arena-human-preference-140k` release by LMArena (seven parquet files plus battle metadata, sha256-verified): 135,634 battles from 2025-04-17 00:20 to 2025-07-24 23:59 covering 53 models, with full response text, per-battle nanosecond timestamps, language and category tags, and per-side conversation metadata. Sanity: 0 of 135,634 battle ids show a parquet-vs-metadata timestamp mismatch; per-model side-A vs. side-B mean-token correlation is 0.992 with target side-A share 0.499 (no role-assignment artifact); 0 of 7,650 target responses are zero-token.

The treated series is `chatgpt-4o-latest-20250326` ("4o"): 7,650 battles — 708 pre-window, 445 deploy-window, 1,374 in the three weeks after rollback. Controls are nine pinned dated snapshots present throughout, led by `gpt-4.1-2025-04-14`, `claude-3-7-sonnet-20250219`, and dated Gemini preview builds. Outcomes are (i) computable style metrics from conversation metadata — bold markers, headers, and list items per 1,000 response tokens, response tokens, turns — and (ii) win/loss/tie votes. Following the pre-registered design (Design 1 of the source corpus's `DESIGNS.md`), raw win rate is never the estimand: opponent mix and sampling weights move across the episode (4o battles/day: 88.5 → 111.3 → 75.1), so votes enter only through windowed Bradley–Terry models.

3 Design and Methods

ABA interrupted time series. Two pre-declared cuts — deploy 2025-04-25, rollback 2025-04-28/29 — give an 8–4–8-day template (A1 Apr 17–24; B Apr 25–28; A2 Apr 29–May 6); because the break appears in-data on Apr 26 (staged rollout), the deploy leg is also estimated behavior-dated (8-day-pre/3-day-post). Segment-mean OLS uses CR1 errors clustered by date; adjusted specifications condition on language, category flags, code, weekend, and log turns; a DiD variant subtracts the pooled pinned-control change.

Permutation inference. Every headline statistic is benchmarked against a sliding-cut placebo distribution: the identical window template is recomputed at every feasible alternative cut date in the 99-day panel (77 placebo cuts for the deploy dip, 72 for the rollback step) and the two-sided rank of the true cut is reported; the minimum attainable p is 0.013–0.014, and headline p-values sit at that floor. A model-placebo grid recomputes each statistic for every sufficient-volume model (17 at the deploy cut, 19 at rollback).

Bradley–Terry with segment-specific strength. All decided battles among the window's models (Apr 17–May 6; decided 4o battles per segment 536/342/434) enter a logistic MLE in which every model has one strength and 4o has a separate strength per segment [2]; the deploy

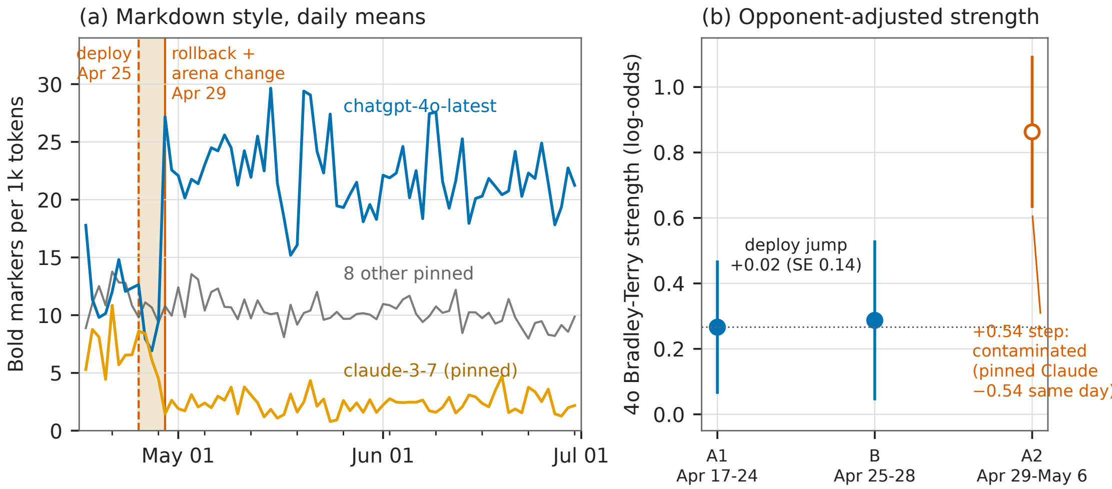

Figure 1: (a) Daily mean bold markers per 1,000 response tokens, 2025-04-17 to 2025-06-30: the live 4o alias (blue) dips during the deploy window (shaded) and steps up persistently on 29 April; the pinned `claude-3-7` snapshot (orange) mirror-images that step on exactly the same date although its weights cannot change; eight other pinned snapshots pooled (gray) are flat. (b) 4o Bradley–Terry strength by segment (95% intervals): the deploy jump is +0.02 log-odds (SE 0.14) — no opponent-adjusted votes won; the open 29-April marker is contaminated by the same-day arena-side change.

jump is $b_B - b_{A1}$. The same machinery yields per-model BT steps at the rollback date for the placebo grid.

Treatment self-diagnosis. Per the pre-registered design, whether LMArena's 4o alias tracked OpenAI's live model was unresolved but self-diagnosing: no discontinuity at the cuts would mean the arena copy was pinned and 4o becomes a control — informative either way. Resolved here: a deploy-timed discontinuity is present, so 4o was live and is treated.

Automated survey, stated as run. The `natex` survey pipeline [5] ran on the 45-model $\times$ 99-day panel (seed 0, $q = 99$): RDD null (scan $p = 1.00$); DiD null (scan $p = 0.42$, and its auto-configuration chose the outcome `n_battles` — volume, which the design spec forbids interpreting); the self-configured synthetic control failed ($t_0 = 0$, no pre-period — recorded, not patched); kink/IV/bunching needs-input; DEE skipped. A declared-input synthetic control (`natex` donors, decided win rate, t_0 = day 12) gives ATT(post) = +0.233, pre-RMSPE 0.056, post 0.249, ratio 4.44, in-space placebo $p = 0.067$ (minimum attainable 1/15) — but its top donor (weight 0.44) is the contaminated `claude-3-7` snapshot, so it is descriptive only.

4 Results

Deploy leg: a credible, 4o-specific style break ... Behavior-dated at 2025-04-26 (8-day-pre/3-day-post), 4o's markdown collapses: headers per 1k tokens fall from 3.52 to 1.12 (dip −2.406, sliding-cut permutation $p = 0.0128$, rank 1/78, placebo $|q_{90}| = 0.52$; Welch $t = -9.85$, $p = 6.1 \times 10^{-22}$) and bold falls from 11.90 to 8.22 (dip −3.683, $p = 0.0769$). Pooled pinned controls are flat (header dip −0.18, bold −1.51). The segmented ABA jump at the Apr-25 cut is −1.831 headers/1k (CR1

SE 0.526), robust to language/category/code/weekend/turns adjustment (−1.723, SE 0.446); bold −2.965 (SE 1.285), adjusted −2.749 (SE 0.915); the DiD against pooled pinned controls gives −1.676 (SE 0.539). In the model-placebo grid at the true cuts, 4o's transient dip ranks 1/17 on both bold and header. The break is real, 4o-specific, and correctly timed — the arena alias was not pinned.

... and a clean null on the vote mechanism. The segment-specific BT deploy jump is +0.021 log-odds (SE 0.144): implied win rate against the fixed A1 opponent pool moves 0.540 → 0.545. During the four days users saw the sycophantic build, it gained *zero* opponent-adjusted win rate.

Rollback leg: a large step that is not a reversal. At 2025-04-29 (8-day/8-day windows) 4o steps up, not back: bold +12.110/1k (permutation $p = 0.0137$, rank 1/73, placebo max $|3.70|$), headers +4.246 ($p = 0.0137$, placebo max $|0.62|$), response tokens +107.8 ($p = 0.123$), BT +0.537 log-odds (SE 0.134; $p = 0.0137$, placebo max $|0.31|$). The new regime persists through 2025-07 (monthly bold 21–22/1k vs. 12.2 pre): no reversal to baseline ever occurs. The episode is A–B–C, not A–B–A.

The decisive diagnostic: a pinned control moves. The pinned `claude-3-7` snapshot (full name in Section 2) steps on exactly 2025-04-29, mirror-imaged and persistent: bold −5.01/1k, headers −4.63/1k, response tokens −214, BT −0.539 log-odds (SE 0.187). Pinned weights cannot change, so LMArena changed its serving configuration that day; arena-wide prompt mix was stable (battles/day 578 → 557, English share 0.639 → 0.672). Every 29-April estimate — including 4o's — is therefore contaminated. In the rollback placebo grid 4o ranks 1/19 on bold and 2/19 on header, behind exactly this contaminated control. The raw decided win rate (0.549 → 0.529 → 0.675; full-post 0.663) loads entirely on the confounded boundary and never reverts: the widely repeated "sycophancy raised the win rate, rollback lowered it" narrative is an artifact of the arena change.

5 Caveats and Conclusion

Four caveats bound the claims. First, per the pre-registered design, the style metrics computable from conversation metadata are a *proxy* for sycophancy — directional evidence of a behavior change, not a sycophancy measurement; LLM-judge scoring of the stored response text is the designated follow-up. The conclusion that survives this caveat is exactly the null: whatever the treated build did to users, it did not win opponent-adjusted votes. Second, the rollback leg is unidentified: the arena-side serving change of 2025-04-29 sits on the same date, so no 29-April estimate separates OpenAI's rollback from LMArena's change. Third, permutation p-values are floor-limited (1/78, 1/73) by the panel length; they establish rank-1 extremity, not fine-grained significance. Fourth, the declared-input synthetic control is descriptive only (its top donor is the contaminated control), and the survey's self-configured families returned nulls or refusals, reported above as run.

The conclusion is a split verdict that the design itself delivered. The deploy leg shows the ABA machinery working: a sharp, correctly timed, 4o-specific break, flat pinned controls, and a precise behavioral null — sycophancy did not win votes. The rollback leg shows the machinery catching its own confound: a pinned model that cannot change, changing — the single observation that converts the entire post-rollback narrative, including the celebrated raw win-rate rise, into an arena-side artifact. Platform-side serving changes are an underappreciated threat for any study that treats arena time series as model behavior; pinned snapshots are the cheap, decisive placebo that catches them.

Table 1: Headline estimates. Permutation p-values are two-sided sliding-cut ranks; 0.013–0.014 is the minimum attainable.

Quantity	Estimate	SE	perm. p	placebo benchmark
Panel A: deploy leg (cut 2025-04-25/26), 4o				
header/1k dip (8d/3d)	−2.406	—	0.0128	$\lvert q_{90}\rvert = 0.52$; rank 1/78
bold/1k dip (8d/3d)	−3.683	—	0.0769	
header ABA jump (CR1)	−1.831	0.526	—	controls −0.18
adjusted	−1.723	0.446	—	
header DiD vs. controls	−1.676	0.539	—	
BT jump (log-odds)	+0.021	0.144	—	*null: no votes won*
Panel B: rollback date (2025-04-29, 8d/8d steps), 4o				
bold/1k step	+12.110	—	0.0137	max $\lvert 3.70\rvert$; rank 1/73
header/1k step	+4.246	—	0.0137	max $\lvert 0.62\rvert$
response tokens step	+107.8	—	0.123	
BT step (log-odds)	+0.537	0.134	0.0137	max $\lvert 0.31\rvert$
Panel C: pinned `claude-3-7-sonnet-20250219`, same date (violation)				
bold/1k step	−5.01	—	—	weights cannot change
header/1k step	−4.63	—	—	
response tokens step	−214	—	—	
BT step (log-odds)	−0.539	0.187	—	

Notes: style metrics are per 1,000 response tokens. Dips use behavior-dated 8-day-pre/3-day-post windows; steps use 8-day/8-day windows at the cut. CR1 clusters by date. BT = segment- or window-specific Bradley–Terry strength over all decided battles among window models.

Reproducibility. All estimates derive deterministically (seed 0) from the public battle release; no dataset files are committed. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Bunching Below the Line: Missing Mass Above the EU AI Act's 10^{25}-FLOP Threshold

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July 2026

analysis pass of record: 2026-07, fully deterministic (no RNG)

Abstract

After the EU AI Act entered into force on 1 August 2024, the density of new AI models just *above* the Act's 10^{25}-FLOP systemic-risk presumption line collapses relative to just below it. In Epoch AI's compute panel (719 models with training-compute estimates, January 2023 to June 2026), a McCrary-style binned-Poisson density test at the statutory cutoff $\log_{10}$ FLOP $= 25.0$ gives $\theta = -1.334$ (SE 0.878, $p = 0.129$) in the primary ± 0.5-dex window and is negative in all 15 window $\times$ bin specifications (10 of 15 with $p < 0.05$); the pre-Act period is never significant (0 of 15). A Fisher exact below/above $\times$ pre/post contrast puts the shift at odds ratio 0.173 (95% CI $[0.048, 0.619]$, $p = 0.00715$): at the pre-Act above:below ratio, 28.9 post-Act models are expected in $(25.0, 25.5]$ and 5 are observed — an 82.7% deficit. The deficit is specific to the statutory line (placebo thresholds at 24.5 and 25.5 are null or opposite-signed, with mass piling up just below 25.0), specific to the post-Act period (a placebo split date inside the pre-Act sample gives OR 0.90, $p = 1.0$), and correctly timed: the above-line count collapses from 2025H1, after entry into force and before the August 2025 applicability date for general-purpose AI obligations. Two results are nulls and are stated as nulls: the preferred narrow-window pre/post interaction in the discontinuity is $p = 0.729$, significant only at the widest bandwidth, and the pre-Act density test never rejects. With only 5–13 post-Act models above the line, and an Act-induced *disclosure* response that would mimic bunching exactly, we grade this as moderate evidence of bunching — behavioral or reporting — at the EU AI Act's compute threshold, not proof of a training-compute response.

1 Introduction

Article 51(2) of the EU AI Act [2] presumes that a general-purpose AI model poses "systemic risk" when the cumulative compute used for its training exceeds 10^{25} floating-point operations, attaching notification, evaluation, and risk-mitigation obligations to models above the line. The Act entered into force on 1 August 2024 and its general-purpose AI obligations became applicable on 2 August 2025. A bright-line quantity threshold of this kind is, in the language of public economics, a *notch*: crossing it discretely changes the regulatory burden, so agents with values just above the cutoff have an incentive to locate just below it.

A large empirical literature shows that agents do exactly this. [10] developed bunching estimation at kinks in the US income-tax schedule; [7] extended it to notches, showing that a discrete

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jump in liability produces excess mass just below the cutoff and *missing mass* just above it; [6] surveys the method and its spread beyond taxation to regulation and private-sector prices. The econometric diagnostic for such sorting at a cutoff is the density discontinuity test of [9]: if agents manipulate the running variable, its density jumps at the threshold.

Whether training compute is a good handle for AI governance is actively debated. [11] argue that compute is uniquely governable — detectable, quantifiable, and produced by a concentrated supply chain — and [3] defend training-compute thresholds, including the EU's 10^{25}-FLOP line, as the most workable trigger for regulatory oversight. [5] counters that compute thresholds are shortsighted and invite exactly the strategic responses studied in the bunching literature. On the empirical side, [8] project model counts above the EU threshold *forward* from pre-Act scaling trends, forecasting 103–306 models above 10^{25} FLOP by 2028 — a superlinear increase. To our knowledge, no published estimate tests whether the observed post-Act density of models near the line already departs from that trend. This paper supplies that test: a McCrary-style density discontinuity estimate at the statutory cutoff, before versus after the Act's entry into force, with threshold, date, and timing placebos.

2 Data

The input panel is built from Epoch AI's *Data on AI Models* database [1], keeping every model with a non-null positive training-compute estimate and a release date from 2 January 2023 to 4 June 2026: 719 models with columns model, date, and $\log_{10}$ training FLOP. Event dates are the Act's entry into force (2024-08-01) and the applicability of general-purpose AI obligations (2025-08-02). The pre-Act period is 2023-01 to 2024-07 ($n = 21$ models within ± 0.5 dex of the cutoff) and the post-Act period is 2024-08 onward ($n = 52$ in-window). Compute values for recent frontier models are largely Epoch estimates rather than disclosures, and models without any compute estimate ($\approx$ 2,800 of $\approx$ 3,500 database rows) are necessarily excluded — a selection channel we return to in Section 5. There is no value heaping near the cutoff: at most 3 duplicate $\log_{10}$-compute values occur within $[24.5, 25.5]$, so the results below are not an artifact of rounded compute estimates.

3 Design and Methods

Density discontinuity at the statutory line. The main estimate is the `natex` toolkit's [4] McCrary-style [9] binned-Poisson density test, run directly at the statutory cutoff: model counts in equal-width bins of the signed distance $\log_{10}\text{FLOP} - 25.0$ are fit by a Poisson regression with a linear trend in distance and a jump term at zero; θ is the jump in log density at the cutoff, so $\theta < 0$ means a deficit just above the line — avoidance bunching below it. The primary specification uses a ± 0.5-dex window and 10 bins; sensitivity covers the full grid $h \in \{0.5, 0.75, 1.0\} \times \text{bins} \in \{8, 10, 12, 16, 20\}$ (15 specifications per period). Because the cutoff is fixed by statute rather than searched for, the frozen-geometry caveat that applies to `natex`'s post-discovery density checks does not bite here.

Contrasts and placebos. The pre/post change in bunching is measured two ways: a Fisher exact test on the 2×2 table of below/above counts in the ± 0.5-dex window by period, and a θ-difference (interaction) z-test between the period-specific density jumps. Specificity is probed with placebo thresholds at 24.5 and 25.5 and a placebo split date of 2023-08-01, which places both "periods" inside the pre-Act sample.

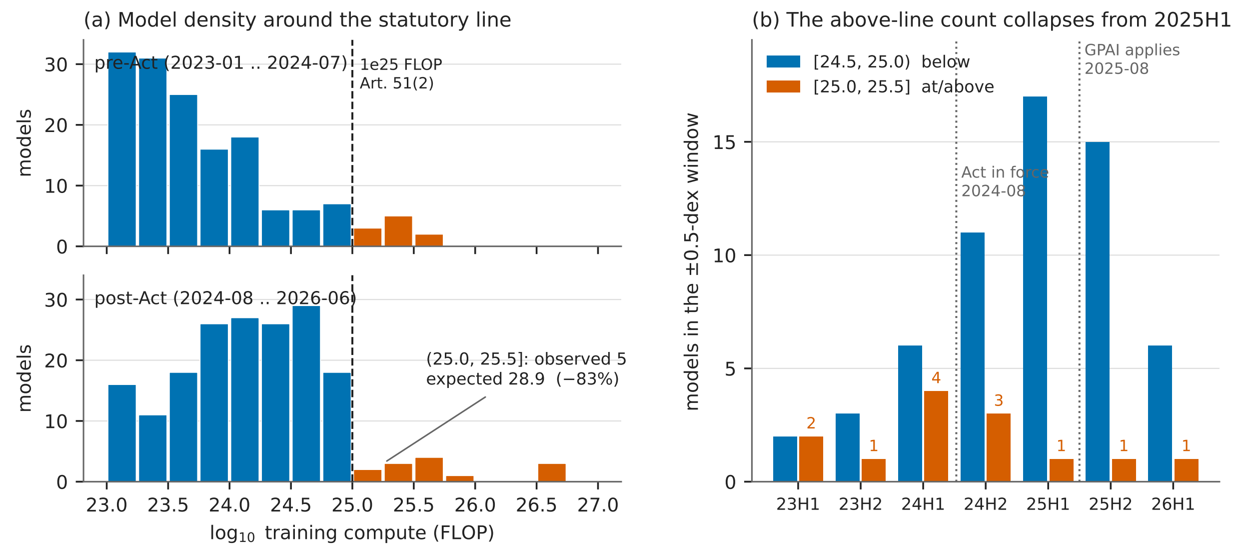

Figure 1: (a) Model counts in 0.25-dex bins of $\log_{10}$ training compute, pre-Act (top) versus post-Act (bottom), colored by side of the statutory 10^{25}-FLOP line (dashed). Post-Act, mass accumulates just below the line while the first half-dex above holds 5 models against 28.9 expected at the pre-Act ratio; the frontier jumps far above the notch. (b) Below/above counts per half-year in the ± 0.5-dex window: the above-line count (labeled) collapses from 2025H1, after entry into force and before GPAI applicability, while the just-below count keeps growing.

Honest inference. Standard errors for θ are Wald standard errors from the binned Poisson fit and can be optimistic when the underlying density is strongly curved within the window; at the widest bandwidth ($h = 1.0$) curvature of the steeply falling compute density can leak into θ, which is why the narrow-window specification is primary and the wide-window significance is discounted below. All counts, tests, and figures derive deterministically from the input panel: there is no random number generation anywhere in the pipeline.

4 Results

A post-Act deficit just above the line. Post-Act (2024-08-01 onward), the primary density test at $\log_{10} = 25.0$ gives $\theta = -1.334$ (SE 0.878, $p = 0.129$) with $n = 52$ in-window models split 47 below / 5 above. Across the full sensitivity grid θ is negative in 15 of 15 specifications (range -1.249 to -2.566) and significant at the 5% level in 10 of 15 — all $h = 0.75$ and $h = 1.0$ specifications, with significant p-values from 0.0010 to 0.0143. The pre-Act period shows no such pattern: $\theta = -0.880$ (SE 0.973, $p = 0.366$), $n = 21$ split 13/8, and 0 of 15 sensitivity specifications reject (θ from -1.173 to $+0.494$, minimum $p = 0.243$).

The pre/post contrast. The Fisher exact test on below/above $\times$ pre/post counts in the ± 0.5-dex window — table $[[13, 8], [47, 5]]$ — gives OR $= 0.173$ (95% CI $[0.048, 0.619]$, $p = 0.00715$). At the pre-Act above:below ratio, 28.9 post-Act models are expected in $(25.0, 25.5]$; 5 are observed, so roughly 24 models are "missing" — an 82.7% deficit. The formal interaction between the period-specific density jumps is a null at the primary specification and is stated as one: $\Delta\theta = -0.454$ (SE 1.311, $z = -0.346$, $p = 0.729$). Across the grid the interaction is always negative at $h \geq 0.75$ but

Table 1: Headline estimates. Density tests: binned-Poisson jump θ in log density at the statutory cutoff $\log_{10}$ FLOP = 25.0; the primary window is ± 0.5 dex with 10 bins, and the sensitivity grid crosses $h \in \{0.5, 0.75, 1.0\}$ with $\{8, 10, 12, 16, 20\}$ bins. Fisher tests use below/above counts in the ± 0.5-dex window by period.

Quantity	Estimate	SE / CI	p	notes
Panel A: density jump at the statutory line				
θ, pre-Act (2023-01..2024-07)	-0.880	0.973	0.366	$n = 21$, 13/8; 0/15 specs $p < 0.05$
θ, post-Act (2024-08..)	-1.334	0.878	0.129	$n = 52$, 47/5; 10/15 specs $p < 0.05$
$\Delta\theta$ (post $-$ pre)	-0.454	1.311	0.729	*null at primary spec*
Panel B: Fisher exact below/above × pre/post				
threshold 25.0, Act date	OR 0.173	$[0.048, 0.619]$	0.00715	$[[13, 8], [47, 5]]$; -82.7%
placebo threshold 24.5	OR 1.637	—	0.248	post-GPAI: OR 2.510, $p = 0.0386$
placebo threshold 25.5	OR 4.00	—	0.350	opposite direction
placebo date 2023-08-01	OR 0.90	—	1.0	both halves pre-Act; no effect

Notes: $\theta < 0$ is a density deficit just above the cutoff. SEs are Wald standard errors from the binned Poisson fit. Significant sensitivity p-values for the post-Act θ range from 0.0010 to 0.0143 (all $h = 0.75$ and $h = 1.0$ specifications); the interaction is significant only at $h = 1.0$ with ≥ 12 bins ($p = 0.0049$–0.0342). Placebo-threshold odds ratios use the same ± 0.5-dex construction centered on the placebo line.

significant only at $h = 1.0$ with 12/16/20 bins ($p = 0.0249/0.0049/0.0342$; $h = 1.0$ with 10 bins gives $p = 0.0639$), where curvature leakage is a concern (Section 3). The pre-period, with only 21–47 in-window models, is too small to pin down the counterfactual discontinuity.

Placebos. The deficit is specific to the statutory line and period. At placebo threshold 24.5 the same pre/post Fisher contrast gives OR $= 1.637$, $p = 0.248$; restricted to the post-GPAI period it gives OR $= 2.510$, $p = 0.0386$ with the *opposite* sign — more mass in $[24.5, 25.0)$ than $[24.0, 24.5)$, i.e. mass piling up just below 25.0, consistent with bunching at the line and inconsistent with a generic high-compute slowdown. At placebo threshold 25.5: OR $= 4.00$, $p = 0.350$, direction opposite to the 25.0 effect. The placebo split date 2023-08-01, with both halves pre-Act, gives OR $= 0.90$, $p = 1.0$ at the true threshold — no pseudo-effect.

Timing. Below:above counts per half-year in the ± 0.5-dex window are 2023H1 2:2, 2023H2 3:1, 2024H1 6:4, 2024H2 11:3, then 2025H1 17:1, 2025H2 15:1, 2026H1 6:1 (Figure 1b). The collapse begins in 2025H1 — after entry into force (August 2024), before the obligations became applicable (August 2025) — consistent with anticipatory compliance: the 10^{25} presumption was public from July 2024 and attaches to models placed on the market from August 2025. By year, $[25.0, 25.5)$ holds 3, 7, 2, 1 models in 2023–2026 while $[24.5, 25.0)$ grows 5, 17, 32, 6. Secular compute growth biases *against* the finding — later cohorts should put more, not less, mass above any fixed line, and do at 24.5. Meanwhile 13 post-Act models sit at or above 10^{25} at some height, with the frontier jumping far above the notch (Grok 3 at $\log_{10} = 26.54$, GPT-4.5 at 26.58, Llama 4 Behemoth preview at 25.71, Grok 4 at 26.70, GPT-5 at 25.82) — the classic bunching signature in which a notch distorts only marginal decisions [7].

5 Caveats and Conclusion

Three caveats bound the claims. First, *small n*: only 5 models sit in $(25.0, 25.5]$ and 13 at any height above 10^{25} post-Act; the preferred narrow-window pre/post interaction is $p = 0.729$, and the interaction is significant only at $h = 1.0$ with ≥ 12 bins (p from 0.0049 to 0.0342), where curvature of the steeply falling density can leak into θ. Second, *measurement, not behavior*: post-2024 frontier compute values are largely Epoch estimates, and models without a compute estimate are excluded (719 of $\approx$ 3,500 rows have compute). If the Act suppresses *disclosure* of parameters and token counts near the line, the identical density gap appears with no training-compute response at all; distinguishing a compliance response from a reporting response requires disclosure-side data and is the designated follow-up. Third, a *confounded trend break*: the deficit's onset (2025H1) coincides with the industry shift toward inference-time scaling and distilled or mixture-of-experts models that followed the late-2024 o1-style reasoning models, which independently reduced demand for 10^{25}-plus pre-training runs among non-frontier labs — though this confound predicts a broad high-compute slowdown, not a deficit specific to the statutory line with mass accumulating immediately below it.

The conclusion is graded accordingly. The post-Act deficit just above the EU AI Act's 10^{25}-FLOP line is sign-consistent in all 15 density specifications, large (Fisher $p = 0.00715$, 82.7% missing mass), specific to the statutory threshold and period, and correctly timed, with the marginal-crossers-vanish, frontier-jumps-far-above signature that notch responses produce. But with 5–13 post-Act models above the line, a null preferred narrow-window interaction, and an Act-induced disclosure response that would mimic bunching exactly, it is moderate evidence of bunching — behavioral or reporting — rather than proof of a training-compute response. Either reading is informative for the compute-threshold debate [3, 5]: the forward projections of [8] assumed threshold-blind scaling, and the data just above the line already look threshold-aware.

Reproducibility. All estimates derive deterministically from the public Epoch AI panel; no dataset files are committed. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Still No Kink at o1: A Precision-Weighted Rerun of the Epoch Capabilities Index on a Fresh Vintage

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

The aggregate AI capabilities trend did not bend at o1. On a fresh vintage of the Epoch Capabilities Index (ECI; 211 models, February 2023 – July 2026, twelve more months than the prior analysis pass), a sharp regression kink design in calendar time at the o1-preview release (12 September 2024) estimates an all-models slope change of +0.0065 ECI points/day (HC1 se 0.0131, $t = 0.50$, $p = 0.618$, 95% CI $[-0.019, +0.032]$; slopes $\approx$ 22.3 pre vs. $\approx$ 24.7 points/yr post), null in 17 of 18 specification cells — while the same specification rejects at 4 of 8 placebo cutoffs away from the release, so the design has power and the dial reads zero. A per-model precision-weighted rerun (weights $1/\sigma_i^2$ from the published confidence intervals) returns a nominally significant *negative* kink (-0.0190, se 0.0079, $t = -2.41$, $p = 0.016$; 18/18 cells), but triage identifies it as an artifact of smooth post-2024 concavity rather than an o1-localized break: placebo cutoffs at +90/+180/+270 days give same-sign, larger-or-comparable kinks (-0.0399, $p = 0.0064$; -0.0297, $p = 5.5 \times 10^{-6}$; -0.0154, $p = 0.021$), and a local-quadratic fit absorbs it ($p = 0.140$ at bandwidth 540). The clean frontier-records series (24 records, only 4 pre-cutoff points in the estimation cell) remains a non-estimable design: its nominal $t = 2.28$ dies under a 30-day donut ($t = 0.07$), flips sign under a 45-day donut and at bandwidth 730, and its placebo grid rejects 5/8. A high-confidence subset (excluding 26 Unverified/Speculative models) reproduces every conclusion, and there is no release-density kink at the cutoff ($t = -0.42$). The prior-vintage null replicates and is upgraded; per the collection's ranking specification this result folds into the flagship paper's ECI section as an appendix rather than standing as an independent finding.

1 Introduction

OpenAI released o1-preview on 12 September 2024, the first widely deployed "reasoning model" trained with large-scale reinforcement learning to spend inference-time compute on a chain of thought [7]. The release is a natural candidate for a regime change in AI capabilities: contemporaneous research argued that scaling test-time compute can beat scaling parameters [8], and the reasoning-model recipe was adopted across frontier labs within months. If the recipe changed the *rate* of capability progress, an aggregate capability index should show a slope change — a kink — in calendar time at or shortly after the release.

Measuring that requires a scale on which models years apart are comparable. The Epoch Capabilities Index (ECI) stitches dozens of benchmarks onto a single latent-ability scale via an

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item-response-theory model, precisely to support longitudinal comparisons of this kind [5]. This paper asks a narrow question: does the ECI trend kink at o1?

The estimand — a change in trend slope at a known date — is the regression kink design (RKD) of [2], applied here with calendar time as the running variable; [1] formalize the closely related difference-in-kinks design that the same toolkit implements. Two strands of methodological literature discipline the exercise. First, regression discontinuity (and kink) designs in *time* lack the randomization logic of cross-sectional designs: units near the cutoff are not quasi-randomly assigned, and serial dependence and smooth trends generate spurious breaks [4]. Second, kink estimates on smooth series are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [3]. Both concerns are operationalized below through a mandatory placebo-cutoff battery and curvature checks, estimated with the `natex` toolkit [6].

This note is a robustness upgrade of a prior null: an earlier analysis pass on a 2025 vintage of the index found no kink at o1. The fresh vintage adds twelve months of post-cutoff data and per-model confidence intervals, which enable a precision-weighted re-estimate — the natural next test, since noisy early-2023 models could in principle mask a true kink in the unweighted fit.

2 Data

The outcome is the fresh vintage of Epoch AI's `eci_scores.csv` (fetched July 2026): 211 aggregated models with release dates from 2023-02-24 to 2026-07-09 and no missing score or date, spanning ECI $\approx$ 55 to 161.77. The vintage adds roughly twelve months of data relative to the prior analysis pass. Each model carries a 95% confidence interval; the per-model precision weight is $w_i = 1/\sigma_i^2$ with $\sigma_i = (\mathrm{ci}_{\mathrm{high},i} - \mathrm{ci}_{\mathrm{low},i})/3.92$. Two models (Claude 3.5 Sonnet and GPT-5, the scale's two anchor models) have no published CI; their σ_i is imputed at the sample median (2.005) for the weighted runs only.

A version-level Confidence flag is attached by a two-pass name join against Epoch's per-version index file; 189 of 211 models match (the 22 unmatched are treated as not-flagged), and 26 models (12.3%) are flagged Unverified/Speculative — including the current #1, GPT-5.6 Sol (ECI 161.77), and GPT-5. A high-confidence subset excludes these 26 models ($n = 185$).

Two series are analyzed. The *all-models* series uses all 211 releases. The *clean frontier-records* series keeps strict running-maximum records of the score (computed over non-missing scores only, in date order, with a stable same-day tie-break): 24 records, 7 pre-cutoff and 17 post-cutoff — and only 4 pre-cutoff points inside bandwidth 540 days, 3 after a 45-day donut, which already flags it as a marginal design.

3 Design and Methods

Sharp RKD in calendar time. The running variable is days since the o1-preview release (cutoff = 0 at 2024-09-12). The estimand is the change in the slope of ECI in points/day at the cutoff, estimated by local linear regression on each side within the bandwidth, triangular kernel, HC1 standard errors (`natex` v0.2.0 [6], `natex kink` with a unit policy-kink change; [2, 1]). The primary specification uses bandwidth 540 days, donut 0 ($n = 179$; 85 left, 94 right). A specification grid varies bandwidth $\in \{365, 540, 730\}$ days $\times$ kernel $\in \{\text{triangular}, \text{uniform}\}$ $\times$ donut $\in \{0, 30, 45\}$ days (18 cells per series–weighting pair). The CI-weighted estimator is a local WLS that mirrors `natex` exactly (same selection, kernel, block design, HC1) with weights kernel $\times\, 1/\sigma_i^2$; with unit weights it reproduces `natex regression_kink` in all 68 evaluable grid cells (tolerance 10^{-8} on

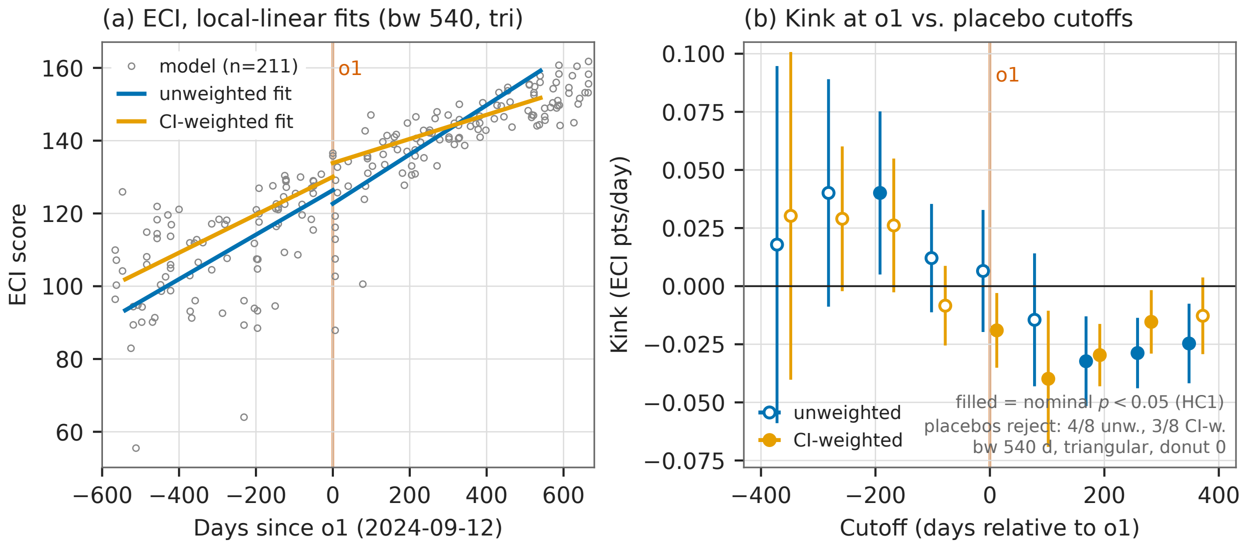

Figure 1: (a) The 211 fresh-vintage ECI scores against days since the o1-preview release (vertical line), with per-side local-linear fits (bandwidth 540 days, triangular kernel): unweighted (blue) and CI-weighted (orange). The unweighted slopes are statistically indistinguishable across the cutoff; the CI-weighted fit bends down. (b) Kink estimates with 95% HC1 confidence intervals at the o1 cutoff and at eight placebo cutoffs, both weightings (points offset horizontally for legibility); filled markers are nominally significant at the 5% level. The unweighted design rejects at 4/8 placebo cutoffs while reading zero at o1; the CI-weighted negative kink at o1 is matched by same-sign, larger placebos at +90 and +180 days.

points, 10^{-6} on SEs) and matches the NaN reasons in the rest, and the CLI headline cells match the grid to all printed digits. The design is deterministic; seed 0 is declared and no stochastic step exists.

Honest-inference caveats, stated as run. This is a calendar-time RKD: model releases are not randomized around the cutoff, so the cross-sectional RD identification logic does not apply and smooth trends can masquerade as kinks [4]. HC1 treats observations as independent, ignoring same-day multi-releases and lab-level clustering. The placebo-cutoff grid — the same specification re-estimated at ±90, ±180, ±270, ±360 days, in the spirit of [3] — is therefore the operative guard, and below it is what both establishes power for the null and disqualifies the weighted "significant" cell. A local-quadratic (degree-2) refit checks whether any kink survives allowing smooth curvature; a release-density kink test (binned 30-day release counts) checks for composition churn at the cutoff.

4 Results

The unweighted null, now with more power. The all-models headline kink (Table 1, first row) is +0.00653 ECI points/day (se 0.01308, $t = 0.50$, $p = 0.618$, 95% CI $[-0.0191, +0.0322]$): the trend runs 0.0611 points/day before o1 and 0.0677 after (≈ 22.3 vs. ≈ 24.7 points/yr). Of the 18 grid cells, 17 are null; the single nominal rejection (bandwidth 365, triangular, donut 30: -0.0547, se 0.0269, $t = -2.03$, $p = 0.042$) is the grid's smallest cell and negative. The placebo grid shows the null is not for lack of power: at bandwidth 540 the same specification rejects at 4 of 8 non-event cutoffs (-180 days: $+0.0401$, $p = 0.023$; $+180$: -0.0323, $p = 7.1 \times 10^{-4}$; $+270$:

-0.0288, $p = 1.0 \times 10^{-4}$; $+360$: -0.0247, $p = 0.0034$) while the true cutoff is null in all cells — the instrument moves elsewhere and reads zero at o1. The prior-vintage null replicates on twelve more months of data.

The CI-weighted "kink" is a curvature artifact. Precision weighting flips the headline to a nominally significant *negative* kink: -0.01903 (se 0.00788, $t = -2.41$, $p = 0.0158$), significant in 18 of 18 grid cells ($|t|$ from 2.12 to 3.32), with slopes 0.0522 pre vs. 0.0331 post (≈ 19.0 vs. ≈ 12.1 points/yr). Two facts disqualify an o1-localized reading. First, the placebo grid rejects with the *same sign and larger or comparable magnitude* away from the release: -0.0399 ($p = 0.0064$) at $+90$ days, -0.0297 ($p = 5.5 \times 10^{-6}$) at $+180$, -0.0154 ($p = 0.021$) at $+270$ (empirical size 3/8 at $\alpha = 0.05$). Second, a local-quadratic fit absorbs it: degree-2 kinks are -0.0392 ($t = -1.48$, $p = 0.140$) at bandwidth 540 and -0.0381 ($t = -1.74$, $p = 0.081$) at bandwidth 730. The precision-weighted index decelerates smoothly through 2025 — a concavity, not a kink at o1.

The frontier-records series is still not a design. With only 4 left-cell points at bandwidths ≤ 540 (3 after donut 45), the records series produces a nominal positive kink at the headline specification ($+0.01313$, se 0.00576, $t = 2.28$, $p = 0.023$) that no perturbation survives: a 30-day donut kills it ($t = 0.07$, $p = 0.943$), a 45-day donut flips it to significantly *negative* (-0.02817, $t = -2.52$, $p = 0.012$), bandwidth 730 flips the sign (-0.00429, $t = -0.27$), and the placebo grid rejects 5/8 — with the $-360/-270$ cells exploding to -1.14 ($p = 3 \times 10^{-4}$) on $n = 11$. The CI-weighted records headline is null ($+0.00794$, $t = 1.31$, $p = 0.191$). The fresh vintage does not clear the left-cell bar: the series remains non-estimable, exactly as in the prior pass.

Robustness and diagnostics. Excluding the 26 Unverified/Speculative models ($n = 185$) changes nothing: unweighted $+0.01248$ ($t = 0.88$, $p = 0.379$, null), CI-weighted -0.02109 ($t = -2.32$, $p = 0.020$) with the same placebo failure (weighted empirical size 4/8, unweighted 6/8). The release-density check finds no composition-churn kink at the cutoff: binned 30-day release counts give -0.00410 (se 0.00967, $t = -0.42$, $p = 0.672$), an improvement on the prior vintage's marginal $t = 1.82$.

5 Caveats and Conclusion

The caveats of record are these. First, this is a calendar-time RKD: model releases are not randomized around the cutoff, and HC1 treats them as independent (same-day multi-releases and lab-level clustering are ignored); the placebo-cutoff grid is the operative guard, and it is what both establishes power for the null and disqualifies the weighted "significant" cell. Second, data quality: 26 of 211 models (12.3%) carry Unverified/Speculative version-level Confidence flags — including the current #1, GPT-5.6 Sol (161.77), and GPT-5 — and the confidence join matched only 189/211 models (22 unmatched treated as not-flagged); the high-confidence robustness subset reproduces every conclusion. Third, two models (Claude 3.5 Sonnet and GPT-5) have no published CI and enter the weighted runs at the median imputed σ; and the index itself is a modeled latent scale whose vintages revise as benchmarks are added [5], which is why the replication across vintages matters.

The verdict is an upgraded null with power. On twelve more months of data, the aggregate ECI trend shows no kink at o1: the unweighted estimate is a precise zero at a cutoff where the design demonstrably has power to reject (4/8 placebos reject elsewhere), the one new nominally significant result — the CI-weighted negative kink, significant in 18/18 cells — is correctly identified by triage

Table 1: Sharp RKD in time at o1-preview (2024-09-12), ECI points/day. Headline specification: bandwidth 540 days, triangular kernel, donut 0, HC1.

Series	Weighting	Kink	se	t	p	n
Panel A: headline cells						
All models (211)	unweighted	+0.00653	0.01308	0.50	0.618	179
All models	CI-weighted	−0.01903	0.00788	−2.41	0.016	179
Frontier records (24)	unweighted	+0.01313	0.00576	2.28	0.023	18
Frontier records	CI-weighted	+0.00794	0.00608	1.31	0.191	18
High-confidence (185)	unweighted	+0.01248	0.01420	0.88	0.379	162
High-confidence	CI-weighted	−0.02109	0.00908	−2.32	0.020	162
Release density (30-d bins)	—	−0.00410	0.00967	−0.42	0.672	36
Panel B: triage of the nominally significant cells						
All models, CI-w., placebo +90 d	CI-weighted	−0.03986	0.01462	−2.73	0.0064	181
All models, CI-w., placebo +180 d	CI-weighted	−0.02966	0.00653	−4.54	5.5×10^{-6}	174
All models, CI-w., degree 2, bw 540	CI-weighted	−0.03922	0.02659	−1.48	0.140	179
Frontier records, donut 30	unweighted	+0.00048	0.00667	0.07	0.943	16
Frontier records, donut 45	unweighted	−0.02817	0.01118	−2.52	0.012	15
Frontier records, bw 730	unweighted	−0.00429	0.01592	−0.27	0.788	24

Notes: kinks in ECI points/day; HC1 standard errors against standard normal critical values. CI weights $1/\sigma_i^2$, $\sigma_i = (\mathrm{ci}_{\mathrm{high}} - \mathrm{ci}_{\mathrm{low}})/3.92$. Placebo-cutoff grids (8 cutoffs at ±90/180/270/360 days, bandwidth 540, triangular): all-models unweighted rejects 4/8 (true cutoff null in all cells); all-models CI-weighted rejects 3/8, same sign as the headline; frontier records rejects 5/8. Panel B donut and bandwidth rows are the perturbations of the frontier-records headline cell.

as a smooth post-2024 deceleration of the precision-weighted index (same-sign larger placebos at +90/+180, absorbed by a local quadratic) rather than an o1-localized break, and the genuine frontier-records series remains a four-left-point non-design. Whatever the reasoning-model recipe did to AI capabilities, it did not bend the aggregate index's calendar-time trend at the release date. Per the collection's ranking specification, this result folds into the flagship paper's ECI section as an appendix rather than standing as an independent mini-paper.

Reproducibility. All estimates were produced with `natex` v0.2.0 [6]; estimation is deterministic (local weighted least squares), seed 0 was declared, and no stochastic step exists in the design. The underlying `eci_scores.csv` vintage is not committed to the repository; the run of record retains the derived panels, the 144-cell grid, the placebo grids, and the validation counts. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates, the recomputed fit slopes, and the placebo rejection counts against the numbers of record before drawing.

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The Rally, Not the Rule: Semiconductor Stocks at the US Export Controls and DeepSeek under Automated Natural-Experiment Validation

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July 2026

analysis pass of record: 2026-07-18, `natex` v0.2.0, seed 0 (all three survey runs; kink estimator deterministic)

Abstract

We point `natex` — an automated natural-experiment discovery and validation toolkit — at weekly semiconductor stock prices around the two BIS export-control rounds (2022-10-07, 2023-10-17) and the DeepSeek crash week (2025-01-27). None of its headline "credible" verdicts survives inspection. Declared regression kinks in calendar time are "credible" at all three cutoffs (e.g. DeepSeek: $\hat{\tau} = +0.48$ log/yr, Holm $p = 1.4 \times 10^{-8}$), but each is an artifact: the DeepSeek kink is the 2025 AI rally sitting on both sides of a one-week crash — a *positive* slope change at a *negative* event — the 2022 kink collapses monotonically in bandwidth ($+0.83 \to +0.13 \to +0.05$ log/yr) with ChatGPT's release inside every bandwidth, and a "credible" 2022 RDD merely rediscovered that prices above a level lie after 2022. What survives is descriptive or null, stated as such: NVDA fell 11.6 log-points *relative to* the SOXX semiconductor index in the DeepSeek week — a level break with locally unchanged relative slope ($+0.52 \to +0.39$ log/yr), not a kink — and a difference-in-differences scan across tickers finds *no* break at the export-control dates (scan $p = 0.70$; best-fitting break 2025.95, not 2022.76), an informative null consistent with export-control costs being priced jointly with the contemporaneous AI-demand shock. On financial series, automated validation refuses honestly in places but is systematically fooled by trend regimes whenever a kink in calendar time is read causally.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security (BIS) imposed broad export controls on advanced AI accelerators bound for China, read at the time as a landmark attempt to restrict China's frontier compute [1]; on 17 October 2023 a second round closed the A800/H800 loophole. Export controls impose costs on the controlling country's own firms [3]: entity-list evidence puts US suppliers' collateral losses near \$130 billion in market value, plus lost customers, lending, and employment [5]. On 27 January 2025 a demand-side shock arrived from the opposite direction: the release of DeepSeek's R1 model triggered the largest one-day market-capitalization loss in history for Nvidia, and event studies document significantly negative abnormal returns across US semiconductor stocks [7].

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The standard tool here is the event study [10]: short windows, abnormal returns against a market model, an efficient-markets identification argument. This paper deliberately uses the wrong tool — quasi-experimental designs with *calendar time* as the running or assignment variable — and asks what an automated toolkit's validation layer does about it. The dangers are known in principle: regression discontinuity in time lacks the cross-sectional no-manipulation logic of true RDDs and is biased by ignored time-series structure [8], and a regression kink [4, 2] estimated on a calendar-time running variable is, mechanically, a before/after slope contrast. The open question is empirical: does a modern validation battery — placebo covariates and cutoffs [6], density tests, bandwidth sensitivity, scan-based localization [9] — actually catch these failure modes on real financial series? We run `natex` [11] over three declared cutoffs and report every verdict, including the refusals. The answer is asymmetric: every "credible" verdict fails inspection, while the refusals, failures, and nulls are the informative outputs.

2 Data

Weekly adjusted-close prices for NVDA, AMD, ASML, TSM, and the SOXX semiconductor ETF, 784 weeks from July 2011 to July 2026 (Yahoo Finance weeklies; not committed). The analysis variable is the log adjusted close, the running variable calendar time in years; the pooled five-ticker panel gives the kink family $n_{\text{used}} = 1960$. Descriptive contrasts use the NVDA-minus-SOXX relative log price, netting out the sector factor. The three declared cutoffs are $t = 2022.7644$ (BIS round one, 2022-10-07), $t = 2023.7918$ (BIS round two, 2023-10-17), and $t = 2025.0712$ (the DeepSeek crash week beginning 2025-01-27). One confound matters throughout: ChatGPT's release (2022-11-30) lands eight weeks *after* the first BIS cutoff, inside every bandwidth the 2022 analyses select.

3 Design and Methods

Three `natex` survey runs (v0.2.0, seed 0) supply the estimates: a pooled five-ticker run declaring the DeepSeek cutoff, a pooled run declaring the BIS-2022 cutoff, and an NVDA-only run declaring the BIS-2023 cutoff. Each survey attempts seven design families on the same panel: a declared regression kink in time (local-linear RKD [4] in the difference-in-kinks tradition of [2], cross-validated bandwidth, Holm-corrected placebo battery in the spirit of [6]), a discovered RDD (candidate role assignment over {treatment, forcing, outcome} columns, scan localization [9], McCrary-style density test, covariate placebos), a sector-by-time DiD scan (SuDDDS, permutation-calibrated), synthetic control, discovered-experiments estimation (dee), IV, and bunching.

Honest-inference notes, stated as run. These caveats are part of the record, not afterthoughts. (i) `natex`'s own kink report carries the warning that "a kink in a calendar-time running variable is a before/after slope contrast" — the causal reading is on the analyst, and this paper's point is what happens when it is taken anyway. (ii) The DeepSeek-week level break below is descriptive: no placebo distribution was run, so it carries no standard error. (iii) The DeepSeek-run DiD *failed* rather than returning a number: the intake ignored the declared numeric time column and picked the string `date` column (error of record: "`time column must be numeric: date`"). (iv) The NVDA-only synthetic control *failed* by construction — "`treated unit has no finite outcome before t0`" — a single-ticker panel has no donors. (v) The dee family returned nulls with zero usable discovered experiments (fewer than its minimum of three); IV and bunching declined as `needs_input` throughout; and where a placebo battery was vacuous we say so and treat the estimate as unvalidated.

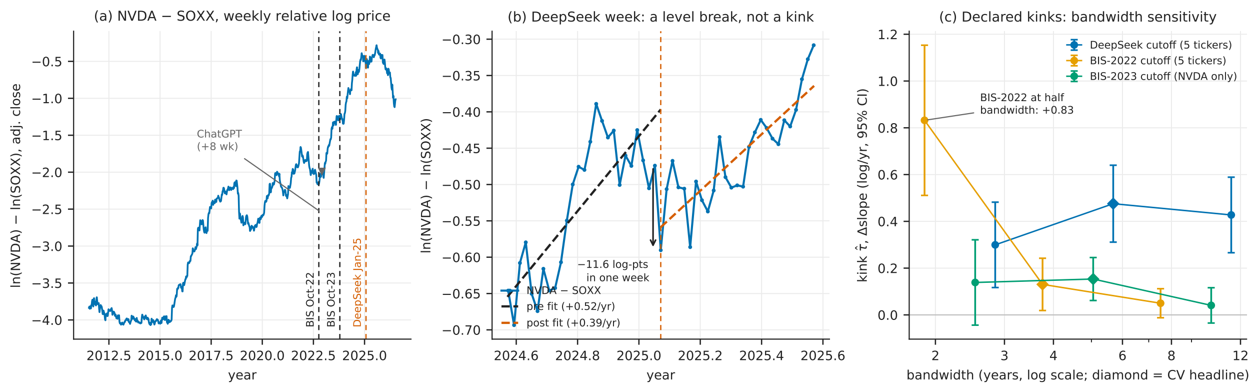

Figure 1: (a) Weekly NVDA-minus-SOXX relative log (adjusted-close) price with the three declared cutoffs; ChatGPT lands eight weeks after the BIS-2022 cutoff, and the 2023–2025 AI rally dominates the series. (b) The DeepSeek week: an 11.6 log-point one-week relative level break with locally unchanged ±26-week slope — a level break, not a kink. (c) Declared-kink estimates (95% CIs) at half, headline (diamond, cross-validated), and double bandwidth: BIS-2022 collapses monotonically, BIS-2023 fades to $t \approx 1.06$, and the DeepSeek "kink" is positive at a crash — the rally on both sides.

4 Results

The real event: a relative level break at DeepSeek. Recomputed exactly from the weekly closes, the week beginning 2025-01-27 took NVDA down -0.17211 in log terms (-17.2 log-points; -15.8% simple return) against -0.05577 for SOXX (-5.6 log-points; -5.4%): an -11.63 log-point one-week fall in NVDA *relative to its own sector*. The local relative trend barely moved: ±26-week OLS slopes of the relative log price are $+0.52$ log/yr before and $+0.39$ log/yr after (Figure 1b). The DeepSeek shock is a *level* break with an approximately unchanged local slope — the signature of a one-off repricing, not a growth-regime change. This estimate is descriptive (no placebo distribution; no standard error).

Declared kinks: "credible" three for three, wrong three for three. Table 1, panel A. At the DeepSeek cutoff the pooled kink is $\hat{\tau} = +0.4756$ log/yr (se 0.0839, 95% CI $[0.3111, 0.6400]$, Holm $p = 1.4 \times 10^{-8}$, cross-validated bandwidth 6.05 yr, $n_{\text{used}} = 1960$). The sign alone refutes the causal reading: DeepSeek was a crash, yet the "effect" is a large *positive* slope change — a six-year bandwidth spans the pre-2023 flat regime on the left and the AI rally on the right, so the kink dates the rally, not the event. At the BIS-2022 cutoff the pooled kink ($+0.1305$, Holm $p = 0.022$) is monotonically bandwidth-sensitive — $+0.832$ (se 0.164) at 1.88 yr, $+0.131$ (se 0.057) at 3.75 yr, $+0.050$ (se 0.032) at 7.50 yr — and ChatGPT's release sits eight weeks post-cutoff inside every one of those windows; nothing localizes the bend to the export controls. The NVDA-only BIS-2023 kink ($+0.1534$, Holm $p = 0.0011$) fades to $+0.0407$ (se 0.0385, $t \approx 1.06$) at double bandwidth. The ±26-week local slopes tell the same story in descriptive form: the relative NVDA-minus-SOXX slope change is $+1.46$ log/yr at BIS-2022 (from -0.46 to $+1.00$), $+0.41$ at BIS-2023 (from $+0.61$ to $+1.02$), and -0.13 ($\approx$ flat, $+0.52 \rightarrow +0.39$) at DeepSeek: the two "export-control kinks" are the AI rally starting, and the one week with a genuine event shows no kink at all.

A "credible" RDD that is a role-assignment artifact. In the BIS-2022 run the discovered-RDD family also returned "credible" (scan $p = 0.010$, density $p = 0.180$). Inspection of the

Table 1: All verdicts across the three survey runs (`natex` v0.2.0, seed 0).

Run (cutoff)	family	$\hat{\tau}$	se	inference	disposition
Panel A: declared kinks in calendar time, log price (log/yr) — all "credible", none survives					
BIS-2022 (pooled)	kink	+0.1305	0.0570	Holm p=0.022; bw 3.75	+0.83 → +0.13 → +0.05 across bw
BIS-2023 (NVDA)	kink	+0.1534	0.0469	Holm p=0.0011; bw 4.77	fades: t≈1.06 at 2×bw
DeepSeek (pooled)	kink	+0.4756	0.0839	Holm p=1.4×10^{-8}; bw 6.05	positive "kink" at a crash
Panel B: every other family disposition					
BIS-2022	rdd	1.581	0.886	scan p=0.010; CI spans 0	role-assignment artifact
BIS-2022	did	—	—	scan p=0.70; best t_0=2025.95	**informative null**
BIS-2022	dee	—	—	$0 < 3$ usable experiments	null
BIS-2023	rdd	—	—	scan p=0.040; placebos vacuous	unvalidated
BIS-2023	did	—	—	scan p=0.76; best t_0=2017.16	null
BIS-2023	sc	—	—	no donors (single ticker)	failed, loudly
BIS-2023	dee	—	—	$0 < 3$ usable experiments	null
DeepSeek	rdd	4.07	2.64	scan p=1.00; CI spans 0	null (placebos worked)
DeepSeek	did	—	—	intake chose string `date` col.	failed, loudly
DeepSeek	iv/sc/bunching	—	—	`needs_input`	refused

Notes: Kink rows: local-linear RKD in calendar time, log adjusted close, cross-validated bandwidth in years, $n_{\text{used}} = 1960$; BIS-2022 disposition: estimate at half/headline/double bandwidth. 95% CIs: BIS-2022 kink $[0.0188, 0.2423]$; BIS-2023 kink $[0.0616, 0.2453]$; DeepSeek kink $[0.3111, 0.6400]$. RDD rows: fuzzy 2SLS at the discovered candidate; CIs BIS-2022 $[-0.156, 3.317]$ and DeepSeek $[-1.11, 9.25]$; density p: BIS-2022 0.180, BIS-2023 0.312, DeepSeek 0.752 with all 8 covariate placebos at Holm $p = 1.0$. DiD rows: SuDDDS permutation scan; point estimates at the (non-rejected) BIS-2022 best t_0: $\hat{\tau}_{\text{DD}} = -2.061$ (se 0.018), $\hat{\tau}_{\text{synth}} = -0.427$ (se 0.025).

selected candidate shows why this is empty: the automated role search assigned treatment = `post_2022_controls`, forcing = `log_adjclose`, and outcome = t — it mechanically rediscovered that prices above a level tend to lie after 2022. The fuzzy estimate $\hat{\tau}_{\text{2SLS}} = 1.581$ (se 0.886, 95% CI $[-0.156, 3.317]$) crosses zero and has no economic content. In the BIS-2023 NVDA-only run the RDD scan rejected at $p = 0.040$ but the covariate-placebo battery was *vacuous* (no testable covariate in the single-ticker numeric panel; density $p = 0.312$): unvalidated, and not reported as a finding.

The informative null: no DiD break at the export controls. The SuDDDS scan over tickers and break dates in the BIS-2022 run finds nothing at the controls: scan $p = 0.70$ (LLR 2.548), with the best-fitting break at $t_0 = 2025.95$ (window 3.76 yr) — not 2022.76 (point estimates at that spurious t_0 in the table notes; scan-level p-values are not computed for a non-rejected scan). The NVDA-only run's DiD is likewise null ($p = 0.76$, best $t_0 = 2017.16$). No ticker-relative break is detectable at either export-control date. This is the substantive result of the paper, and it is a null: whatever costs the controls imposed on these five firms were priced jointly with — and are not separable in these data from — the contemporaneous AI-demand shock lifting the same stocks.

The validation layer's scorecard. Table 1, panel B. Credit: at the DeepSeek cutoff the discovered-RDD family correctly returned *null* (scan $p = 1.00$, density $p = 0.752$, all eight covariate placebos at Holm $p = 1.0$; the fuzzy $\hat{\tau}_{\text{2SLS}} = 4.07$, se 2.64, was discarded), and the pipeline refused rather than fabricated where inputs were missing (`needs_input`; dee nulls; two loud failures quoted in Section 3). Debit: every "credible" verdict on these series — three kinks and one RDD — fails inspection, and cross-validated bandwidth selection actively favors the artifact, choosing windows wide enough to straddle regimes.

5 Caveats and Conclusion

Five caveats bound the claims. First, a unit correction to the working notes of record: the DeepSeek-week figures −17.2% / −5.6% are one-week *log* returns (log-points); the simple returns are −15.8% and −5.4%. Second, the level break and the ±26-week slopes are descriptive — a check on magnitudes against the formal event-study machinery [10, 7], not a substitute for it. Third, all causal-design caveats for time-as-running-variable apply with full force [8], and `natex`'s own recorded warning — a kink in a calendar-time running variable is a before/after slope contrast — should be read as governing every panel-A row. Fourth, the event window is confounded by construction: ChatGPT lands eight weeks after the BIS-2022 cutoff, and the 2023–2025 AI rally overlaps everything after it; the DiD null is therefore a statement about non-separability in these data, not evidence that the controls were costless. Fifth, five tickers are a narrow panel; the SuDDDS permutation floor and the vacuous placebo batteries both reflect that thinness.

The conclusion is a methods verdict with one substantive corollary. Methods: on trending financial series, an automated validation layer is asymmetric — its refusals, failures, and nulls were all correct and informative, while every one of its "credible" verdicts was an artifact of trend regimes, role search, or vacuous batteries; cross-validated bandwidths made the kink artifacts *more* confident, not less; "credible" on a calendar-time design is a prompt for the checks in Figure 1, not a result. Substance: the one clean event in these data is DeepSeek — an 11.6 log-point one-week repricing of NVDA against its own sector with no local trend change — while the export-control rounds left no detectable ticker-relative break at all, consistent with their costs having been priced against the contemporaneous AI-demand shock rather than as standalone news.

Reproducibility. All estimates come from three `natex` v0.2.0 survey runs at seed 0 (2026-07-18; the kink estimator is deterministic) on frozen weekly price extracts not committed to the repository. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the level break, the local slopes, and the three headline kink estimates against the numbers of record before drawing.

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An Honest Miss: Chinchilla as an Adoption Ramp in a Discontinuity Scan of Tokens per Parameter

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

A LoRD3 continuous-treatment discontinuity scan of $\log_{10}$ training tokens per parameter over calendar time, run blind on 549 language models (2019–2024) from Epoch AI's notable-models panel, produces a single significant top discovery: a break centered on 2022-09-22 (LLR 8.687, scan $p \leq 0.01$, robust to background degree and to neighborhood size $k \geq 50$). The known truth — the Chinchilla compute-optimal scaling result, posted 2022-03-29, after which the global median tokens-per-parameter ratio rose from 1.7 to 29.2 — is *not* localized: the best candidate near the announcement ranks 158 of 547 (LLR 1.39). The discovered break is six months late and points the wrong way: a downward local contrast of 1.63 $\log_{10}$ units ($\times 43$) at the Sep/Oct 2022 boundary, where the downstream edge of the Chinchilla adoption wave collides with a measurement artifact — fine-tuned releases whose recorded token counts cover only the fine-tuning run over full base parameters. Nulls are reported as nulls: the placebo test is vacuous (no non-forcing covariates), the density test fails ($p = 0.0041$) but is uninformative for a calendar-time forcing, post-selection 2SLS is weak-instrument, and an independent kink audit rejects a slope-change reading. The verdict is an artifact — an honest miss, correctly diagnosed — with two transportable lessons: diffusion-style treatments need a ramp/slope statistic, not only a level-break statistic, and fine-tune rows must be filtered from tokens-per-parameter panels before any scan.

1 Introduction

On 29 March 2022, Hoffmann et al. posted the "Chinchilla" result: for compute-optimal training, model size and training tokens should scale in equal proportion, implying that the then-standard practice — descended from the earlier scaling-law fits of [7] — left frontier models substantially undertrained on data [6]. Release practice responded within months, and by 2023 the industry had moved past compute-optimality into deliberate over-training of small models for inference economy, exemplified by LLaMA's trillion-token training runs [12] and rationalized by inference-adjusted scaling laws [10]. Chinchilla itself has since been re-examined: [2] replicate its parametric fit and find inconsistencies in the reported estimates, while the broader quantitative history of training-run inputs is documented by [11] and maintained in Epoch AI's notable-models dataset [4].

This gives a rare, clean testbed for *automated natural-experiment discovery*: a panel where the event, its date, its sign, and its approximate magnitude are known ex ante. This paper runs the

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`natex` toolkit's LoRD3 scan [9, 5] on tokens per parameter over calendar time, blind to the event date, and asks: does the scanner find Chinchilla? The answer is no — and the anatomy of the miss is the contribution. Event-dated diffusion processes enter publication data as ramps, and a local level-break statistic rewards sharp edges, not ramps; dating the cause is an interrupted-time-series problem [1], not a discontinuity-scan problem.

2 Data

The panel is a frozen extract of Epoch AI's notable-models database [4]: 549 language models released 2019-01 through 2024-12 with non-missing parameter and training-token counts. Columns: release date (the forcing variable, as days since 2019-01-01), $\log_{10}$ tokens per parameter (the treatment), $\log_{10}$ parameters (the outcome), $\log_{10}$ training compute (21% missing) and an open-weights flag (3.6% missing), both excluded from the scan to avoid listwise deletion. The known truth: Chinchilla was posted 2022-03-29 (day ≈ 1183); across that date the global median tokens per parameter moves from 1.69 to 29.17 ($n = 185$ pre, 364 post), a median-based shift of $+1.24$ $\log_{10}$ units ($+1.12$ on group means, $2.2 \to 29$; Mann–Whitney $p = 1.6 \times 10^{-7}$ on an earlier extract, medians $2.2 \to 29.1$, $n = 133/307$). At monthly resolution the shift is a ramp, not a step: window medians run 1.7 (pre-April 2022) $\to$ 5.7 (mid-April to mid-May) $\to \approx 28$ (mid-May to mid-July 2022).

3 Design and Methods

LoRD3 continuous-treatment scan. Following [5] as implemented in `natex` v0.2.0 [9], the scan fits a background model of the treatment on the forcing variable (OLS, degree 1; degree 2 as robustness), then scores every k-nearest-neighbor neighborhood ($k = 50$) of every data point for the best split of its members into two half-spaces, using a normal likelihood ratio on background residuals with locally estimated variances. The top discovery is the neighborhood-split with the maximum LLR; its p-value comes from a fitted-null Monte Carlo with $q = 99$ replicas and a +1-rank rule, so the minimum attainable p is 0.010 — reported throughout as $p \leq 0.01$, never as an exact value. The specification is treatment = $\log_{10}$ tokens per parameter, forcing = days since 2019-01-01 (forcing influence 1.0), outcome = $\log_{10}$ parameters, no covariates; pipeline `natex study` $\to$ `natex discover --k 50 --q 99 --seed 0`, with localization detail and robustness via the Python API, seed 0 throughout.

Validation battery, stated as run. The battery's nulls and refusals are reported, not patched over. (i) Randomization: $p = 0.010$ at both background degrees — the $q = 99$ floor. (ii) Placebo covariates: `placebo_passed = True` but *vacuous* — `placebo_holm = {}`, because the design has no non-forcing covariates to test. (iii) Density: $p = 0.0041$, a *fail*; this is expected and uninformative here, because the forcing is calendar time with strongly nonstationary publication intensity, and McCrary-style manipulation logic [8] does not apply to a time forcing — the test is retained as a falsification check only. (iv) Post-selection effects on $\log_{10}$ parameters (2SLS and Wald at the discovered cutoff) are *advisory*: estimated on the discovery sample with no honest split and instrumented by a partly artifactual break, they must not be read as causal scaling-law parameters. (v) An independent kink-design audit of the same panel (a difference-in-slopes design [3]) cross-checks any slope-change reading.

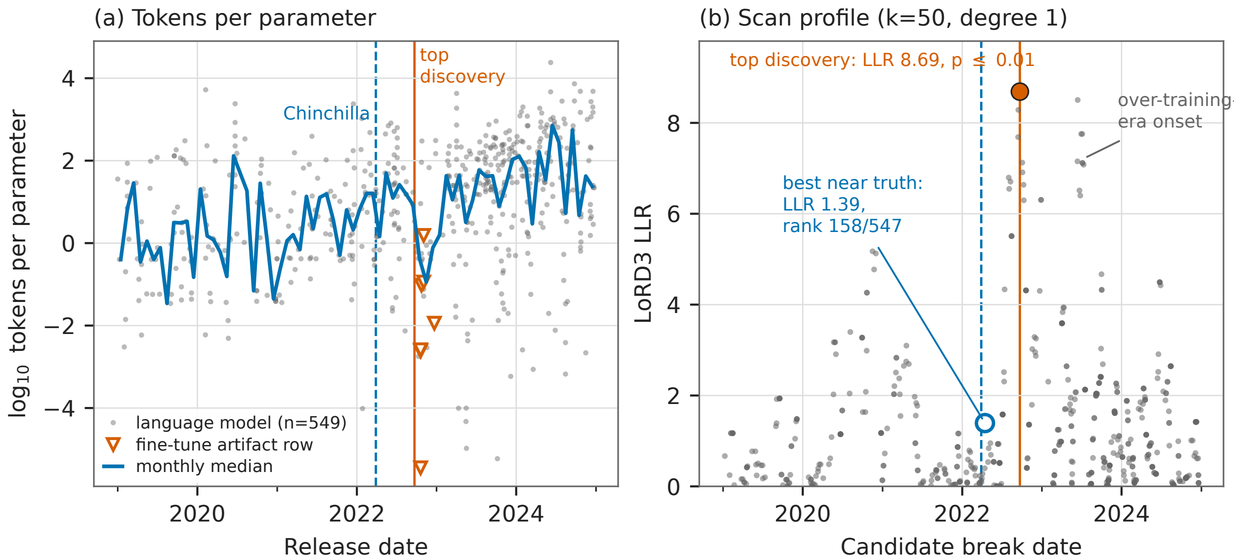

Figure 1: (a) $\log_{10}$ tokens per parameter for all 549 models, with the monthly median (line), the Chinchilla posting date (dashed), the scan's top discovery (solid), and the seven verified fine-tune token-counting artifact rows (open triangles; Table 1, panel C). (b) LoRD3 scan profile: each candidate neighborhood's LLR at its center date ($k = 50$, degree 1, $n = 547$ candidates). The top discovery (filled marker, 2022-09-22, LLR 8.69, $p \le 0.01$) sits at the Sep/Oct 2022 artifact edge; the best candidate within ±45 days of the truth (open marker, 2022-04-15) scores LLR 1.39, rank 158/547. The 2023-06/07 cluster is the onset of the deliberately over-trained era.

4 Results

The discovery. The scan's top discovery (Table 1, panel A; Figure 1b) is centered on day 1360 = 2022-09-22 (split normal −1 along the forcing axis): LLR = 8.687 (8.687416 unrounded in the run report), scan $p = 0.010$ — the $q = 99$ Monte Carlo floor, i.e. $p \le 0.01$ — over 547 candidate neighborhoods from $n = 549$ rows. A degree-2 background reproduces the same top center with LLR 8.836 (randomization $p = 0.01$; null max LLRs 1.8–7.6 against observed 8.69/8.84), and $k = 60$ moves it only to 2022-08-15; but $k = 30/40$ latch onto a 2020-10 composition boundary instead (Table 1, panel B), so localization is stable only for $k \ge 50$.

The known truth is not localized. Within ±45 days of 2022-03-29 the best candidate neighborhood is centered 2022-04-15 with LLR 1.39 — rank 158 of 547. The reason is visible in Figure 1a: at daily resolution the Chinchilla shift enters the published record as a 2–3 month adoption ramp, and a local level-break statistic rewards sharp edges, not ramps.

What the discovered break actually is. The top neighborhood splits exactly at the Sep/Oct 2022 boundary into group A (2022-06-21 to 2022-09-22, $n = 25$, median tokens/param 13.6 — the post-Chinchilla adoption wave) versus group B (2022-10-03 to 2022-12-22, $n = 25$, median 0.4): a *downward* local group-mean contrast of 1.63 $\log_{10}$ units (×43), against the known *upward* global shift of +1.12 $\log_{10}$ (+1.24 median-based). Local-linear first-stage fits jump −1.7 to −2.0 $\log_{10}$ at bandwidths of 120–240 days. Group B is dominated by fine-tuned and continued-pretraining releases whose Epoch token field records only fine-tuning tokens over full base parameters (Table 1,

Table 1: Chinchilla scan: numbers of record (`natex` v0.2.0, seed 0).

Quantity	Value	Quantity	Value
Panel A: top discovery ($k = 50$, $q = 99$, degree 1; $n = 549$, 547 candidates)			
center	2022-09-22 (day 1360)	LLR	8.687
scan p	0.010 = floor; $p \leq 0.01$	local contrast	-1.63 $\log_{10}$ ($\times 43$, down)
degree-2 center	2022-09-22, LLR 8.836	first stage	-1.7 to -2.0 $\log_{10}$ (bw 120–240d)
Panel B: k-sensitivity and truth localization			
$k = 30$	2020-09-29 (LLR 7.2)	$k = 40$	2020-10-21 (LLR 9.0)
$k = 50$	2022-09-22 (LLR 8.7)	$k = 60$	2022-08-15 (LLR 9.8)
truth ±45d best	2022-04-15, LLR 1.39	truth rank	158 / 547
global shift (true)	$+1.12$ $\log_{10}$ ($2.2 \to 29$, up)	medians	$1.69 \to 29.17$ ($n = 185/364$)
Panel C: verified fine-tune artifact rows (tokens/param; group B median 0.4 *vs. A* 13.6*)*			
Flan-PaLM 540B	0.0026	U-PaLM	0.0024
LMSI-PaLM	3.6×10^{-6}	OPT-IML 175B	0.011
Tk-Instruct	0.095	BLOOMZ-176B	0.114
mT0-13B	1.54		
Panel D: diagnostics and post-selection effects (advisory)			
randomization	$p = 0.010$ (deg. 1 and 2)	placebo	passed, *vacuous* (no covariates)
density	$p = 0.0041$, fails†	kink audit	rejected; $t \in [-0.91, +1.48]$
2SLS $\hat{\tau}$	1.017 (se 0.900), weak	CI	$[-0.747, 2.781]$, 1st-st. $t = 1.89$
Wald $\hat{\tau}$	0.557 (se 0.257)	CI	$[0.053, 1.062]$, 1st-st. $t = 4.01$

Notes: †expected and uninformative for a calendar-time forcing with nonstationary publication intensity (McCrary manipulation logic does not apply). Scan p-values from a $q = 99$ fitted-null Monte Carlo with a $+1$-rank rule; 0.010 is the minimum attainable. Effects are estimated on the discovery sample with no honest split. Artifact rows verified in the source panel: the token field records fine-tuning tokens only, over full base parameters.

panel C) plus small academic models. The top discovery is therefore the adoption wave's downstream edge colliding with a token-accounting artifact, ≈6 months after the event and with the opposite sign. The secondary discovered cluster (2023-06-13 to 2023-07-06, LLR 6.7–8.5) is the onset of the deliberately over-trained era [12, 10]; the December-2022 monthly median is already 148 tokens/param, and the median of the top-25 centers is day 1371 = 2022-10-03.

Effects and cross-checks, all null or advisory. Post-selection effects on $\log_{10}$ parameters at the discovered cutoff (Table 1, panel D): the 2SLS estimate is weak-instrument (first-stage $t = 1.89$; CI $[-0.747, 2.781]$); the Wald estimate is 0.557 (CI $[0.053, 1.062]$, first-stage $t = 4.01$). Both are discovery-sample estimates with no honest split, instrumented by a partly artifactual break — diagnostics, not causal scaling-law parameters. The independent kink audit *rejects* a Chinchilla regime change read as a slope change: a sign-unstable null with t from -0.91 to $+1.48$ across bandwidths.

5 Caveats and Conclusion

Five caveats bound the claims. First, the value of this exercise is purely methodological and diagnostic: no causal adoption claim is licensed; the scan is a regime-edge detector, and dating the cause of a diffusion process needs an interrupted-time-series or difference-in-differences framing [1] plus publication-lag domain knowledge. Second, the scan $p = 0.010$ is the minimum attainable with $q = 99$ replicas and must be reported as $p \leq 0.01$, not as an exact value. Third, the density test fails ($p = 0.0041$) but is uninformative for a calendar-time forcing with nonstationary publication intensity, and the placebo pass is vacuous (`placebo_holm = {}`; no non-forcing covariates to test). Fourth, the post-selection 2SLS/Wald effects are estimated on the discovery sample with no honest split and are instrumented by a partly artifactual break; they must not be read as causal scaling-law parameters. Fifth, "robust to k" holds only for $k \geq 50$: at $k = 30/40$ the scan latches onto a 2020-10 composition boundary (tiny academic LMs against BERT-era corpus models), so the robustness claim must be stated conditionally.

The verdict is an artifact — an honest miss, correctly diagnosed: the scan found real, significant structure whose two regimes bracket the truth, but it dated the adoption wave's sharp downstream *edge*, not the event, which entered the record as a ramp. Two lessons transport beyond this dataset. For method builders: scan batteries aimed at event-dated diffusion treatments need a ramp/slope statistic — a kink-style scan in the sense of [3] — alongside the level-break statistic, or announcements will systematically lose to their own adoption edges. For users of the Epoch panel [4]: fine-tuned and continued-pretraining rows manufacture a spurious low-tokens-per-parameter cluster in late 2022 and must be filtered before tokens-per-parameter data are used for scaling-law or event-study claims [6, 2].

Reproducibility. All estimates: `natex` v0.2.0 [9], seed 0, on a frozen extract of the Epoch panel [4] (not committed). Figure 1 regenerates deterministically from `figures/make_fig.py` (committed), which asserts the headline numbers of record before drawing.

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Recovering Proposition 99 Blind: Validating the `natex` Pipeline Against the Abadie–Diamond–Hainmueller Benchmark

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July 2026
run of record re-verified 2026-07-18, `natex` v0.2.0, seeds 0/1/2 (donor path rng-free)

Abstract

Run blind on the Abadie–Diamond–Hainmueller (ADH) California smoking panel (39 states × 31 years), the `natex` pipeline reproduces the canonical Proposition 99 benchmark end to end. All three SuDDDS scan methods (greedy, wcc, single_delta; normal model) recover (California, $t_0 =$ 1989) exactly — precision = recall = 1.0 over the 31 California records, LLR 13.82; the Bernoulli-model scan is also exact (LLR 13.86), a y-blindness test proves the scan bitwise independent of the outcome, and a Bernoulli($\hat{p}$)-null Monte Carlo ($Q = 99$, seed 1) gives $p = 0.010$ (the dependence-preserving AR(1) null is structurally conservative on a policy dummy, pinning at $p = 1.0$; documented, not claimed as evidence). Deterministic full-pool simplex weights put summed weight 0.955 on ADH's five published donors, and the synthetic-control ATT over 1989–2000 is −19.514 packs per capita, matching ADH's canonical average gap of roughly −19. Nulls are reported as nulls: the exact in-space RMSPE-ratio placebo ranks California 3/39 ($p = 0.077$; ADH's 1/39 requires their poor-pre-fit exclusion, kept opt-in here), and under the corrected two-sided studentized placebo statistic the source thesis's claim that all three effect estimates are significant at 5% does *not* survive (ranks 13/39, 7/39, 9/39). By construction this exercise has zero novelty; its value is credibility transfer — external-validity anchoring for the `natex` studies on novel data.

1 Introduction

California's Proposition 99, passed in November 1988 and effective January 1989, raised the state cigarette excise tax by 25 cents per pack and funded a large anti-smoking campaign. Taxation and media spending each measurably reduced cigarette sales [9], and the program is associated with accelerated declines in consumption and heart-disease mortality [6]. It is also the canonical testbed of the synthetic control method: building on the comparative case-study estimator of [4], Abadie, Diamond, and Hainmueller [2] constructed a "synthetic California" from a weighted combination of donor states and estimated an average gap of roughly −19 packs per capita over 1989–2000. That example anchors a large methodological literature [3, 1, 5] and is arguably the most-replicated benchmark in comparative case studies.

This paper uses it the other way around: not to learn about Proposition 99, but to validate a pipeline. `natex` [11] is an automated natural-experiment discovery-and-estimation toolkit in the LoRD3 lineage [7, 10]; its difference-in-differences arm implements SuDDDS (Subset Discovery of

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Difference-in-Differences) from chapter 6 of [8], whose printed Table 6.1 evaluates the method on this same smoking panel. We run the full pipeline *blind* — the scan is never told the treated unit or the intervention year, and provably never reads the outcome — and ask whether it rediscovers (California, 1989) and reproduces the ADH estimate. It does — exactly, deterministically, and with every deviation from the thesis's printed values investigated and reconciled. Because the exercise replicates a known benchmark, we write it up short, structurally as validation: its only role is to transfer credibility to the `natex` studies on novel data.

2 Data

The data are the ADH California tobacco panel [2] as shipped in the `natex` dataset registry (dataset `prop99`; 1,209 rows): annual state-level per-capita cigarette sales (packs, from tax revenues) for 39 states, 1970–2000 — California plus ADH's 38 donor states — with auxiliary covariates (log income, beer consumption, share aged 15–24, retail price). The panel is balanced (39×31, no missing outcome cells). Treatment is the policy dummy: California from 1989 onward. (The source thesis [8] reports "30 US states" and does not document its panel; ours is ADH's 39-state pool.)

3 Design and Methods

Blind discovery (SuDDDS). SuDDDS [8], as implemented and corrected in `natex` [11], scans panel records (x_i, t_i, θ_i) — covariates, time, treatment — for a subset of units and an intervention time T_0 at which θ jumps, scoring candidates by a log-likelihood ratio and never reading the outcome y. Protocol, fixed before calibration: windows $(5, 8, 10)$, 4 time bins, 8 restarts, seed 0, and background degree 0 (unit effects only). Degree 0 is required, not tuning: θ is a pure policy dummy whose only time variation is the candidate discontinuity, so any global time polynomial can only fit the jump itself; with degree 1 the leaked slope manufactures a spurious whole-panel optimum (the thesis never reports its background — an unreported-hyperparameter risk). All three scan methods (greedy, wcc, single_delta) are run under the normal model, plus a wcc scan under the Bernoulli model, the corrected default for a binary treatment. A y-blindness test re-runs the scan with the outcome deleted and requires bitwise-identical discoveries.

Calibration. The top discovery's LLR is calibrated by a fitted-null Monte Carlo with $Q = 99$ replicas and a +1-rank rule (minimum attainable $p = 0.010$), drawing nulls from Bernoulli($\hat{p}$) — the correct null family for a binary θ. As a diagnostic we also run the dependence-preserving AR(1)-unit null, which is *structurally conservative* on a deterministic policy dummy: 38 of 39 units have identically-zero residuals, so the pooled AR(1) fit absorbs California's step as noise and hands it to every replica; its p pins at 1.0, documented as a regression, never claimed as evidence.

Estimation conditioned on the discovery. Three effect estimators are run on the recovered discovery over the full post period 1989–2000: two-way difference-in-differences (DD), synthetic control (simplex-weighted donor fit on pre-period outcomes, as in [2]), and GESS (greedy control-group expansion, thesis ch. 6). Separately, the `natex` synthetic-control donor path is run on the raw (state, year, cigsale) matrix with the full donor pool and with an $n = 8$ pre-fit-ranked variant; this path is deterministic — no rng anywhere — and bitwise-identical across runs.

Honest inference. Significance is assessed by exact, enumerated placebo tests, with the caveats stated plainly. (i) The in-space RMSPE-ratio placebo [2, 3] refits every one of the 38 control states as

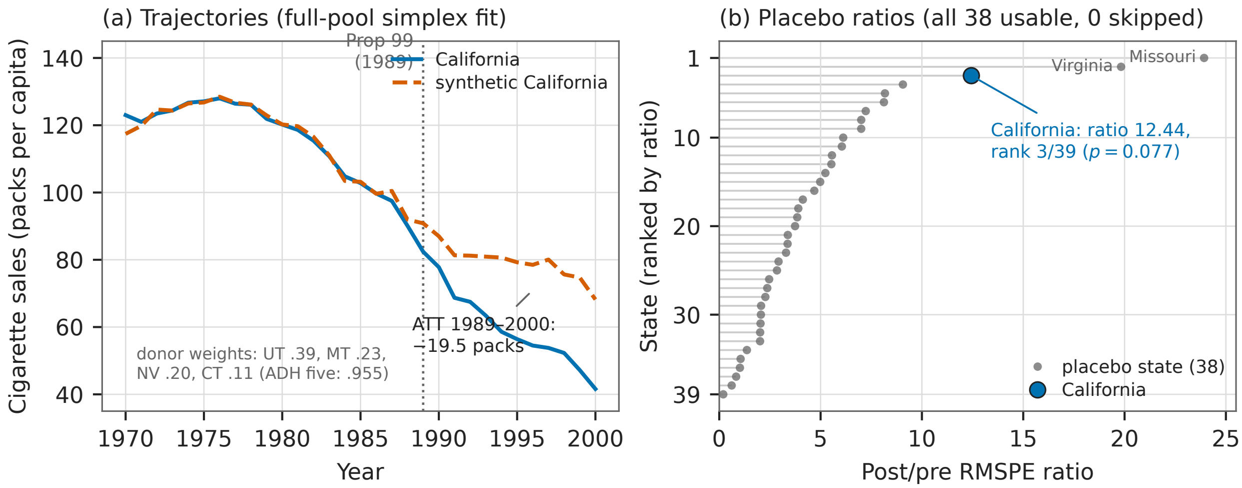

Figure 1: (a) California vs. synthetic California (deterministic full-pool simplex weights), 1970–2000: pre-RMSPE 1.656, post-period gap averaging −19.514 packs per capita. (b) In-space placebo: post/pre RMSPE ratios for all 39 states (all 38 placebos usable, 0 skipped). California ranks 3/39 ($p = 0.077$); the two out-ranking states, Missouri and Virginia, are poor-pre-fit placebos that ADH's exclusion rule would discard.

a pseudo-treated unit (`exclude_poor_fit = None`: all 38 usable, 0 skipped) and ranks California's post/pre RMSPE ratio among all 39; ADH's $p = 1/39$ requires their poor-pre-fit exclusion, kept opt-in in `natex`. (ii) The corrected effect-level test is a two-sided studentized statistic $|\hat{\tau}/\mathrm{se}|$ with +1-rank p-values over the 38 enumerated placebo states — under it, genuinely non-parallel placebo states (flat post-shifted gaps with tiny standard errors) can out-t California's trending gap. (iii) The validation battery's degenerate cases report "cannot test" (NaN, failed) rather than fabricating passes: composition reduces to a one-row table for a single-unit discovery, and anticipation has zero pre-period treatment variance.

4 Results

Blind discovery is exact. All three SuDDDS scan methods under the normal model return (California, $t_0 = 1989.0$) as the top discovery, with the discovered subset exactly the 31 California records: precision = recall = 1.0, LLR 13.82. The Bernoulli-model scan is also exact, LLR 13.86. The y-blindness test passes bitwise (Table 1, panel A).

The discovery is significant under the correct null. Against Bernoulli($\hat{p}$) nulls ($Q = 99$, seed 1) the observed max LLR 13.86 exceeds the null max 7.95 (null q90 4.67): $p = 0.010$, the +1-rank floor. The AR(1)-unit diagnostic pins at $p = 1.0$, as expected for a deterministic policy dummy (Section 3); recorded as a structural limitation of dependence-preserving nulls, not as evidence against the discovery (Table 1, panel B).

Effects match ADH; the printed thesis numbers reconcile. On the full 1989–2000 post period, DD gives $\hat{\tau} = -27.349$ packs (pre-MSE 51.23), synthetic control gives $\hat{\tau} = -19.514$ (pre-MSE 2.74), and GESS gives $\hat{\tau} = -26.653$ (control: Montana, pre-MSE 18.35). Signs and the pre-fit ordering match the thesis's Table 6.1 (−10.94 / −8.96 / −6.67): synthetic control's pre-MSE is far below DD's, which is exactly ADH's argument for it. The magnitudes differ by a factor of ∼2–2.5 because the thesis's printed values correspond to an unreported effective post window of about five years, while the gap accumulates over time; restricting `natex` to 1989–1993 gives DD −18.8 and synthetic −12.3, bracketing the printed values (Table 1, panel C).

Donor selection recovers ADH's pool. The deterministic full-pool simplex fit puts nonzero weight on six donors, with summed weight 0.955 on ADH's five published donors, and the four heavyweights are exactly ADH's own {Utah, Montana, Nevada, Connecticut} (weights in Table 1, panel D). $\mathrm{ATT}_{\mathrm{post}} = -19.514$ packs per capita, with pre-RMSPE 1.656 against post-RMSPE > 10 (the ADH-style pre/post contrast); the 8-donor variant gives ATT −22.648, pre-RMSPE 3.671 (Figure 1a).

Honest inference: what survives and what does not. The exact in-space RMSPE-ratio placebo ranks California 3 of 39 in both variants (treated ratio 12.440 full pool, 6.570 top-8): $p = 3/39 = 0.077$ (Figure 1b). ADH's 1/39 arises only under their poor-pre-fit exclusion, which `natex` keeps opt-in. The corrected two-sided studentized τ placebos over the 38 enumerated states give DD rank 13/39 ($p = 0.333$), synthetic control 7/39 ($p \leq 8/39 = 0.205$), and GESS 9/39 ($p = 0.231$): the thesis's claim that all three are significant at 5% does *not* survive the corrected statistic, and we report that as the finding it is (Table 1, panel E).

Run of record and reproducibility. The entire benchmark is executable: running `uv run pytest -m backtest` with `NATEX_DATA` set to the data root, on the modules `test_prop99.py` and `test_prop99_donors.py` under `tests/backtests/`, gives 16 passed, 1 xfailed in 79.74 s (re-verified 2026-07-18). The single xfail is a non-blocking instrument-selection stretch goal on a *different* dataset, pinned as a documented finding; all 15 Proposition 99 tests pass. All estimates: `natex` v0.2.0 [11] on the registry copy of the ADH panel (not committed). Figure 1 regenerates deterministically from `figures/make_fig.py` (committed), which asserts the headline numbers of record before drawing.

5 Caveats and Conclusion

Three caveats bound the claims. First, this exercise has zero novelty by construction: Proposition 99 on the ADH panel is the most-replicated synthetic-control benchmark in economics [2, 1], and a pipeline that failed it would be disqualified rather than informative; its role is external-validity anchoring for the `natex` studies on novel data, and it should be read as methods validation, not as a finding about tobacco policy. Second, effect magnitudes here are ∼2–2.5× the source thesis's printed Table 6.1 values because `natex` estimates on the full 1989–2000 post period while the thesis's numbers match an unreported ∼5-year effective window; the thesis also never reports its panel ("30 US states" vs. ADH's 39) or its treatment background. Signs, ordering, and pre-fit ranking agree, and the synthetic-control estimate matches ADH's canonical ≈ -19 — but the reconciliation must be stated, not glossed. Third, the honest-inference results are genuinely weaker than the textbook telling: without ADH's poor-pre-fit exclusion the in-space placebo p is 0.077, not $1/39 = 0.026$, and under the corrected two-sided studentized placebo statistic none of the three effect estimators clears 5% on this panel. These are properties of the inference conventions, faithfully surfaced.

Within those bounds, the conclusion is simple: run blind, `natex` rediscovers (California, 1989) exactly by every scan method, calibrates it significantly under the correct null, selects ADH's donors with 0.955 of the simplex weight, and reproduces the canonical ≈ -19 synthetic-control estimate

Table 1: Proposition 99 backtest: numbers of record (`natex` v0.2.0; seeds 0/1/2, donor path rng-free).

Quantity	Value	Quantity	Value
Panel A: blind discovery (SuDDDS; windows (5,8,10), bins 4, restarts 8, degree 0, seed 0)			
greedy / wcc / single_delta	(California, 1989.0)	precision, recall	1.0, 1.0 (31 CA records)
normal-model LLR	13.82	Bernoulli wcc	exact; LLR 13.86
y-blindness	bitwise identical		
Panel B: discovery calibration ($Q = 99$, +1-rank, seed 1)			
Bernoulli($\hat{p}$) null	$p = 0.010$	observed vs null max	13.86 vs 7.95 (q90 4.67)
AR(1)-unit null	$p = 1.0$, conservative†		
Panel C: effects on the discovery, post 1989–2000 (thesis Table 6.1: $-10.94/-8.96/-6.67$)			
DD $\hat{\tau}$	-27.349 (pre-MSE 51.23)	synthetic $\hat{\tau}$	-19.514 (pre-MSE 2.74)
GESS $\hat{\tau}$	-26.653 (MT, pre-MSE 18.35)	1989–1993 restriction	DD -18.8, synthetic -12.3
Panel D: synthetic-control donor path (deterministic, bitwise-reproducible)			
weights	UT .394, MT .232, NV .205	ADH-five summed weight	0.955
	CT .109, NH .045, CO .015	ATT_{post}	-19.514 packs (ADH ≈ -19)
pre- / post-RMSPE	1.656 / > 10	8-donor variant	ATT -22.648, pre 3.671
Panel E: exact placebo inference (enumerated; no poor-pre-fit exclusion)			
RMSPE ratio (treated)	12.440 full / 6.570 top-8	rank, p	3/39 both; $p = 3/39 = 0.077$
studentized τ: DD	13/39 ($p = 0.333$)	synthetic	7/39 ($p \leq 8/39 = 0.205$)
GESS	9/39 ($p = 0.231$)	thesis "all sig. at 5%"	does *not* survive

Notes: †structural, not evidential: a deterministic policy dummy leaves 38 of 39 units with identically-zero residuals, so the pooled AR(1) null absorbs the step as noise; pinned as a documented regression. Scan p-values use a $Q = 99$ fitted-null Monte Carlo with a +1-rank rule; 0.010 is the minimum attainable. The synthetic-control placebo rank carries one rank of slack because each placebo refits SLSQP weights.

deterministically. The pipeline earns the benefit of the doubt it needs on data where the answer is not known.

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Exact to the Quarter, Lost in the Generation: US AI-Chip Export Controls in Product-Level Shipment Panels

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

At product granularity, the US export-control rounds on AI accelerators localize in time but not in attribution. In chip $\times$ quarter panels of shipment flows built from Epoch AI's chip-sales and chip-components estimates, all six known-treatment difference-in-differences scans put the maximum-likelihood-ratio break at the declared policy quarter *exactly*, and the October-2023 A800/H800 ban clears significance (Bernoulli scan $p = 0.030$; the banned SKUs' shipments go to literal zero within one quarter), while the April-2025 H20 licensing round localizes exactly but insignificantly ($p = 0.44$–0.60 on sales, 0.13–0.23 on components). The H20 effect magnitude is large and sign-correct — $\hat{\tau} = -2.756$ asinh units under two-way fixed effects, CR1 $t = -4.23$, 0 of 6 matched placebos exceeding, at the exact placebo-in-space floor $p = 0.143$ — but cannot be causally attributed: an unsupervised rediscovery scan shows that the panel's sharpest structure at 2025Q2 is the *entire Hopper-generation wind-down* with the H20 inside it, and the placebo battery generates same-size effects from ordinary product-lifecycle churn, with the matched donor pool either contaminated (3 of 6 controls are Chinese substitutes whose demand the ban plausibly boosted) or too small (3 clean controls, placebo $p = 0.75$). The date localization of the October-2023 round is the headline positive result; the correctly identified lifecycle confound and the pipeline's clean refusals are the rest of the story.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security (BIS) imposed the first broad export controls on advanced AI accelerators destined for China, a policy contemporaries called a landmark in US–China technology competition [1]. Nvidia responded within weeks with China-market SKUs — the A800 and H800 — engineered just under the control thresholds. On 17 October 2023 BIS closed that loophole: the updated advanced-computing rule [2] adjusted the chip-level performance parameters so that the workaround SKUs themselves became controlled. Nvidia iterated once more with the H20, a Hopper-generation part compliant with the 2023 thresholds — until 9 April 2025, when the US government informed the company that H20 exports to China (and any chip matching its bandwidth profile) would require a license, indefinitely; Nvidia's securities filing disclosed up to \$5.5 billion of associated charges [13].

A growing empirical literature evaluates these controls. On the supplier side, [6] show that US firms subject to export controls halt sales to their Chinese customers as intended but largely fail to

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replace them, losing market value, profitability, and employment. On the evasion side, [9] document underground markets for controlled accelerators and estimate the 2023 smuggling flow at hundreds to low thousands of chips. A companion country-level study in this collection [11] found that the October-2023 round *bent* China's legal chip-stock growth downward while aggregate compute was substantially bypassed. What that literature leaves untested is the sharpest granularity at which the policy is written: the named SKU. Each control round names specific products; if the rounds bind, product-level shipment panels should carry a break at the declared quarter in exactly the named SKUs — and nowhere else.

This paper runs that test three ways with the `natex` toolkit [12]. Leg 1 asks whether a scan-based difference-in-differences that searches over break dates [10], *knowing* which chips were treated, recovers the declared policy quarter from the data. Leg 2 removes the treatment labels entirely and asks what the sharpest structure in each panel actually is. Leg 3 estimates effect magnitudes with the placebo-based inference appropriate to a handful of treated clusters [5, 3]. The answer pattern — exact localization everywhere, significance only for October 2023, attribution defeated by product-lifecycle churn — is documented below, refusals included.

2 Data

Two Epoch AI datasets are used. *Data on AI Chip Sales* [8] provides quarterly shipment-flow estimates by chip model in thousands of H100-equivalents (kH100e); the extract's estimates were generated in January 2026 with 2025Q4 the last full-coverage quarter. *Data on AI Chip Components* [7] provides quarterly CoWoS advanced-packaging wafer estimates by chip for 2024Q1–2025Q4. From these, three chip $\times$ quarter panels are built (time index $t = 4\,\text{year} + \text{quarter} - 1$, so $8095 = 2023\text{Q4}$ and $8101 = 2025\text{Q2}$): (i) *sales Oct-23*, 7 chips $\times$ 16 quarters (2022Q1–2025Q4), treated {A800, H800}, declared $t_0 = 2023\text{Q4}$; (ii) *sales H20*, 20 chips $\times$ 8 quarters (2024Q1–2025Q4), treated {H20}, declared $t_0 = 2025\text{Q2}$; and (iii) *components H20*, 15 chips $\times$ 8 quarters of CoWoS wafers, treated {H20, MI308X}, declared $t_0 = 2025\text{Q2}$.

The outcome is asinh of the flow. Zeros are true zeros (no shipments), not missingness; chips born after $t_0 - 1$ are excluded at build time so that treatment cannot be mechanically confounded with birth; and the ragged 2026 tail is dropped at the 2025Q4 snapshot. Because the treatment effect here operates on the extensive margin — banned SKUs' shipments go to approximately zero — asinh magnitudes are unit-dependent and should not be read as percentage effects [4]; the qualitative reading (shipments driven to zero) is the transform-robust statement. One measurement caveat is carried through the paper: Huawei Ascend and Cambricon Siyuan flows in the source data are annual allocations smeared uniformly across quarters, which makes those series smooth by construction.

3 Design and Methods

Leg 1: known-treatment scans (date localization). For each panel, the `natex` SuDDDS scan [12, 10] maximizes a likelihood-ratio statistic for a declared-treatment DiD over candidate break quarters t_0 and window widths $W \in \{2, 4, 8\}$. Because the unit column is the chip itself, no subset search is available to the known-treatment scan: it localizes over (t_0, W) only, so *date localization is the contest* — does the maximum-LLR break land on the declared quarter? The primary model is Bernoulli with within-cluster calibration (wcc); a normal-model cross-check is calibrated under a dependence-preserving AR(1)-by-unit null. Permutation calibration uses $Q = 99$ draws at seed 0, so the smallest attainable scan p is 0.01.

Leg 2: unsupervised discovery rematch. The same scan machinery is then run with the treatment column deleted: θ is the observed flow, a categorical chip-name dimension is admitted to the subset search (model normal, method single_delta, restarts 8, panel-null $Q = 99$, seed 0), and the scan reports whatever subset $\times$ window is sharpest. Discovery never reads treatment labels; the question is whether the policy set is the panel's sharpest structure or merely sits inside something larger.

Leg 3: effects. A manual two-way fixed-effects (TWFE) DiD with chip and quarter fixed effects gives $\hat{\tau}$ in asinh units with CR1 standard errors clustered by chip. With at most 7 clusters, CR1 t-statistics are known to over-reject [3] — and demonstrably do here — so the *inference of record* is an exact placebo-in-space test in the spirit of [5]: every same-size subset of control chips is pseudo-treated (true treated units dropped) and p is the rank of $|\hat{\tau}|$ in that exact distribution. The preferred specification is lifecycle-matched: donor chips alive (nonzero) in all 8 quarters, removing births and deaths from the donor pool. Robustness: dropping 2025Q1 (a visible pull-forward spike of 54.2k H100e before the April license), restricting to non-Chinese controls (substitution-contamination check), a placebo-in-time at pseudo-t_0 = 2024Q4 on pre-period data only, and an event study relative to 2025Q1.

Honest refusals, stated as run. `natex`'s built-in dd/synthetic-control/GESS effect estimators refused on all three panels (0 usable placebo units; ≥ 5 required) — hence the manual TWFE leg. The scan's anticipation test refused on all known-treatment runs (no usable pre-quarters at these window widths; reported as not-passed with p = None, a refusal rather than a failure). And the lifecycle-matched Oct-23 effect leg refused outright: 0 of 5 control chips are alive in all 16 quarters, so no matched donor pool exists. Refusals are reported as refusals, not patched over.

4 Results

Localization: six for six, one significant. Every known-treatment scan puts the maximum-LLR break at the declared quarter exactly (Table 1, panel A). For the Oct-2023 round (7 chips $\times$ 16 quarters, treated A800+H800) the Bernoulli scan discovers $t_0 = 8095 =$ 2023Q4 with $W = 4$ and LLR = 4.896, scan $p = 0.030$ ($Q = 99$); the normal/ar1_unit cross-check finds the same t_0 and W with LLR = 4.085, $p = 0.14$. The banned SKUs' shipments go to literal zero within one quarter (Figure 1a). For the H20 round the localization is equally exact but never significant: sales panel $p = 0.44$ (Bernoulli) and 0.60 (normal); components panel (treated H20+MI308X) $p = 0.13$ and 0.23. The composition check passes ($p = 1.0$) in all six scans, and role-assignment sanity passes throughout: the scans are dating the declared treatment, not rediscovering an artifact.

Discovery: the generation, never the policy set. With treatment labels removed (Table 1, panel B), the Oct-23 panel's sharpest structure is {A800, H100/H200, H800, TPU v4, TPU v5e} at *2022Q4* (LLR = 16.16, $p = 0.01$ floor) — the Hopper production ramp and the birth of the A800, itself the workaround SKU created by the October-*2022* round. The ban quarter 2023Q4 does not appear in the top five discoveries; the closest is {A100, A800, H800} at 2024Q1 (rank 3) — the banned SKUs plus the A100's end-of-life. On the H20 sales panel, discovery puts the sharpest break at 2025Q2 *exactly* (LLR = 17.32, $p = 0.05$), but the discovered subset is {H100/H200, H20, MI300A, TPU v5e, TPU v5p, Trainium1} — the H20 merged into the entire Hopper-generation wind-down, with the complementary Blackwell-generation upshift as the runner-up. The components panel repeats the pattern one quarter early (7-chip old-generation set at 2025Q1, LLR = 16.41, p =

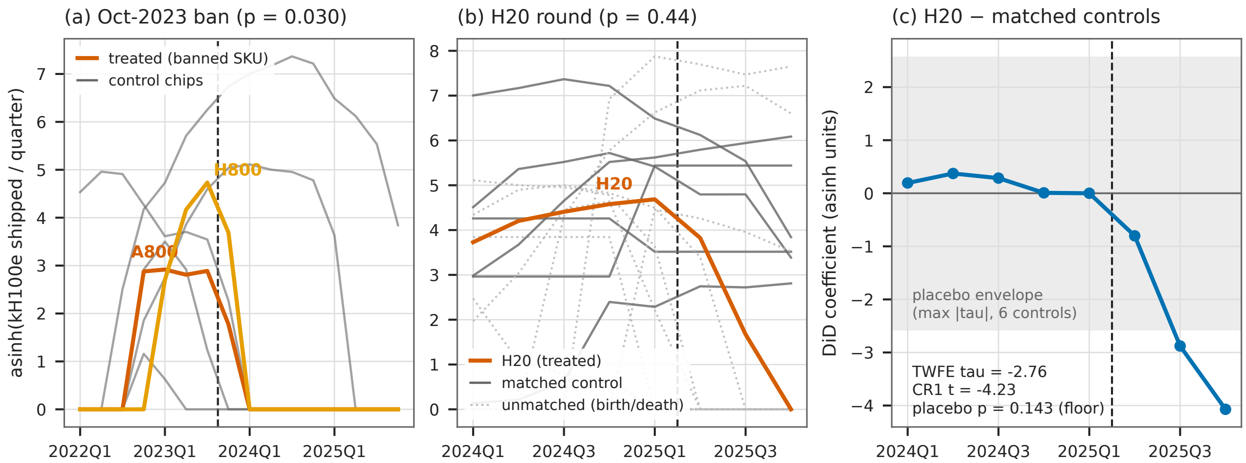

Figure 1: (a) Oct-2023 round: asinh shipment flows for the banned SKUs (A800, H800) against the five incumbent control chips; the scan puts the break exactly at the declared quarter (dashed line, 2023Q4 boundary) with $p = 0.030$, and treated shipments hit literal zero within one quarter. (b) H20 round: the H20 against 6 lifecycle-matched controls (solid) and 13 unmatched chips entering or exiting the market (dotted) — the churn that defeats attribution; the break again lands exactly on the declared quarter but at $p = 0.44$. (c) Event study, H20 minus matched controls relative to 2025Q1: flat pre-trend, monotone post-collapse to -4.07 asinh units by 2025Q4; the shaded band is the placebo-in-space envelope (max $|\tau|$ over the 6 matched controls), which the pooled estimate exceeds only at the $p = 1/7$ floor.

0.02). Every autonomously discovered subset is a chip *generation*, never the policy set: at product granularity, the H20 ban and the Hopper-to-Blackwell transition are the same event time.

Effects: large, sign-correct, unattributable. The lifecycle-matched TWFE estimate for the H20 (Table 1, panel C) is $\hat{\tau} = -2.756$ asinh units (CR1 se 0.652, $t = -4.23$) with 0 of 6 matched placebos exceeding — but 6 controls cap the exact test at its $p = 1/7 = 0.143$ floor. The event study is clean in shape: pre-period coefficients +0.19, +0.37, +0.29, +0.01, 0; post-period -0.80, -2.88, -4.07 (Figure 1c). The naive (unmatched) legs reproduce the lifecycle failure and calibrate how oversized CR1 is on these panels: the naive H20 spec has $t = -4.09$ yet 7 of 19 placebo chips produce larger pseudo-effects ($p = 0.400$); naive Oct-23 gives $p = 0.727$; naive components $p = 0.203$. Robustness on the matched leg: dropping the 2025Q1 anticipation spike leaves $\hat{\tau} = -2.799$ ($t = -3.73$) but costs one placebo exceedance ($p = 0.286$); restricting to the 3 non-Chinese controls gives $\hat{\tau} = -2.018$ with placebo $p = 0.750$ — the contamination check fails its placebo test, with the H100/H200's own Blackwell wind-down (-2.12) and Trainium2 (+2.88) both comparable to the estimate. The placebo-in-time is clean ($\hat{\tau} = -0.280$, $t = -0.67$).

5 Caveats and Conclusion

Four caveats bound the claims. First, *product-lifecycle churn is a competing sharp treatment*: unsupervised discovery finds only chip-generation subsets at the policy quarters — the Hopper wind-down with the H20 merged into it at 2025Q2, the Hopper ramp with the A800's birth at 2022Q4 — so at product granularity the H20 ban and the Blackwell transition are colinear in event time and cannot be separated inside this dataset. Second, *control contamination*: 3 of the 6

Table 1: Headline numbers of record, all three legs. Panel A: known-treatment SuDDDS scans; the declared quarter is 2023Q4 for the Oct-23 panel and 2025Q2 for both H20 panels, and $\hat{t}_0$ is the scan's maximum-LLR break quarter. Panel B: unsupervised discovery (no treatment column; $\theta =$ the flow). Panel C: TWFE effects in asinh units; the exact placebo-in-space p is the inference of record.

Panel A: known-treatment scans (date localization)	$\hat{t}_0$	W	LLR	scan p
sales Oct-23 (7×16; A800+H800), Bernoulli/wcc	2023Q4 exact	4	4.896	0.030
normal/ar1_unit cross-check	2023Q4 exact	4	4.085	0.14
sales H20 (20×8; H20), Bernoulli/wcc	2025Q2 exact	4	1.532	0.44
normal/ar1_unit cross-check	2025Q2 exact	2	0.914	0.60
components H20 (15×8; H20+MI308X), Bernoulli/wcc	2025Q2 exact	4	2.331	0.13
normal/ar1_unit cross-check	2025Q2 exact	2	1.949	0.23
Panel B: unsupervised discovery (top subset; every one a chip generation)	$\hat{t}_0$	W	LLR	p
sales Oct-23: {A800, H100/H200, H800, TPU v4, TPU v5e}	2022Q4	4	16.16	0.01 (floor)
sales H20: {H100/H200, H20, MI300A, TPU v5e/v5p, Trainium1}	2025Q2 exact	4	17.32	0.05
components H20: 7-chip old-generation set (incl. H20, MI308X)	2025Q1	4	16.41	0.02
Panel C: TWFE effects (asinh units; exceedances = placebos $\geq \lvert\hat{\tau}\rvert$)	$\hat{\tau}$	CR1 se	t	exceed., p
matched sales H20 (6 controls alive all 8 q)	−2.756	0.652	−4.23	0/6, 0.143 (floor)
drop 2025Q1 (anticipation spike)	−2.799	0.751	−3.73	1/6, 0.286
non-Chinese controls only (3)	−2.018	0.939	−2.15	2/3, 0.750
naive sales Oct-23 (all 5 controls)	−1.070	1.160	−0.92	7/10, 0.727
naive sales H20 (all 19 controls)	−2.347	0.574	−4.09	7/19, 0.400
naive components H20 (all 13 controls)	−2.059	0.876	−2.35	15/78, 0.203
placebo-in-time (pseudo-t_0 2024Q4, pre-only)	−0.280	0.420	−0.67	clean
matched Oct-23 analog	refused: 0 of 5 control chips alive all 16 quarters			

Notes: all runs `natex` v0.2.0, seed 0. Scan p-values from permutation calibration with $Q = 99$ (floor 0.01); Bernoulli scans use within-cluster calibration (wcc), normal scans a dependence-preserving AR(1)-by-unit null. Composition checks pass ($p = 1.0$) in all six panel-A scans; the anticipation test refused (no usable pre-quarters). CR1 clusters by chip; with $G \leq 7$ the exact placebo-in-space p is the inference of record, and the naive H20 row ($t = -4.09$, placebo $p = 0.400$) calibrates CR1's oversizing on these panels. Matched controls: Ascend 910B, Ascend 910C, H100/H200, MI300X, Siyuan 590, Trainium2 (placebo $\hat{\tau}$: −1.03, +2.06, −2.58, −1.50, +1.63, +1.42).

lifecycle-matched controls are Chinese accelerators (Ascend 910B/910C, Siyuan 590) that plausibly *gained* demand from the ban — a SUTVA violation biasing $\hat{\tau}$ negative (their placebo $\hat{\tau}$'s are +2.06, +1.63, −1.03) — while the no-China robustness leg (3 controls) gives $\hat{\tau} = -2.02$ with placebo $p = 0.75$: the matched leg's 0/6 leans on contaminated donors. Third, *measurement*: Huawei and Cambricon flows are annual allocations smeared over quarters — smooth by construction, understating placebo dispersion — and CR1 t-statistics are demonstrably oversized here (the naive H20 leg pairs $t = -4.09$ with placebo $p = 0.400$), which is why the exact placebo-in-space p is the inference of record. Fourth, *resolution floors*: 6 matched controls cap the placebo test at $p = 1/7 = 0.143$ and the scan permutation floor is 0.01 ($Q = 99$); no p-value below its floor is claimed. Relatedly, all treated units share one policy date per round, so scan p-values come from the fitted-null panel calibration rather than cross-unit timing variation — the Bernoulli $p = 0.030$ for Oct-23 is the defensible headline, with the normal cross-check ($p = 0.14$) reported beside it.

The verdict is deliberately asymmetric. Date localization at product granularity *works*: all six known-treatment scans land on the declared quarter exactly, the October-2023 round clears its calibrated null ($p = 0.030$), and the banned SKUs' shipments hit literal zero within one quarter — a policy written against named products is visible in product-level data on the dot. Attribution of effect *size* at this granularity does not work, and the pipeline says so rather than manufacturing a significant number: the H20 collapse is large ($\hat{\tau} \approx -2.8$ asinh units) and correctly signed, but placebo chips generate ban-sized effects from ordinary lifecycle churn, the clean-donor robustness fails its own placebo test, and the matched Oct-23 leg refuses for want of donors. For an automated natural-experiment pipeline, the refusals and the correctly identified confound are the point: the machinery distinguished what these panels can establish — *when* — from what they cannot — *how much*.

Reproducibility. All estimates were produced with `natex` v0.2.0 [12] at seed 0 from frozen extracts of the Epoch AI datasets [8, 7] (not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline scan and effect estimates against the numbers of record before drawing.

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A Sharp Bend, Now Gone: The DeepSeek-R1 Adoption Kink Fails Out of Sample in the Extended BTOS Panel

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analysis pass of record: 2026-07, `natex` v0.2.0; fully deterministic (the kink CLI has no RNG)

Abstract

The DeepSeek-R1 adoption bend does not survive nine more months of data. In the Census Bureau's Business Trends and Outlook Survey (BTOS), a companion analysis through October 2025 found a sharp acceleration of measured AI use in AI-exposed sectors at the first survey wave after R1's release (20 January 2025), with a group difference-in-kinks (DiK) estimate of +7.3 pp/yr (cluster-robust $p = 0.042$) — nominally significant, though already failing attribution placebos. Extending the sector panel to June 2026 (71 waves, spliced across the December-2025 questionnaire rewording at the companion measurement-RDD ratio of 1.553), the prior estimate replicates exactly at bandwidth 0.5 (+7.27 pp/yr, $p = 0.042$) — but its estimation window ends in July 2025 and contains zero new data. Every specification that sees the extended panel is null to negative at R1: bandwidth 1.0 gives $\hat{\tau} = -0.16$ pp/yr (CR1 se 3.44, 95% CI $[-8.57, +8.24]$, $p = 0.963$, $n = 244$); bandwidth 1.5 gives -2.09 ($p = 0.543$); per-sector splice factors give +1.9–2.0 ($p > 0.3$). The placebo battery is now two-sided: non-event 2024 cutoffs produce positive "kinks" as large as the old R1 estimate (+11.50 at $t = 2024.30$, $p = 0.018$) and non-event 2025H2 cutoffs produce significant *negative* ones (-9.08 at $t = 2025.60$, $p = 0.011$; -4.06 at the splice boundary, $p = 0.019$), so a generically convex-then-plateauing diffusion gap explains every cutoff. Descriptively, the exposed-minus-unexposed gap slope *decelerated* across R1, from +4.4 to +2.3 pp/yr. The lone remaining significant R1 estimate (raw outcome plus an additive wording dummy at bandwidth 1.5: +8.04, $p = 0.023$) is a measurement-splice artifact that fails its own placebos. The verdict is an honest reversal: the January-2025 "descriptive acceleration" was a short-window artifact, and nothing in the extended panel distinguishes DeepSeek-R1 from ordinary curvature in a diffusion curve.

1 Introduction

When an event-study estimate is nominally significant but its falsification battery is ambiguous, the cleanest arbiter is data that did not exist when the estimate was made. This paper supplies that arbiter for one specific claim. A companion analysis [8] of the Census Bureau's Business Trends and Outlook Survey (BTOS) through October 2025 found that measured AI use in AI-exposed sectors accelerated sharply at exactly the first survey wave after the release of the DeepSeek-R1 reasoning model on 20 January 2025 [6]: a data-driven scan localized the break at that wave, and a group difference-in-kinks (DiK) estimate at the release date was +7.3 pp/yr (cluster-robust se 2.8). That paper already graded the finding descriptive-only, because placebo cutoffs at non-event 2024 dates

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produced slope breaks as large as the R1 estimate. Nine more months of BTOS data now exist. They support a sharper test: if a genuine R1 response bent the adoption curve, the kink should persist — and its precision improve — as the estimation window widens to include the new data; if the "bend" was local curvature in a steep diffusion path, it should dissolve.

Running that test requires carrying the BTOS series across a measurement break. The BTOS has asked US firms since September 2023 whether they used AI in the last two weeks [2]; on 3 December 2025 the Census Bureau announced that the AI questions had been reworded — from AI use "in producing goods or services" to AI use "in any of its business functions" — and started a new time series [11]. A second companion paper [9] estimates the rewording jump in a known-cutoff regression-discontinuity-in-time design and calibrates a splice factor: the new-wording national level is 1.553 times the old-wording level at the cutoff. This paper divides new-wording values by that ratio (with per-sector and covariate-adjusted variants as robustness), which is what makes the extended panel usable at all.

Methodologically the paper applies the DiK design of [1], which builds on the regression kink design of [4], using the `natex` toolkit [10]. Kink estimates with calendar time as the running variable inherit the known pathologies of regression discontinuity in time [7], and slope breaks in time series are notoriously easy to "find," which is why placebo estimates at non-event dates — the permutation logic of [5] — are the decision criterion throughout, rather than any single cluster-robust p-value resting on 6–7 sector clusters [3]. The result is a clean reversal: the prior estimate replicates exactly on its own window, every extended-window specification is null to negative, and the placebo landscape now brackets the R1 date with significant breaks of *both* signs at dates where nothing happened.

2 Data

Two BTOS vintages are combined. The old-wording segment is the frozen AI core workbook of the companion paper (DRB approval CBDRB-FY25-0425): sector-level biweekly estimates and standard errors for Question 7 ("In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services?"), 54 waves 202319–202520, collection periods 2023-09-11 through 2025-10-05. The new-wording segment is a fresh census.gov sector download (fetched 2026-07-19): the reworded question ("…in any of its business functions?"), 17 waves 202524–202614, December 2025 through June 2026. Waves 202521–202523 were never collected (the October–November 2025 federal government shutdown), leaving a 7-week no-data gap at $t \in [2025.78, 2025.92]$ immediately before the rewording. The running variable t is the fractional year of each wave's collection-period midpoint.

The spliced panel divides new-wording values by 1.553, the national new/old measurement-RDD ratio at the 2025-12-03 rewording [9], and contains 1,121 sector-wave cells across 19 NAICS sectors, $t = 2023.710$–2026.507, of which 294 cells are new-wording. The exposed group is {51 information, 52 finance and insurance, 54 professional/scientific/technical services}; the comparison group is {11, 21, 23, 72}, thin as in the companion paper: sectors 11 and 21 have only ∼3 usable old-wording waves each, so the unexposed side is effectively construction (23) and accommodation and food services (72) before 2025. Two robustness panels vary the measurement handling: (i) per-sector splice factors estimated at the rewording boundary (old-side local linear trend extrapolated to the first new-wording waves; factors range 1.19–4.08, larger in low-adoption sectors, national fallback where fewer than two boundary observations exist); and (ii) the raw unspliced outcome with an additive new-wording dummy as a covariate.

3 Design and Methods

Group difference-in-kinks. Following [1], the DiK estimand is the change in the slope of the outcome in calendar time at a known cutoff for the exposed group minus the same slope change for the comparison group, estimated by local linear regression on each side of the cutoff within each group (`natex` v0.2.0 [10]). All specifications use a triangular kernel, a one-wave donut (0.0384 yr), and CR1 standard errors clustered by sector, with p-values from $t(G-1)$ with $G \in \{6, 7\}$ clusters [3]. The R1 cutoff is $t_0 = 2025.055$ (2025-01-20; first post wave 202503). Bandwidth 0.5 yr reproduces the companion paper's primary specification — its window $[2024.56, 2025.56]$ ends in July 2025 and therefore contains *zero* post-extension data — while bandwidths 1.0 and 1.5 yr are the out-of-sample test: their windows extend to January and July 2026 respectively and see the new waves.

Falsification battery. The same estimator runs at non-event placebo cutoffs (2024.30, 2024.50, 2024.85, 2025.30, 2025.60), at the splice boundary itself (2025.845, a pure measurement date), and at OpenAI's o1 release (2024-09-12, $t = 2024.6967$) — 36 deterministic `natex` kink runs in total across the three measurement panels (the kink CLI has no RNG). In the spirit of [5], the placebo grid, not any single p-value, is the decision criterion: with calendar time as the running variable, any macro or sector-specific shock aliases as a kink [7], so an R1 effect must stand out *against* the placebo landscape, not merely reject zero.

Measurement handling, stated in advance. A national splice ratio applied to all sectors is wrong in detail whenever the rewording effect is sector-heterogeneous — and it is (boundary factors 1.19–4.08). The per-sector-splice panel addresses this directly. The covariate variant (raw outcome plus an additive new-wording dummy) is included for completeness but is structurally misspecified: the rewording is a multiplicative rescale ($\sim$1.55$\times$ nationally), and a single additive dummy common to both groups cannot absorb a shift that lifts the exposed group ($\sim$24 pp raw) far more than the unexposed ($\sim$5 pp). The symptom to watch for is the covariate spec "finding" significant kinks at placebo cutoffs; it does (Section 4), so its R1 estimates are discounted.

4 Results

The replication is exact; the extension is null. At bandwidth 0.5 the R1 DiK is $\hat{\tau} = +7.27$ pp/yr (se 2.81, 95% CI $[0.38, 14.16]$, $p = 0.042$, $n = 123$) — to the digit the companion paper's primary estimate, as it must be: the spliced panel is byte-identical to the old panel on that window. The moment the window widens to include any of the nine new months, the estimate collapses: bandwidth 1.0 gives $\hat{\tau} = -0.16$ pp/yr (se 3.44, 95% CI $[-8.57, +8.24]$, $p = 0.963$, $n = 244$, 7 clusters) and bandwidth 1.5 gives -2.09 (se 3.24, 95% CI $[-10.03, +5.85]$, $p = 0.543$, $n = 373$). This is not bandwidth dilution of a real kink — a persistent slope change measured with more data on both sides would gain, not lose, precision-weighted magnitude; it is the signature of local curvature that a short window mistook for a bend.

Robust to how the splice is done. With per-sector splice factors the extended-window R1 estimates are $+1.93$ (se 2.82, $p = 0.520$) at bandwidth 1.0 and $+2.05$ (se 1.90, $p = 0.324$) at bandwidth 1.5 — positive but small and nowhere near significant. The covariate variant gives $+4.73$ (se 3.15, $p = 0.185$) at bandwidth 1.0 and $+8.04$ (se 2.64, $p = 0.023$) at bandwidth 1.5 — the only significant extended R1 estimate in the grid, and it is discounted for the reason stated

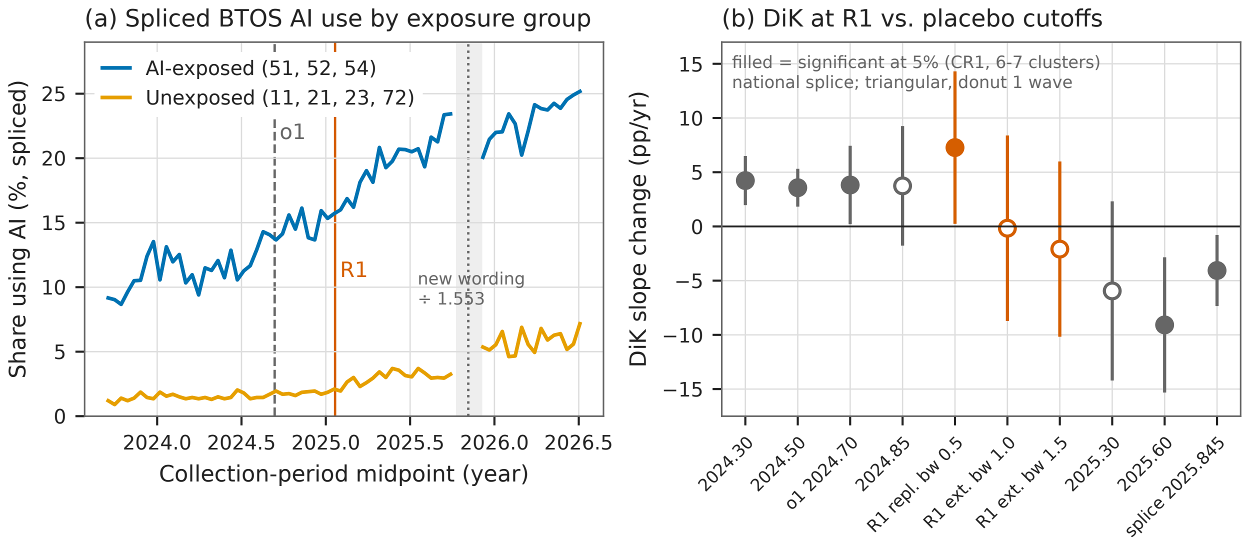

Figure 1: (a) Spliced BTOS AI-use share by exposure group across all 71 waves (new-wording waves divided by 1.553); vertical lines mark o1, DeepSeek-R1, and the rewording splice (dotted), with the shaded band the 7-week shutdown no-data gap. (b) Group DiK estimates with 95% CR1 confidence intervals at the R1 cutoff (vermillion: the bandwidth-0.5 replication, whose window ends in July 2025, and the extended bandwidth-1.0/1.5 estimates, which see the new data) and at placebo cutoffs (gray, bandwidth 1.0); filled markers are significant at the 5% level. The extended R1 estimates are null while non-event cutoffs on both sides are significantly positive (2024) and negative (2025H2).

in advance: the same specification "finds" significant positive kinks at non-event placebo cutoffs (+8.52, $p = 0.014$ at $t_0 = 2025.30$; +5.35, $p = 0.009$ at $t_0 = 2025.60$), because an additive dummy cannot absorb a multiplicative $\sim$1.55$\times$ measurement rescale. A specification that fails its own placebo battery cannot rescue the effect it was built to test.

The placebo landscape is now two-sided. On the extended national-splice panel, non-event cutoffs in 2024 still produce positive DiKs as large as the old R1 estimate — +11.50 (se 3.58, $p = 0.018$) at $t_0 = 2024.30$ bandwidth 0.5, +4.23 ($p = 0.003$) at bandwidth 1.0; +3.98 (se 0.90, $p = 0.007$) and +3.57 ($p = 0.002$) at $t_0 = 2024.50$ — and non-event cutoffs in 2025H2 produce significant *negative* DiKs: -11.77 ($p = 0.007$) at $t_0 = 2025.30$ bandwidth 0.25, -16.70 ($p = 0.019$) and -9.08 (se 2.49, $p = 0.011$) at $t_0 = 2025.60$, and -4.06 ($p = 0.019$) at the splice boundary itself (Table 1; Figure 1b). A gap curve that is convex through 2024 and plateauing from mid-2025 generates exactly this pattern at *any* cutoff; R1 is nowhere special on it. The o1 contrast remains bandwidth-aliased, as in the companion paper: -5.91 ($p = 0.099$) at bandwidth 0.5, +3.82 ($p = 0.036$) at 1.0, +2.73 ($p = 0.192$) at 1.5.

Descriptively, the gap decelerated. The SE-weighted exposed-minus-unexposed gap slope is +4.38 pp/yr in the year *before* R1 and +2.27 pp/yr in the year *after* — a deceleration, not an acceleration. The spliced gap rises from 8.0 pp at the panel start (September 2023) to roughly 18 pp by mid-2026 and is approximately flat over February–June 2026 (Figure 1a). The companion paper's "descriptive acceleration" — a gap slope tripling measured over ±13 waves — was a short-window artifact of this convex-then-plateauing path.

Table 1: Group difference-in-kinks estimates on the extended panel, BTOS AI-use share (pp/yr).

Specification	$\hat{\tau}$	se	p	sig. 5%
Panel A: DeepSeek-R1 cutoff ($t_0 = 2025.055$), national splice				
bw 0.5 (replication; window ends 2025.56)	+7.27	2.81	0.042	$*$
bw 1.0 (extended)	−0.16	3.44	0.963	
bw 1.5 (extended)	−2.09	3.24	0.543	
Panel B: R1 cutoff, alternative measurement handling				
per-sector splice, bw 1.0	+1.93	2.82	0.520	
per-sector splice, bw 1.5	+2.05	1.90	0.324	
raw + wording dummy, bw 1.0	+4.73	3.15	0.185	
raw + wording dummy, bw 1.5	+8.04	2.64	0.023	$(*)$
Panel C: placebo and secondary cutoffs, national splice				
$t_0 = 2024.30$ (no event), bw 0.5	+11.50	3.58	0.018	$*$
$t_0 = 2024.30$ (no event), bw 1.0	+4.23	0.87	0.003	$*$
$t_0 = 2024.50$ (no event), bw 0.5	+3.98	0.90	0.007	$*$
$t_0 = 2024.50$ (no event), bw 1.0	+3.57	0.65	0.002	$*$
$t_0 = 2025.30$ (no event), bw 0.25	−11.77	2.68	0.007	$*$
$t_0 = 2025.60$ (no event), bw 0.5	−16.70	5.27	0.019	$*$
$t_0 = 2025.60$ (no event), bw 1.0	−9.08	2.49	0.011	$*$
splice boundary $t_0 = 2025.845$, bw 1.0	−4.06	1.28	0.019	$*$
o1, $t_0 = 2024.6967$, bw 0.5	−5.91	2.92	0.099	
o1, $t_0 = 2024.6967$, bw 1.0	+3.82	1.42	0.036	$*$
o1, $t_0 = 2024.6967$, bw 1.5	+2.73	1.85	0.192	

Notes: CR1 standard errors clustered by sector; p-values against $t(G-1)$ with $G \in \{6, 7\}$ clusters (5–7 at bandwidth 0.25). Triangular kernel, one-wave donut, bandwidths in years. New-wording waves divided by 1.553 (national splice), by boundary-estimated per-sector factors (Panel B rows 1–2), or entered raw with an additive new-wording dummy (Panel B rows 3–4; discounted — see text: this specification also finds +8.52, $p = 0.014$ and +5.35, $p = 0.009$ at the 2025.30 and 2025.60 placebo cutoffs). Outcome: sector-level share answering Yes to the BTOS AI-use question, percentage points.

5 Caveats and Conclusion

Five caveats bound the reversal, none of which rescues R1. First, the national splice ratio applied to every sector is wrong in detail — the boundary-estimated per-sector factors range from 1.19 to 4.08, larger in low-adoption sectors — but the R1 conclusions are robust to using those factors, which are themselves noisy boundary extrapolations across the 7-week shutdown gap. Second, the covariate variant is structurally misspecified (additive dummy against a multiplicative shift); its lone significant R1 estimate fails its own placebos and should not be cited as evidence. Third, CR1 inference with 6–7 clusters is fragile [3] — but the headline here is a null, and the significant placebos argue *against* the effect, so any over-rejection makes the case against R1 stronger, not weaker. Fourth, the significant negative kink at the splice boundary mixes a genuine gap plateau with possible splice fragility; slopes after $t = 2025.9$ should be treated as measurement-provisional either way. Fifth, BTOS measures extensive-margin adoption ("used AI in the last two weeks"); a null here does not rule out intensive-margin or within-firm R1 effects that an any-use share cannot see.

The verdict is an honest reversal, and a clean one. The companion paper's R1 kink replicates exactly on its own window — nothing about the old computation was wrong — and fails completely out of sample: every specification that sees the nine new months of data is null to negative at R1, under two independent splice treatments; the placebo battery now brackets the R1 date with significant breaks of both signs at non-event dates; and the descriptive gap slope decelerated across the release. What looked like an adoption response to DeepSeek-R1 was the local curvature of a diffusion gap that grew convexly through 2024 and plateaued in late 2025 — a pattern that manufactures "significant" kinks at almost any cutoff a researcher might try. The companion paper withheld causal attribution; the extended panel withdraws even the descriptive acceleration.

Reproducibility. All estimates were produced with `natex` v0.2.0 [10] from the frozen BTOS workbooks (published at `https://www.census.gov/hfp/btos`; not committed to the repository); the 36 kink runs are fully deterministic (the CLI has no RNG), and the spliced kink panel is byte-identical to the scout build of record. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Teaching to the Test, Undetected: No Release-Localized Contamination Premium on Public versus Held-Out Benchmarks

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July 2026

analysis pass of record: 2026-07, seed 0 (wild-cluster bootstrap only; randomization inference is exhaustive)

Abstract

In Epoch AI's cross-benchmark capability panel, models released after a benchmark's publication score above their general-capability (ECI) expectation on *public* benchmarks relative to four *held-out* benchmarks — but the premium does not survive cluster-honest inference, and the design's own placebo battery shows it is not localized at benchmark release. In 896 model × benchmark cells spanning the 19 benchmarks that straddle their own release date, the post × public level contrast is $\tau = +0.127$ z (CR1 SE 0.079); every level specification is positive, and late models underperform their ECI expectation on held-out benchmarks by -0.19 z — exactly the sign pattern a teaching-to-the-test story predicts. But with only 4 held-out benchmark clusters the inference of record is a wild-cluster bootstrap ($p = 0.145$) and exact randomization inference over all $\binom{19}{4} = 3{,}876$ held-out labelings (one-sided $p = 0.213$), neither of which rejects. The premium collapses under IRT-difficulty controls (+0.04) and model fixed effects (+0.03) — held-out status is confounded with benchmark difficulty — and it fails the placebo-cutoff localization test: the contrast at a -0.5-year placebo cutoff (+0.267) *exceeds* the true-cutoff estimate, the signature of a smooth group trend difference in event time rather than a step at release. Ability-proxy variants that avoid ECI's mechanical attenuation (the index is trained on the public benchmarks) triple the estimate (+0.35 to +0.39, wild-cluster $p = 0.047$–0.082) but inherit the same localization failure. A difference-in-kinks secondary is a correctly identified artifact (its placebo cutoff beats every true-cutoff estimate), and the automated `natex` SuDDDS run recovered the declared contrast but honestly *refused* a placebo p (no in-space placebos with a single binary group dimension); the manual bootstrap/randomization battery fills exactly that gap and does not reject. The verdict is a suggestive-sign null: no release-localized contamination premium is detectable in this panel.

1 Introduction

Train–test contamination — benchmark answers leaking into the training corpora of the models being benchmarked — is the standing objection to public AI leaderboards. [9] show that test-set membership can be proven from a model's preference for the canonical ordering of an exchangeable benchmark; [11] exploit the natural experiment of GPT training cutoffs to show that performance on code benchmarks is systematically related to whether the problems predate the cutoff; [13] commission a fresh clone of GSM8k and find accuracy drops of up to 13% for model families that apparently overfit the public original. [4] generalize the worry beyond verbatim leakage: *training*

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on the test task — benchmark-shaped data and formats entering training pipelines — confounds model comparisons even when no test item was ever seen.

The benchmark-building community's institutional response has been to hold answers back. FrontierMath keeps private problem tiers scored only via Epoch AI [6]; Humanity's Last Exam withholds a private split against overfitting [10]; ARC-AGI-2 maintains a private evaluation set [3]; LiveBench rotates fresh questions monthly precisely to stay ahead of training corpora [12]. This design pattern implies a cross-benchmark test of contamination in the wild. If public-benchmark scores are inflated by leakage or test-task training, then models released *after* a public benchmark's publication should overperform their general capability on it, and the premium should be absent on held-out benchmarks whose answers cannot be downloaded — a difference-in-differences in event time around each benchmark's release date, with held-out benchmarks as the comparison group.

This paper runs that test on Epoch AI's benchmarking panel [5, 8], treating benchmark release as a dated event inside a model × benchmark panel and demanding that any premium be *localized* at the event: estimated with cluster-honest inference (only 4 of 19 benchmark clusters are held out), stress-tested against difficulty controls, and benchmarked against placebo release dates, with the automated `natex` toolkit's discovery-and-refusal machinery [7] reported as run. The sign pattern survives everything; the localized premium survives nothing.

2 Data

Three public Epoch AI files are joined [5]: the ECI benchmark score table (2,010 model × benchmark performance cells with benchmark release dates), the ECI model index (211 models with release dates and ECI capability scores [8]), and the IRT benchmark parameter table (52 benchmarks with estimated difficulty `edi` and discrimination). Three release dates missing from the source are hand-filled as declared inputs: HLE 2025-01-23, ARC-AGI-2 2025-03-24, and FrontierMath-Tier-4-Private 2025-07-01. The held-out classification is likewise declared, by whether answers are publicly downloadable: FrontierMath-2025-02-28-Private, FrontierMath-Tier-4-2025-07-01-Private, HLE, and ARC-AGI-2 (4 benchmarks); all other benchmarks are public.

A cell is a model × benchmark pair. The running variable is event time, x = model release date − benchmark release date, in years. The identifying sample keeps the 19 benchmarks with cells on both sides of $x = 0$: $n = 896$ cells across 150 models, split 227 pre-release / 668 post-release (one cell at exactly $x = 0$ is coded pre). Outcomes are built within benchmark: z is the within-benchmark z-score of performance, and the primary outcome r residualizes z linearly on the model's ECI score, so r measures performance relative to what the model's general capability predicts on that benchmark. Variants used below: a logit-performance residual (48 zero and 3 one scores clipped at $[0.01, 0.99]$), and two ability proxies that avoid ECI itself — the model's mean z across all *other* benchmarks (leave-one-out) and across held-out benchmarks only. The panel builder asserts bit-identical reproduction of the scout-pass r before writing any file.

3 Design and Methods

Level contrast. The primary estimand is the post-release premium of public over held-out benchmarks in r, from the cell-level regression

$$r_{mb} = \alpha + \beta x_{mb} + \gamma\,\mathrm{post}_{mb} + \tau\,(\mathrm{post}_{mb} \times \mathrm{public}_b) + \mu_b + u_{mb},$$

with benchmark fixed effects μ_b and a common linear event-time trend; γ is the held-out post-release baseline and τ the contamination premium. Errors are CR1 cluster-robust by benchmark with a

$t(18)$ reference. With 19 clusters of which only 4 are held out, CR1 alone is not trustworthy [2], so the inference of record is (i) a null-imposed Rademacher wild-cluster bootstrap (B = 9,999, seed 0) and (ii) *exact* randomization inference over all $\binom{19}{4}$ = 3,876 assignments of the held-out label to 4 of the 19 benchmarks, FWL-accelerated (the observed labeling's coefficient is ranked in the full permutation distribution; no sampling anywhere).

Localization. A contamination premium must be a step at benchmark release, not a generic divergence of the two groups in event time. The identical regression is re-estimated with the cutoff moved to $x_0 \in \{-1.0, -0.5, +0.5, +1.0\}$ years; a true release-localized effect should peak at the true cutoff. Side composition (straddling benchmarks and held-out benchmarks per placebo) is reported because extreme placebo cutoffs thin one side.

Robustness. Seven re-estimates: the logit-residual outcome; post × IRT-difficulty and post × discrimination controls (standardized `edi` and slope); a two-way model + benchmark fixed-effects specification on z; the two ability-proxy outcomes; dropping saturated cells (performance > 0.95); and model-release-year fixed effects, which isolate cross-benchmark stagger from calendar time. A per-benchmark jackknife re-estimates τ dropping one cluster at a time.

Automated discovery (SuDDDS). The `natex discover` pipeline [7] ran on the benchmark × half-year-bin panel (94 cells; `--design did`, declared treatment path θ = public × post, seed 0). Its subset scan, anticipation gate, and placebo machinery either validate the declared event or refuse; refusals are reported as run, per the toolkit's honest-inference conventions.

Difference-in-kinks secondary. As a slope-based cross-check, a group difference-in-kinks in the tradition of [1] — `natex kink` with the held-out group as the comparison dimension, triangular kernel, CR1 by benchmark — estimates the change in the event-time *slope* contrast at $x = 0$, over bandwidths 0.5–2.0 years, with a placebo cutoff at $x_0 = +0.5$.

4 Results

A positive premium with a contamination-consistent sign pattern. The primary contrast is $\tau = +0.1267$ z (CR1 SE 0.0786, $t = 1.61$, $p_{t(18)} = 0.124$) on $n = 896$ cells in 19 benchmark clusters. The held-out post-release baseline is $\gamma = -0.1929$ z (SE 0.0526): after a benchmark's release, models underperform their ECI expectation on held-out benchmarks, while public benchmarks absorb the offsetting premium — the two-sided pattern teaching-to-the-test predicts. The premium is not driven by any single cluster: the per-benchmark jackknife keeps $\tau \in [+0.085, +0.171]$, positive in all 19 leave-one-out samples.

Cluster-honest inference does not reject. The wild-cluster bootstrap gives $p = 0.145$. Exact randomization inference over all 3,876 held-out labelings gives one-sided $p = 0.213$ (two-sided 0.384): more than a fifth of all possible "held-out" labelings produce a premium at least as large as the observed one, and the permutation distribution's 5th/50th/95th percentiles (-0.224 / $+0.028$ / $+0.200$) comfortably bracket $+0.127$. With 4 treated-side clusters, this is the honest finite-sample yardstick, and the observed contrast is unremarkable against it.

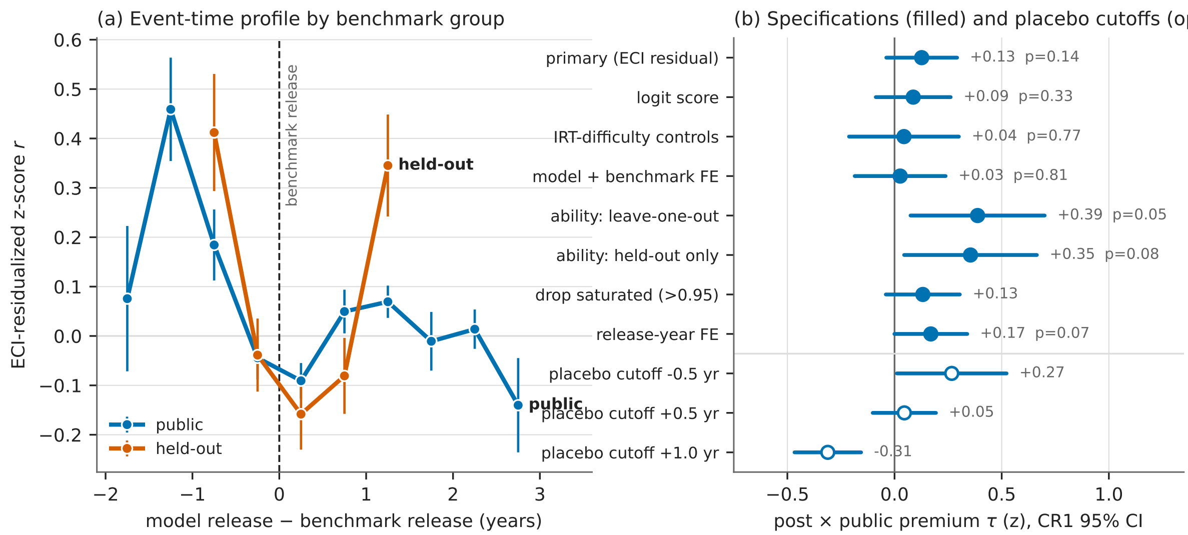

Figure 1: (a) Mean ECI-residualized within-benchmark z-score r by 0.5-year event-time bin (whiskers ± 1 SE; bins with ≥ 3 cells), public versus held-out benchmarks, vertical rule at benchmark release. In the first year after release, held-out cells fall below their capability expectation while public cells sit above it — but the raw profiles show no step at $x = 0$, and the two groups already differ before release. (b) The post $\times$ public contrast τ across specifications (filled circles; CR1 95% CIs, $t(18)$ critical value; wild-cluster bootstrap p annotated where computed) and at placebo event-time cutoffs (open circles). Every specification is positive, but the -0.5-year placebo cutoff exceeds the true-cutoff estimate — the localization failure that grades this a null.

The premium is difficulty in disguise. The held-out set — FrontierMath tiers, HLE, ARC-AGI-2 — is also the hardest-benchmark set. Adding post $\times$ IRT-difficulty and post $\times$ discrimination controls collapses τ from $+0.127$ to $+0.044$ (wild-cluster $p = 0.769$); two-way model + benchmark fixed effects give $+0.026$ ($p = 0.814$); the logit outcome gives $+0.087$ ($p = 0.334$). Two specifications move the other way: the ability-proxy outcomes, which replace ECI — itself fit *on the public benchmarks* [8] and therefore mechanically absorbing part of any public-benchmark inflation into measured capability — with leave-one-out or held-out-only mean z. These triple the estimate ($+0.388$, wild-cluster $p = 0.047$; $+0.354$, $p = 0.082$), confirming the attenuation channel is real, but they inherit the localization failure below and the 4-cluster fragility. Dropping saturated cells ($+0.132$) and release-year fixed effects ($+0.169$, $p = 0.072$) leave the picture unchanged.

Localization fails. Moving the cutoff half a year *before* benchmark release yields $\tau = +0.267$ (SE 0.122, 15 straddling benchmarks, all 4 held-out) — *larger* than the true-cutoff estimate. The grid declines monotonically through the true cutoff: $+0.800$ at -1.0 (uninterpretable: 11 pre cells, 4 straddling benchmarks), $+0.267$ at -0.5, $+0.127$ at 0, $+0.046$ at $+0.5$, -0.311 at $+1.0$. A release-localized contamination premium should peak at release; a smooth difference in the two groups' event-time trends produces exactly this monotone drift. The design therefore identifies the positive contrast as a trend difference, not a step at benchmark release.

Automated runs. The `natex` SuDDDS pipeline discovered the declared subset {group: public} at $t_0 = 0$ with window 2 (treatment-path scan $p = 0.010$ — mechanical given the declared θ) and estimated DD $\tau = +0.232$ (SE 0.088) on the 94-cell binned panel, but *refused* its placebo

Table 1: Headline estimates. Outcome r is the within-benchmark z-score of performance residualized on model ECI; τ is the post $\times$ public coefficient with benchmark FE and a linear event-time trend, CR1 clustered by benchmark (19 clusters, 4 held-out). WCB: null-imposed Rademacher wild-cluster bootstrap, $B = 9{,}999$, seed 0. RI: exact randomization inference over all $\binom{19}{4} = 3{,}876$ held-out labelings.

Quantity	τ	CR1 SE	p	notes
Panel A: primary contrast and inference of record				
post $\times$ public (primary)	+0.127	0.079	0.145	WCB; $p_{t(18)} = 0.124$
exact RI, one-sided	—	—	0.213	two-sided 0.384
held-out post baseline γ	−0.193	0.053	—	contamination-consistent sign
Panel B: robustness (WCB p where computed)				
logit score	+0.087	0.083	0.334	$n = 896$
IRT-difficulty controls	+0.044	0.122	0.769	post $\times$ edi, post $\times$ slope
model + benchmark FE	+0.026	0.101	0.814	outcome z
ability: leave-one-out	+0.388	0.149	0.047	$n = 895$
ability: held-out only	+0.354	0.147	0.082	$n = 661$
drop saturated (perf > 0.95)	+0.132	0.082	—	$n = 878$
release-year FE	+0.169	0.081	0.072	$n = 896$
Panel C: placebo event-time cutoffs (straddling / held-out)				
cutoff −1.0 yr	+0.800	0.078	—	4/1; 11 pre cells, uninterpretable
cutoff −0.5 yr	+0.267	0.122	—	15/4; *exceeds true cutoff*
cutoff +0.5 yr	+0.046	0.070	—	19/4
cutoff +1.0 yr	−0.311	0.074	—	10/3
Panel D: automated `natex` runs				
SuDDDS DD (94-cell panel)	+0.232	0.088	refused	0 in-space placebos; gate closed
DiK slope, cutoff 0, bw 0.75	+0.015	0.443	—	drifts to −0.843 (0.218) at bw 2.0
DiK placebo +0.5, bw 0.75	−1.467	0.644	—	exceeds all true-cutoff estimates

Notes: jackknife over benchmarks keeps $\tau \in [+0.085, +0.171]$ (positive in all 19 leave-one-out samples). RI permutation quantiles (5%/50%/95%): −0.224 / +0.028 / +0.200. The SuDDDS scan $p = 0.010$ localizes the declared treatment path and is mechanical given the declared θ; its refusal of a placebo p is the design working as intended with one binary group dimension.

randomization p: with a single binary group dimension there are 0 usable in-space placebos (≥ 5 required), and the anticipation gate failed closed (null Holm p-values). The manual wild-cluster-bootstrap and exact-randomization battery above is precisely the placebo inference the refusal requests, and it does not reject. The difference-in-kinks secondary is an identified artifact: the slope contrast at the true cutoff is +0.015 (SE 0.443) at bandwidth 0.75, drifting to −0.843 (SE 0.218) at bandwidth 2.0 as saturated public post-release profiles compress, while the +0.5 placebo cutoff gives −1.467 (SE 0.644) — larger in magnitude than every true-cutoff estimate — so the slope design carries no evidential weight here.

5 Caveats and Conclusion

Six caveats bound the claims, none softened. First, *four held-out clusters*: CR1/$t(18)$ inference alone is not trustworthy at this cluster count [2]; the wild-cluster bootstrap and the exact label-randomization test are the inference of record, and they do not reject — but randomization inference assumes label exchangeability, which the composition differences below strain; it remains the most honest finite-sample check available here. Second, *difficulty confounding*: held-out status is not randomly assigned — the held-out set is the hardest-benchmark set, and post $\times$ difficulty controls absorb most of the premium. Third, *mechanical attenuation*: ECI is trained on the public benchmarks, so contamination inflates the capability control itself; the ability-proxy variants quantify this (tripling τ) but inherit every other weakness. Fourth, *localization failure*: the -0.5-year placebo contrast exceeds the true-cutoff contrast, so the design cannot distinguish a release-localized step from a smooth group trend difference — and reads the data as the latter. Fifth, *time confounding*: within a benchmark, event time and calendar time are collinear; the release-year-FE specification leans entirely on cross-benchmark stagger. Sixth, *declared inputs and ceilings*: three hand-filled release dates and the held-out membership are declared, not estimated; saturation ceilings compress public post-release residuals *against* the hypothesis, and the logit and drop-saturated variants only partially derate this, so a true premium could be somewhat attenuated — but ceilings cannot manufacture the placebo-cutoff pattern.

The verdict is a suggestive-sign null. Every level specification is positive; the held-out post-release shortfall (-0.19 z) is exactly the shape leakage or test-task training [4] would leave; and the ability-proxy estimates show the premium is larger once the contaminated capability index is removed from the control set. But the premium is not significant under any cluster-honest test, collapses when held-out difficulty is controlled, and — decisively — is not localized at benchmark release. What these data support is at most a slow divergence of public-benchmark performance from held-out performance across model generations, consistent with gradual test-task adaptation [4] and with [13]'s finding that frontier models show little sharp overfitting; what they do not support is a detectable step premium that switches on when a benchmark's answers become available. Benchmark designers' held-out and rotating designs [6, 10, 3, 12] are the reason this test was possible at all; more held-out clusters would make it decisive.

Reproducibility. All estimates derive from the public Epoch AI tables plus three declared release dates; no dataset files are committed. The only stochastic step is the wild-cluster bootstrap (single generator, seed 0); randomization inference is exhaustive and `natex` runs declared seed 0. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates, the placebo grid, the jackknife range, and the panel composition against the numbers of record before drawing.

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A Break Made of Growth: The 2025Q1 Datacenter Surge as Smooth Exponential Expansion, Not a Stargate Regime Change

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July 2026

analysis pass of record: 2026-07-19, seeded (`--seed 0`), `natex` v0.2.0 at commit `cce6186`

Abstract

In Epoch AI's Frontier Data Centers panel (73 sites, 2023Q4–2026Q2), tracked quarterly IT-power additions rise roughly fivefold after 2025Q1 — the quarter of the Stargate announcement and DeepSeek-R1 — from a 296.8 MW/q pre-period mean to 1,516.9 MW/q, with complete pre/post separation (minimum post quarter 537 MW > maximum pre quarter 448 MW). We show this "Stargate-quarter industry break" is a correctly-identified artifact, and that attribution to "neocloud" operators is a clean null. On a placebo break-date grid (all 6 feasible dates), 2025Q1 is *never* the best break date — rank 3–6 of 6 on all three statistics, with 2025Q4 the argmax on two — and log tracked capacity grows near-linearly with mildly *decelerating* growth ($0.40 \to 0.33$ log/q); the log-level kink is negative at every candidate date. The levels "break" is what smooth ∼40%-per-quarter exponential growth looks like. A seeded `natex` SuDDDS scan does return $t_0 = 2025\text{Q1}$ *exactly* with scan $p = 0.010$, but discovery reads only the design columns (x, t, θ), never the outcome: the hit mechanically recovers the constructed 21-of-73-site treatment coding and contributes no outcome evidence — a role-assignment triage that corrects the scout pass's interpretation. Attribution fails directly: a two-way fixed-effects contrast of neocloud versus other sites gives $\tau = +0.098$ asinh-MW ($t = 0.699$) with a 199-draw label-permutation $p = 0.56$. The surge is instead broad-based and hyperscaler-led: the neocloud group carries 23.7% of post-2025Q1 additions, and the top site is the Anthropic–Amazon New Carlisle campus at 10.0%. All conclusions survive panel-end truncation and an incumbent-only variant, and the panel is provably invariant to the projection-snapshot filter.

1 Introduction

January 2025 was the most crowded month in the short history of AI infrastructure policy: the US diffusion rule was published on 13 January, DeepSeek-R1 was released on 20 January [3], and the $500-billion Stargate project was announced on 21 January [8]. A natural reading of subsequent buildout data is that 2025Q1 marks a regime change — the quarter the industry shifted from ordinary expansion to a land rush — and that the shift is concentrated in the "neocloud" operators (CoreWeave, xAI/Colossus, the Stargate sites, and peers) whose business is AI compute rather than general cloud. This paper tests both claims at the site level and rejects both, in an instructive way: the industry-wide "break" is real in levels but is not a break, and the neocloud attribution is a clean null.

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The buildout itself is well documented. Data centers are the engine rooms of AI: industrial facilities whose construction, power, and cooling determine where frontier compute can exist [9]. The leading AI supercomputers have doubled in performance roughly every nine months, with hardware cost and power needs doubling yearly [10], and the International Energy Agency projects global data-centre electricity demand to more than double by 2030, driven chiefly by AI [7]. What has been missing is site-level measurement: Epoch AI's Frontier Data Centers hub [4] now tracks individual campuses from satellite imagery, permits, and disclosures, making the timing question empirically addressable.

Methodologically, the exercise sits where two cautionary literatures meet. Breaks located in *time* lack the cross-sectional randomization logic of standard quasi-experimental designs, and naive inference on a single temporal cutoff is fragile [5]; when identification comes from one calendar event hitting one group, asymptotic cluster inference is unreliable and permutation-style inference over placebo assignments is the appropriate discipline [2]. We add a third, automation-specific caution: when a discovery algorithm is handed a panel whose treatment coding was *constructed* from the hypothesis under test, an exact recovery of the hypothesized date is confirmation of the coding, not of the outcome. The scout pass that preceded this study made exactly that misreading; the role-assignment triage below corrects it.

2 Data

The input is Epoch AI's Frontier Data Centers hub timelines [4]: 411 dated site snapshots covering 73 sites (2018-12-18 to 2030-01-01), extracted 2026-07-19. The 73 rows dated on or after 2026-07-25 are stated projections and are excluded; Section 4 shows this filter is provably a no-op for the analysis panel. Each site's "IT power (MW)" is carried forward (LOCF) as a step function evaluated at quarter ends 2023Q4 to 2026Q2, with a site at 0 MW before its first snapshot; quarterly *additions* are first differences with negative revisions floored at 0; the site-level outcome is asinh(additions/10). Time is indexed $t = 4\,\text{year} + (\text{quarter} - 1)$, so 2025Q1 = 8100. The neocloud group comprises 21 sites with hypothesized treatment onset 2025Q1: Colossus 1/2, seven CoreWeave sites, Fluidstack Lake Mariner, eight OpenAI Stargate sites, Oracle Batam, Start Campus Sines, and VNET Bayin Ulanqab. The known-treatment comparison panel codes Colossus 1 alone as treated. Baseline panels are bitwise identical to the scout pass's.

3 Design and Methods

Seeded discovery scans. Two `natex` [6] SuDDDS difference-in-differences scans (Bernoulli model, within-cluster-copula inference, $q = 99$ permutations, seed 0) search the panels for the best-supported treatment date t_0 and window W: one on the neocloud coding, one on the Colossus known-treatment coding. Crucially, SuDDDS discovery reads only the design columns — unit, time, and the treatment indicator θ — and never the outcome. Both panels *construct* $\theta = \text{group} \times \mathbf{1}[t \geq t_0]$, so an exact recovery of t_0 certifies that the constructed assignment is scannable, not that MW additions break there. Outcome evidence must come from the two designs below.

Aggregate break-date placebos. On total tracked additions we compute, at every feasible break date c (six dates with at least three quarters per side), three statistics: S1, the post−pre shift in mean log additions; S2, the slope change (kink) in the MW-level series at c; and S3, the kink in *log* capacity at c, i.e. growth acceleration. S2 and S3 are single-series analogues of the kink contrasts formalized in the difference-in-kinks literature [1] and implemented in `natex` [6]. If

2025Q1 is a genuine regime break it should be the argmax of the grid; its rank across placebo dates is the test, in the spirit of randomization inference over assignments [2], sized for a coarse grid (placebo-p floor 1/6).

Neocloud attribution. A two-way fixed-effects (TWFE) regression of asinh(additions/10) on neocloud $\times$ $\mathbf{1}[t \geq 2025\text{Q1}]$ with site and quarter fixed effects (73 sites $\times$ 11 quarters) estimates the neocloud-specific post-2025Q1 shift τ. Because the contrast is a single calendar event hitting one group, inference is by label permutation [2]: 199 draws (seed 0) reassigning the 21-site label among all 73 sites, plus a chip-protocol variant reassigning among the 52 controls. CR1 cluster-by-site standard errors are reported alongside and agree.

Decomposition and sensitivity. Post-2025Q1 additions are decomposed by site and group. Sensitivity variants truncate the panel end at 2026Q1 and 2025Q4, and an incumbent-only variant keeps the 59 sites first tracked on or before 2024-12-31 (12 treated) to separate real expansion from satellite-coverage entry [4, 5].

4 Results

The surge is real in levels. Tracked additions average 296.8 MW/q over 2023Q4–2024Q4 and 1,516.9 MW/q over 2025Q1–2026Q2 — a ratio of 5.11 — with complete separation: the smallest post-2025Q1 quarter (537 MW) exceeds the largest pre-quarter (448 MW). Nothing below disputes this; the question is whether it is a *localized break at 2025Q1*.

But 2025Q1 is never the best break date. On the six-date placebo grid (Table 1, Panel A), 2025Q1 is the argmax of none of the three statistics. The S1 log-additions shift at 2025Q1 is 1.488, rank 3 of 6 (placebo $p = 0.50$); the argmax is 2025Q4 at 1.665. The S2 level kink at 2025Q1 is 1,354 MW/q, dead last — rank 6 of 6 ($p = 1.00$), argmax 2025Q4 at 2,119. The S3 log-level kink is -0.100 at 2025Q1 and *negative at all six candidate dates*: quarterly log growth of tracked capacity decelerates from 0.404 to 0.327 log/q. A series growing smoothly at $\sim$40% per quarter mechanically makes every later quarter's additions exceed every earlier one's; the "break" is the exponential, not a regime change. The largest single jumps are 2025Q4 (+1,936 MW) and 2026Q2 (+3,610 MW). Across all sensitivity variants, 2025Q1 is the argmax in 1 of 12 variant-statistic cells (the 2025Q4-truncated S3, placebo $p = 0.25$ on a four-date grid).

The SuDDDS 2025Q1 hit is mechanical. The seeded neocloud scan returns max LLR = 24.114, $t_0 = 8100 = $ 2025Q1 *exactly*, $W = 3$, scan $p = 0.010$ (the $q = 99$ permutation floor), and passes composition validation. This is the scout pass's headline — and it is a recovery of the constructed treatment coding, not outcome evidence: discovery never reads the outcome (Section 3), and the scan's in-scan effect estimators refused in both runs (0 usable placebo units of the required 5), so the scans contribute no outcome estimate. The Colossus known-treatment scan illustrates the power ceiling with 1 treated site of 73: LLR = 2.537, $t_0 = $ 2024Q3, $W = 5$, scan $p = 0.230$, deterministic on re-run (the scout logged $p = 0.28$ from a marginally earlier code state with identical LLR/t_0/W).

Neocloud attribution is a clean null. The TWFE contrast gives $\tau = +0.0976$ asinh-MW (CR1 SE = 0.1396, $t = 0.699$). The 199-draw label permutation puts the observed τ well inside the null: $p = 0.560$ relabeling among all 73 sites, $p = 0.555$ under the chip protocol, with a permutation

Figure 1: (a) Tracked quarterly IT-power additions, stacked by group: the post-2025Q1 rise is roughly fivefold and occurs in *both* groups; the largest single quarters are 2025Q4 and 2026Q2, not 2025Q1. (b) Log total tracked capacity is near-linear ($\sim$0.4 log points per quarter) with mildly decelerating growth — the levels "break" is smooth exponential growth read in levels. (c) Placebo break-date grid for S1, the post−pre shift in mean log additions: 2025Q1 (vermilion) ranks 3 of 6; the argmax is 2025Q4. (d) Top-10 site contributions to post-2025Q1 additions: broad-based and hyperscaler-led, with the neocloud group (orange) carrying 23.7% overall.

τ SD of 0.150 — the estimate is $\sim$0.65 null SDs from zero. Truncating the panel at 2026Q1 or 2025Q4 leaves the null intact ($p = 0.605$, $p = 0.505$). The incumbent-only variant is larger but still not significant: $\tau = +0.258$, $t = 1.49$, permutation $p = 0.155$.

Broad-based and hyperscaler-led. Post-2025Q1 additions total 9,101 MW, of which the 21-site neocloud group carries 2,158 MW — 23.7%. Concentration is moderate: the top 5 sites carry 33.9% and the top 10 carry 49.1%. The leaders are the Anthropic–Amazon New Carlisle campus at 910 MW (10.0%), Microsoft Fairwater Atlanta (636), Meta Prometheus (631), Colossus 2 (490, neocloud), and OpenAI Stargate Abilene (421, neocloud). Inside the scan window 2025Q1–2025Q3 (2,245 MW) the largest contributors are New Carlisle (398 MW, 17.7%) and Amazon Madison (284 MW, 12.7%) — hyperscaler campuses, not neoclouds.

Sensitivity and the projection filter. Excluding stated projections is provably a no-op: all 73 projection rows are dated on or after 2026-07-25, later than the last quarter-end 2026-06-30, and every site has a real snapshot on or before 2026-06-30, so the 2023Q4–2026Q2 panel is bitwise invariant to any snapshot-filter date from 2026-06-30 on (including no filter). The incumbent variant

Table 1: Headline estimates. Panel A: aggregate break-date placebo grid (six feasible dates, $\geq$ 3 quarters per side); the test is 2025Q1's rank. Panel B: seeded `natex` SuDDDS scans (Bernoulli/WCC, $q = 99$, seed 0). Panel C: TWFE neocloud attribution with 199-draw label-permutation inference (seed 0).

Quantity	Estimate	Rank / t	p	notes
Panel A: is 2025Q1 the best break date? (placebo grid, $n = 6$ dates)				
S1 shift in mean log additions	1.488	3/6	0.50	argmax 2025Q4 (1.665)
S2 level kink (MW/q)	1,354	6/6	1.00	argmax 2025Q4 (2,119)
S3 log-level kink (log/q)	−0.100	4/6	0.67	negative at all 6 dates
post/pre mean additions ratio	5.11	—	—	min post 537 > max pre 448
log growth, pre → post	0.404 → 0.327	—	—	deceleration
Panel B: SuDDDS scans (discovery reads (x, t, θ) only — never the outcome)				
neocloud scan, max LLR	24.114	$t_0 = $ 2025Q1, $W = 3$	0.010	recovers the coding
Colossus scan, max LLR	2.537	$t_0 = $ 2024Q3, $W = 5$	0.230	1 treated of 73
Panel C: neocloud attribution, TWFE τ (asinh-MW), site+quarter FE				
baseline (21/73 treated)	+0.098	$t = 0.699$	0.560	τ SD 0.150; chip $p = 0.555$
truncate 2026Q1 / 2025Q4	0.088/0.112	—	0.605/0.505	conclusions unchanged
incumbent-only (12/59 treated)	+0.258	$t = 1.49$	0.155	larger, still n.s.

Notes: Panel A placebo p is the rank of 2025Q1 divided by the grid size (floor 1/6). Panel B scan p-values are $q = 99$ within-cluster-copula permutation floors; the neocloud scan's exact t_0 recovery certifies the constructed 21/73-site assignment is scannable, not that the outcome breaks there (in-scan effect estimators refused: 0 usable placebos of ≥ 5 required). Panel C p-values are 199-draw label permutations, seed 0; CR1 cluster-by-site t-statistics agree with the permutation verdicts.

shows the surge is not satellite-coverage entry: the 59 sites first tracked by 2024-12-31 carry 3,074 of the 3,610 MW of 2026Q2 additions (∼85%).

5 Caveats and Conclusion

Six caveats bound the claims. First, *LOCF timing*: snapshots are sparse (median 4 per site), so carried-forward power smears true addition dates by up to one inter-snapshot gap; quarter-level timing is approximate, and non-monotone estimate revisions are floored at 0. Second, *tracked-universe growth*: the hub's coverage grows from 4 to 54 sites with nonzero power over the panel, so aggregate levels lower-bound industry capacity and early growth rates are inflated by coverage catch-up — a bias *toward* finding late acceleration, which strengthens the no-2025Q1-break reading, though absolute MW shares remain conditional on Epoch's tracking choices [4]. Third, *coarse placebo grids*: with 4–6 feasible dates the placebo p has a floor of 1/6–1/4; statements are about rank, not tight p-values. Fourth, a *crowded quarter*: Stargate (21 January), DeepSeek-R1 (20 January), and the diffusion rule (13 January) all land in 2025Q1 [8, 3], so even a real 2025Q1 break could not separate these candidate causes. Fifth, *calendar-time inference*: the neocloud contrast is one calendar event hitting one group; we therefore rest on permutation-in-space inference [2], and the asymptotic CR1 statistics happen to agree. Sixth, *scale*: asinh(additions/10) compresses large sites, and τ lives on that scale.

The conclusion is a double negative with positive content. The post-2025Q1 surge in tracked datacenter additions is real, large, and completely separated in levels — but it is what smooth ∼40%-per-quarter exponential growth of tracked capacity looks like, with 2025Q1 never the best break date on any statistic, growth mildly decelerating, and the largest jumps arriving in 2025Q4 and 2026Q2. The one seemingly sharp result — a seeded discovery scan returning 2025Q1 exactly at $p = 0.010$ — is a mechanical recovery of the constructed treatment coding, a failure mode worth naming for any pipeline that scans panels whose design columns encode the hypothesis. And the group story fails directly: neocloud attribution is a clean permutation null, with the buildout led by hyperscaler campuses — first among them the Anthropic–Amazon New Carlisle site at 10% of all post-2025Q1 additions. The Stargate quarter did not change the growth regime; the regime was already exponential [10, 7].

Reproducibility. All estimates derive from the public Epoch AI Frontier Data Centers hub timelines [4]; no dataset files are committed. The only stochastic steps are the seeded scan and permutation draws (seed 0 throughout). Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Manufactured Gradients: State AI Exposure and US Business AI Adoption at DeepSeek-R1

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July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

US states more exposed to AI — measured by the Anthropic Economic Index (AEI) usage-per-capita index — did not detectably accelerate business AI adoption relative to less-exposed states at the release of DeepSeek-R1 on 20 January 2025. In Census Bureau Business Trends and Outlook Survey (BTOS) state panels (2,727 state-wave cells, 51 states, 71 biweekly waves, 2023–2026), the change in each state's adoption-trend slope at R1 rises by +1.635 pp/yr per AEI index unit (HC1 se 1.458) — reproducing the scout estimate to the third decimal — but is statistically null: exposure-permutation $p = 0.257$. The same statistic at non-event placebo cutoffs brackets the headline (+1.56 at $t = 2024.30$, +2.24 at 2024.50), and a placebo *outcome* — remote work, whose federal return-to-office executive order was signed the same day as R1 — matches it (+1.97), with the weighted placebo version itself nominally significant ($p = 0.045$): the design manufactures gradients. Every nominally significant estimate in the battery traces to a known mechanism. Long-window negative gradients (-4.31 at $w = 1.0$) vanish under old-wording-only data — a splice artifact, because the nationally calibrated 1.553 rewording ratio misprices high-adoption, high-AEI states (corr +0.676). The two-way fixed-effects level effect (+0.527 pp, permutation $p = 0.011$) is a pre-trend: the high-minus-low gap was already growing at +0.600 pp/yr before R1, and state-specific trends flip the estimate to -0.38. The scan's localization at the first post-R1 wave is a mechanical rediscovery of the constructed treatment column. There is no credible evidence that high-AEI-exposure states accelerated AI adoption differentially at DeepSeek-R1: the BTOS state Bartik-style design, excluded from an earlier screen for want of exposure weights, stays dead even with real ones.

1 Introduction

The release of the DeepSeek-R1 reasoning model on 20 January 2025 [7] is the natural candidate event for a break in US business AI adoption: a companion sector-level analysis [10] of the Census Bureau's Business Trends and Outlook Survey (BTOS) found a sharp acceleration in AI-exposed sectors at exactly the first post-R1 survey wave, but attribution to R1 failed a placebo battery. This paper asks the cross-sectional version of the same question: did *states* whose workforces use AI more intensively accelerate measured business AI adoption differentially at R1? The design — state-level BTOS adoption interacted with an external AI-exposure measure — was excluded from an earlier automated screen for want of exposure weights; here we revive it with real ones.

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Both sides of the design rest on new measurement. The outcome is the BTOS AI use rate, a biweekly probability-sample estimate of the share of US firms using AI, introduced and validated by [3]. The exposure is the Anthropic Economic Index (AEI), which maps Claude conversations to economic tasks [8]; its geographic report [1] publishes a usage-per-capita index by US state, which we take as the cross-state exposure measure. Interacting differential exposure with a common shock is the logic that shift-share (Bartik) designs formalize [6], and its known failure modes discipline the analysis: exposure here is measured *after* the outcome window, so every estimate below is a descriptive gradient, never a causal Bartik coefficient; pre-existing differential trends can masquerade as treatment effects [13]; and slope changes at a single date in smooth aggregate series are spuriously significant often enough that placebo distributions at non-event dates, in the spirit of [5], are the appropriate yardstick. The per-state slope-change contrast is a kink-style estimand in the sense of the difference-in-kinks design [2], estimated and falsified with the `natex` toolkit [12], whose scan-based DiD [9] supplies a localization check. The battery converts one reproduced headline gradient and three nominally significant side results into a clean null plus three identified artifacts.

2 Data

The outcome panel combines three BTOS sources. The state sheets of the AI core workbook (DRB approval CBDRB-FY25-0425) give biweekly estimates and standard errors of the share answering "Yes" to Question 7 ("In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services?"), waves 202319–202520. A fresh census.gov download extends the panel under the December-2025 rewording ("…in any of its business activities?"), waves 202524–202614; new-wording values are divided by 1.553, the national new/old measurement-RDD ratio estimated at the rewording cutoff by a companion paper [11], with an additive -6.04 pp variant as a diagnostic. The archived remote-work Question 6 (share of businesses with paid employees working from home at least one workday), waves 202417–202514, supplies a placebo outcome (1,221 state-wave cells). The assembled panel has 2,727 state-wave cells across 51 states (50 plus DC) and 71 waves, $t = 2023.710$–2026.507 in fractional years of collection-period midpoints; 24.7% of the state-by-wave grid is disclosure-suppressed, concentrated in small states.

Exposure is the AEI `usage_per_capita_index` per state [1], averaged over the 2026-04-01..05-01 and 2026-05-01..06-01 data windows (verified: CA $= (1.56 + 1.62)/2 = 1.59$); it ranges from 3.52 (DC) to 0.245 (WV). Two features bind the interpretation. First, the exposure window (April–June 2026) lies entirely *after* the outcome window, so exposure is endogenous by construction and no estimate below is causal, whatever its p-value. Second, the index measures Claude usage specifically, proxying all-AI exposure with attenuation and tech-hub confounding both plausible.

3 Design and Methods

Exposure gradient at a known cutoff. For each state s, local OLS slopes of the adoption share in calendar time are fit on each side of $t_0 = 2025.055$ (DeepSeek-R1; first post wave 202503) within $\pm w$ years, and the slope change $\Delta\kappa_s$ is the post-minus-pre difference — the right-minus-left kink convention of [2], with the cross-state gradient replacing the group contrast. The headline statistic is the cross-state regression of $\Delta\kappa_s$ on AEI exposure, b in pp/yr per index unit. The spec of record is $w = 0.5$, unweighted, restricted to states with at least 6 usable waves per side ($n = 36$; DC never enters — it is suppressed below the wave minimum). Inference is primary by exposure permutation (AEI values reassigned across states, $B = 10{,}000$, `default_rng(0)`), with

HC1 standard errors reported alongside [4]; windows $w \in \{0.5, 0.75, 1.0, 1.4\}$ are examined, and an SE-weighted variant fits $1/\mathrm{se}^2$ WLS slopes from the published State SE sheets and combines them by inverse-variance meta-analysis.

Falsification battery. (i) placebo cutoffs at the non-event dates 2024.30 and 2024.50, at o1 (2024.6967, a real secondary event), at 2025.30, and at 2025.555 (a window that straddles the rewording splice); (ii) the placebo outcome (remote work) at the R1 cutoff — noting that the federal return-to-office executive order was signed on 2025-01-20, the *same day* as R1, so remote-work breaks at that date are expected for non-AI reasons; (iii) leverage checks dropping DC and CA, and a Spearman rank gradient; (iv) splice diagnostics rerunning the long-window gradients under the multiplicative (/1.553) splice, the additive (−6.04 pp) splice, and old-wording-only data ($t < 2025.8$).

Panel estimators. A tercile two-way fixed-effects (TWFE) level DiD interacts the top AEI tercile (17 states, cut 1.002) with post-R1, with wave and state fixed effects, CR1 standard errors by state [4], a placebo-in-space permutation of the treated set ($B = 2{,}000$), and a state-specific linear-trends variant. The `natex` survey pipeline's scan-based DiD (SuDDDS) [9] searches break dates and windows on the same panel (seed 0, normal model, AR(1)-by-unit permutation null) as a localization and role-assignment check.

Honest-inference notes, stated as run. HC1/CR1 intervals on smooth biweekly aggregates are potentially oversized, so permutation p is the primary inference throughout. The `natex` refusals are reported, not patched over: the SuDDDS anticipation test refused (Holm p all NaN), and all three effect estimators (dd, synthetic control, GESS) refused with "only 0 usable placebos; ≥ 5 required" on this unbalanced panel — the manual TWFE and permutation estimates below are the effect numbers of record.

4 Results

The scout number reproduces exactly, and it is null. At the spec of record ($w = 0.5$, unweighted, 36 states), the R1 slope-change gradient is $b = +1.635$ pp/yr per AEI index unit (HC1 se 1.458; classical se 1.428, matching the scout's "se 1.43, t 1.15"; $t_{\mathrm{HC1}} = 1.12$; correlation 0.193), with exposure-permutation $p = 0.257$ (Table 1, panel A; Figure 1a). The SE-weighted version is +1.877 (HC1 se 1.922, $p_{\mathrm{perm}} = 0.360$); the Spearman rank gradient is $\rho = 0.098$ ($p = 0.568$); dropping DC and CA gives +1.54 ($p = 0.326$) unweighted and +0.53 ($p = 0.804$) weighted. DC — the highest-exposure unit at AEI 3.52 — never enters the gradient sample in the first place, so no single unit drives the (non-)result.

Placebos bracket the headline. The identical statistic at non-event cutoffs returns +1.56 ($p = 0.277$) at $t = 2024.30$ and +2.24 ($p = 0.162$) at $t = 2024.50$ — magnitudes that *bracket* the R1 estimate — and −0.23 ($p = 0.889$) at o1 (Table 1, panel B; Figure 1b). The remote-work placebo outcome at R1 is +1.97 ($p = 0.288$), the same magnitude as the headline; its SE-weighted version is +4.39 (HC1 se 1.895) with permutation $p = 0.045$ — a nominally significant *placebo.* A design that produces R1-sized gradients at dates where nothing happened, and a significant gradient for an outcome whose own policy shock (the federal return-to-office executive order, signed 2025-01-20) merely shares R1's date, manufactures gradients; the null at R1 carries no information beyond that.

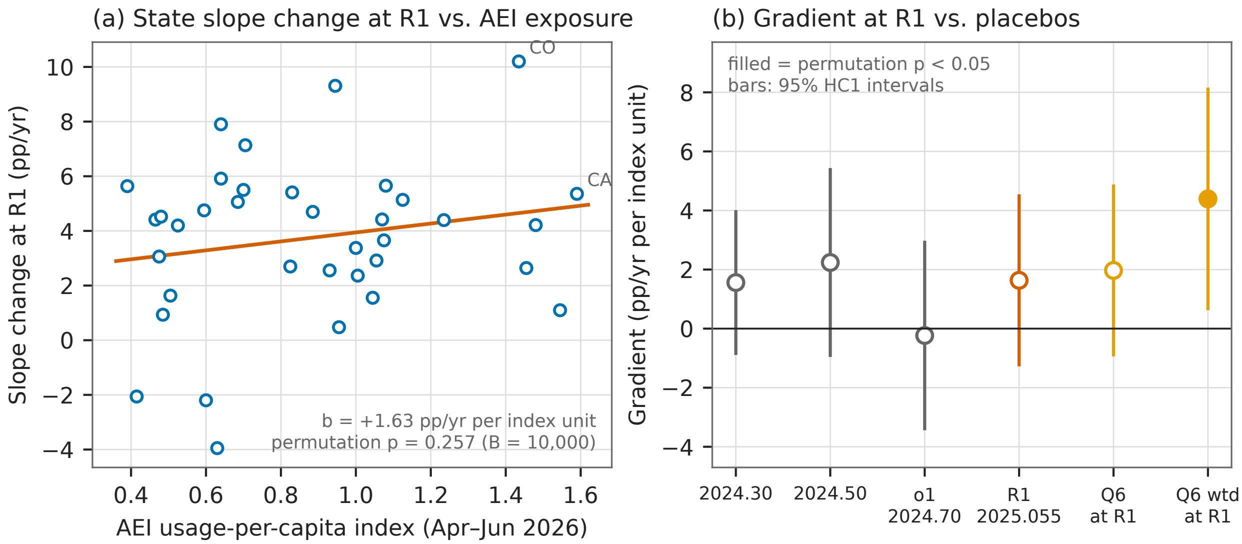

Figure 1: (a) State-level slope change in BTOS AI use at DeepSeek-R1 ($w = \pm 0.5$ yr, $n = 36$ states with ≥ 6 waves per side) against AEI usage-per-capita exposure, with the unweighted OLS fit: $b = +1.635$ pp/yr per index unit, exposure-permutation $p = 0.257$. (b) The same gradient statistic at non-event placebo cutoffs, at o1, at R1, and for the remote-work placebo outcome at R1, with 95% HC1 intervals; filled markers are nominally significant by permutation at the 5% level. The only filled marker in the battery is a *placebo* — the SE-weighted remote-work gradient at R1 (+4.39, $p = 0.045$).

Artifact 1: the long-window sign flip is the splice. Once the window reaches across the December-2025 rewording, the gradient turns significantly negative: -4.31 ($p = 0.011$) at $w = 1.0$ and -2.53 ($p = 0.011$) at $w = 1.4$ (Table 1, panel C). This is a measurement artifact, identified three ways. AEI exposure correlates +0.676 with the pre-R1 adoption level, so the nationally calibrated multiplicative ratio 1.553 misprices high-level (equivalently high-AEI) states; the additive -6.04 pp splice — wrong in the opposite direction — attenuates the same estimates to -2.92 and -1.07; and old-wording-only data collapse both to null (-0.87, $p = 0.279$ at $w = 1.0$; +0.44, $p = 0.513$ at $w = 1.4$). A placebo cutoff at 2025.555, placed so the window straddles the splice with no event nearby, yields -9.16 ($p = 0.00025$) multiplicative and still -5.81 ($p = 0.015$) additive: *neither* national correction is valid cross-state, so state-level gradients through the rewording — including any June-2026 extension of this design — are unidentified.

Artifact 2: the TWFE level effect is a pre-trend. The tercile TWFE DiD gives $\hat{\tau} = +0.527$ pp (CR1 se 0.210) with placebo-in-space permutation $p = 0.011$ (+0.547, $p = 0.038$ on the full panel) — nominally the strongest result in the study (Table 1, panel D). But the high-minus-low AEI gap was already growing at +0.600 pp/yr before R1 (HC1 se 0.180, 36 pre-R1 waves), which mechanically predicts roughly +0.3 pp of the estimate over the post window, and adding state-specific linear trends flips the $w = 0.5$ estimate to -0.38 (se 0.31). This is the textbook pre-trend failure [13]: the level DiD is descriptive only, and the trend-robust estimand — the slope-change gradient — is exactly the null of panel A.

Artifact 3: the scan localization is treatment rediscovery. The SuDDDS scan (seed 0, both panels) maximizes at LLR 154.5 for the empty subset — *all* states — at $t_0 = 2025.0877$,

Table 1: Exposure gradients, placebos, and panel estimates. Gradient rows: slope-change gradient b (pp/yr per AEI index unit) with HC1 se and exposure-permutation p ($B = 10{,}000$; splice diagnostics $B = 4{,}000$). TWFE rows: level effect (pp) with CR1 se by state and placebo-in-space permutation p ($B = 2{,}000$).

Specification	Estimate	se	p_{perm}	n
Panel A: AEI gradient at DeepSeek-R1 ($w = 0.5$ yr)				
unweighted (spec of record)	+1.635	1.458	0.257	36
SE-weighted (WLS + IV meta)	+1.877	1.922	0.360	36
unweighted, drop DC+CA	+1.544	1.651	0.326	35
SE-weighted, drop DC+CA	+0.531	2.316	0.804	35
Panel B: placebo cutoffs and placebo outcome ($w = 0.5$, unweighted)				
placebo cutoff $t_0 = 2024.30$	+1.560	1.224	0.277	33
placebo cutoff $t_0 = 2024.50$	+2.238	1.604	0.162	34
o1 cutoff $t_0 = 2024.6967$	−0.233	1.610	0.889	33
remote work (Q6) at R1	+1.970	1.458	0.288	51
remote work (Q6) at R1, SE-weighted	+4.392	1.895	0.045*	51
Panel C: long windows across the 2025-12 rewording (unweighted)				
$w = 1.0$, spliced (/1.553)	−4.305	1.676	0.011*	39
$w = 1.0$, additive splice (−6.04 pp)	−2.922	1.885	0.020*	39
$w = 1.0$, old wording only	−0.873	1.000	0.279	39
$w = 1.4$, spliced	−2.535	0.945	0.011*	40
$w = 1.4$, old wording only	+0.439	0.645	0.513	40
splice-window placebo $t_0 = 2025.555$, spliced	−9.161	2.150	0.00025*	39
splice-window placebo, additive	−5.806	2.639	0.015*	39
Panel D: tercile TWFE level DiD (pp)				
TWFE, $w = 0.5$	+0.527	0.210	0.011*	964
TWFE, full panel	+0.547	0.236	0.038*	2727
TWFE + state trends, $w = 0.5$	−0.382	0.315	—	964
pre-R1 high-minus-low gap slope (pp/yr)	+0.600	0.180	—	36 waves

Notes: * nominally significant at the 5% level by permutation. The $w = 1.4$ additive-splice gradient is −1.069 ($p = 0.095$). Panel D's TWFE p-values are placebo-in-space; the state-trends row and the pre-trend slope row report CR1 and HC1 standard errors respectively, with no permutation test run. Every nominally significant row is attributed to an identified artifact in the text: splice mispricing (panel C), a pre-existing differential trend (panel D), or a same-day policy shock to the placebo outcome (panel B).

the first post-R1 wave, with scan $p = 0.010$ under the AR(1)-by-unit null. Because the declared treatment column is the constructed `high_aei`×`post` indicator, a scan that lights up for all states at the post boundary is a mechanical rediscovery of that column, not outcome evidence — the same role-assignment failure documented in the companion sector study [10]. The composition check passes ($p = 0.707$); the anticipation test and all three effect estimators refused, as stated in Section 3.

5 Caveats and Conclusion

Four caveats bound the claims. First, *exposure is post-outcome*: the AEI index is measured April–June 2026, after every outcome wave used at the R1 cutoff, so even a robust gradient would have been descriptive, not Bartik-causal — the null is about this gradient design, not proof of zero differential adoption. Second, *exposure is Claude-specific*: as a proxy for all-AI exposure it is attenuated and plausibly confounded with tech-hub geography. Third, *coverage*: 24.7% of the state-wave grid is disclosure-suppressed, concentrated in small states; estimation samples run 33–40 states, and DC — the most exposed unit — is never estimable at the primary window. Fourth, *the calendar*: R1's release week was inauguration week, sharing its date with the federal return-to-office executive order (2025-01-20) and adjoining the Stargate announcement (2025-01-21), the AI-diffusion export rule (2025-01-13), and a BTOS sample-year rollover, so even a surviving gradient could not have been attributed to R1 alone.

The conclusion is a disciplined null. The headline gradient reproduces exactly and is statistically indistinguishable from the same statistic at dates where nothing happened; the one nominally significant gradient in the $w = 0.5$ battery belongs to a placebo outcome with its own same-day shock; and all three seemingly significant side results — the long-window negative gradients, the TWFE level effect, and the scan localization — dissolve under, respectively, old-wording-only data, state trends, and role-assignment scrutiny. These data provide no credible evidence that high-AEI-exposure states accelerated business AI adoption differentially at DeepSeek-R1, and the state-level splice problem implies that extending the design through the rewording cannot rescue it.

Reproducibility. All estimates were produced with `natex` v0.2.0 [12] at seed 0 from the frozen BTOS workbooks (published at `https://www.census.gov/hfp/btos`) and the public AEI state release [1]; no dataset files are committed. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline gradient and the significant-placebo numbers against the numbers of record before drawing.

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No Kink at R1: The CVE-Publication Surge Localizes to Late 2025, Not the DeepSeek-R1 Release

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analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

Monthly CVE publications did not bend at DeepSeek-R1. On 90 months of National Vulnerability Database publication counts (January 2019 – June 2026), a sharp regression kink design in calendar time at the R1 release (20 January 2025) estimates a slope change in ln monthly publications of +0.038 ln-points/yr (HC1 se 0.128, $z = 0.30$, $p = 0.764$, 95% CI $[-0.213, +0.289]$; 18 pre / 17 post months, bandwidth 1.5 years, triangular kernel). A scout-flagged +0.18 ln/yr break — identical to the uniform-kernel cell (+0.184, se 0.104, $p = 0.077$; Newey–West $p = 0.129$ at 6 lags) — is correctly identified as an artifact of action far from the cutoff: the estimate grows with bandwidth and vanishes under triangular weighting; a 49-cutoff placebo grid shows HC1 is oversized on this smooth serially correlated series (empirical size 0.14 at nominal 0.05) and puts the placebo-calibrated p at 0.80 (0.76 on the 37-cutoff clean subgrid), with 38 of 49 placebos exceeding the observed |kink| and the largest (+0.385, $z = 3.5$) at November 2020, nowhere near any AI release; a local quadratic flips the contrast negative (−0.90, $p = 0.015$ at bandwidth 2.0, uniform); and sub-window slopes place the first nine post-R1 months at +0.103 ln/yr — *shallower* than the +0.280 pre-trend — with all steepening from November–December 2025 onward (+1.19 ln/yr through June 2026). Blind LoRD3 localization (seed 0, $k \in \{12, 24, 36\}$, discovery reads only the series and time) independently ranks November/December 2025 (scan $p = 0.02/0.01$ at $k = 24/36$) and February–April 2026 first; R1's month appears in no top-20 at any k. An o1-preview contrast is a clean null (−0.035, $p = 0.844$). The post-2025 acceleration is real — 3,116 publications in November 2025, 5,643 in December, 8,001 by June 2026 — but it arrives 10–17 months after the release, consistent with NVD operational disruption and backlog catch-up on top of CNA-program expansion. On this series, attribution of the surge to AI-assisted vulnerability discovery is unsupported; the result stands as a calibrated null plus an artifact-identification case study, and reserve-date robustness via the `cvelistV5` history is required before any attribution claim.

1 Introduction

DeepSeek released R1, the first open-weights frontier reasoning model, on 20 January 2025 [4]. The release is a natural candidate for a regime change in the economics of vulnerability discovery: contemporaneous work showed that LLM agents can autonomously exploit real one-day vulnerabilities from their CVE descriptions [5], and an open-weights reasoning model removes the API chokepoint that had kept such capabilities meterable. If AI assistance materially accelerated vulnerability

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discovery *at scale*, the aggregate flow of published CVEs should steepen — kink — at or shortly after the release. This paper tests that hypothesis directly.

The series demands discipline before any such reading. CVE publication counts are heavily process-driven: annual volumes are forecastable to within roughly 8% a year ahead from disclosure-pipeline variables alone [9]; the publication dates and other fields of the National Vulnerability Database have documented quality problems [1]; and NVD processing itself broke and recovered inside the sample window — the February 2024 enrichment slowdown and subsequent backlog, followed by operational changes that NIST announced in April 2026 in response to record CVE growth [11]. Publication-date slopes therefore conflate discovery with processing, and any slope break must be localized before it is attributed.

The estimand — a change in trend slope at a known date — is the regression kink design of [3], applied with calendar time as the running variable; [2] formalize the closely related difference-in-kinks design that the same toolkit implements. Two methodological strands discipline the exercise. Regression discontinuity and kink designs in *time* lack the randomization logic of cross-sectional designs: serial dependence and smooth trends generate spurious breaks [7]. And kink estimates on smooth series are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [6]. Both concerns are operationalized below — a 49-cutoff placebo grid whose calibrated p-value is the inference of record, curvature checks, HAC variants, and a blind localization scan — estimated with the `natex` toolkit [8].

The verdict is a clean, well-powered null at the cutoff, plus a correctly identified artifact: the post-2025 acceleration in CVE publications is real, but every localization tool places it in late 2025 and 2026, ten to seventeen months after R1, with the first nine post-release months *shallower* than the pre-trend.

2 Data

The outcome series is monthly counts of CVEs by NVD publication date, obtained from the NVD API 2.0 [10] as `totalResults` per month-long `pubStartDate`/`pubEndDate` window (fetched July 2026): 90 months, January 2019 through June 2026, with no missing months. The running variable is mid-month decimal time $t = \text{year} + (\text{month} - 0.5)/12$, and the outcome is ln of the monthly count. The R1 cutoff is $t = 2025.055$ (20 January 2025); the o1-preview contrast cutoff is $t = 2024.70$ (12 September 2024). Levels anchor the story told below: November 2025 has 3,116 publications — the series' anomalous dip — followed by 5,643 in December 2025 and 8,001 by June 2026. The series records the NVD *published* date, not the CVE reserve date; the distinction carries the paper's principal caveat (Section 5).

3 Design and Methods

Sharp RKD in calendar time. The estimand is the change in the slope of ln monthly publications, in ln-points per year, at the cutoff, estimated by local linear regression on each side within the bandwidth (`natex` v0.2.0 [8], `natex kink` with a unit policy-kink change; [3, 2]). The headline specification uses bandwidth 1.5 years, triangular kernel, degree 1, donut 0 ($n = 35$; 18 left, 17 right). A 48-cell grid varies cutoff $\in$ {R1, o1} $\times$ bandwidth $\in \{1.0, 1.5, 2.0\}$ years $\times$ kernel $\in$ {triangular, uniform} $\times$ donut $\in \{0, 1\}$ months $\times$ degree $\in \{1, 2\}$. The uniform bandwidth-1.5 cell reproduces the wave-2 scout's confirmed side-OLS slope delta (+0.280 $\rightarrow$ +0.464 ln/yr) exactly, and a fully interacted local WLS mirror of `natex` matches its uniform kink to 10^{-8}. The design is deterministic; seed 0 is declared and no stochastic step exists in estimation.

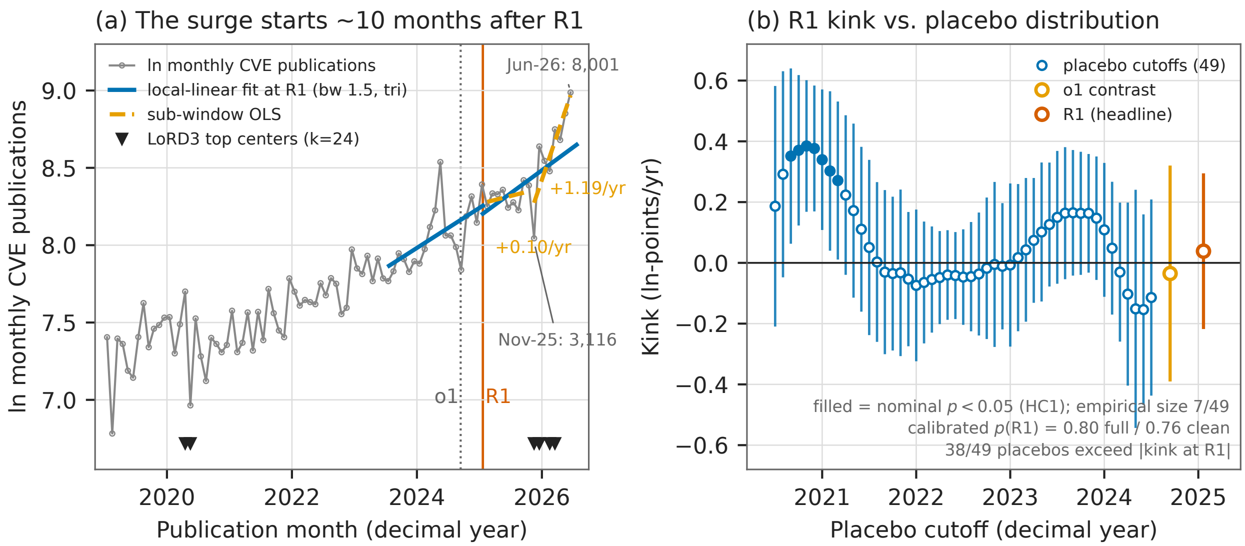

Figure 1: (a) ln monthly CVE publications (gray), 2019-01 – 2026-06, with per-side local-linear fits at the R1 cutoff (blue, bandwidth 1.5 years, triangular kernel): the slope change is +0.038 ln/yr ($p = 0.764$). Orange dashed segments are sub-window OLS fits: the first nine post-R1 months run at +0.103 ln/yr, *shallower* than the +0.280 pre-trend; November 2025 – June 2026 runs at +1.19 ln/yr. Markers along the bottom are the top blind LoRD3 discovery centers ($k = 24$): November/December 2025 and February–April 2026, plus the April/May 2020 COVID dip — never R1. The November 2025 dip (3,116) and December spike (5,643) are labeled. (b) The same specification re-estimated at 49 monthly placebo cutoffs (2020.5–2024.5), with 95% HC1 intervals; filled markers are nominally significant at the 5% level (7 of 49 — empirical size 0.14). The R1 estimate (vermillion) sits well inside the placebo distribution: 38 of 49 placebos exceed it in magnitude and the placebo-calibrated p is 0.80; the largest placebo (+0.385, $z = 3.5$, November 2020) is nowhere near any AI release. The o1 contrast cell (orange) is a clean null.

Inference on a smooth serially correlated aggregate. HC1 treats months as independent; on this series it is demonstrably oversized (residual lag-1 autocorrelation 0.23 in the uniform cell), so three guards are run. *(i) HAC:* Newey–West standard errors (Bartlett, lags 3/6/12) on the fully interacted local fit, with and without month-of-year dummies. *(ii) Placebo calibration*, in the spirit of [6]: the headline specification re-estimated at 49 monthly placebo cutoffs (2020.5–2024.5), reporting the empirical size at nominal 0.05 and the placebo-calibrated $p = (1 + \#\{|z_{\text{placebo}}| \geq |z_{\text{R1}}|\})/(1 + 49)$; a clean 37-cutoff subgrid keeps only placebo windows that end before R1. The calibrated p is the inference of record. *(iii) Curvature:* degree-2 refits test whether any kink survives allowing a smooth quadratic [7]. *Blind localization:* `natex discover` (LoRD3, seed 0, $k \in \{12, 24, 36\}$) scans the series with discovery reading only $(t, \ln n)$ — never the cutoff — and ranks candidate break locations; if the R1 date does not surface, the case for an R1-localized break must come from priors, not the data.

4 Results

Null at the cutoff, with power. The headline kink at R1 (Table 1, Panel A) is +0.038 ln-points/yr (HC1 se 0.128, $z = 0.30$, $p = 0.764$, 95% CI $[-0.213, +0.289]$): the local trend in ln publications does not change measurably at the release. The o1-preview contrast cell — same specification at $t = 2024.70$ — is likewise null (-0.035, se 0.179, $z = -0.20$, $p = 0.844$), matching the scout's confirmed side-OLS delta of -0.028. The null is not for lack of power: the same specification moved across 49 placebo cutoffs rejects nominally at 7 of 49 and detects a $+0.385$ break ($z = 3.5$) at November 2020, so the instrument registers breaks of the size at issue when they exist.

The uniform $+0.18$ **is an edge artifact, and triage identifies it as one.** The uniform-kernel cell — numerically identical to the scout's confirmed side-OLS delta ($+0.280 \to +0.464$) — gives $+0.184$ (se 0.104, $p = 0.077$), and is not significant under HAC (Newey–West $p = 0.129$ at 6 lags, 0.077 at 12). Four facts disqualify an R1-localized reading. First, the kernel/bandwidth pattern: the estimate vanishes under triangular weighting (which concentrates mass near the cutoff) and *grows* with bandwidth under uniform weighting ($+0.068$ at bandwidth 1.0, $+0.184$ at 1.5, $+0.217$ at 2.0) — the signature of action far from the cutoff, not at it. Second, placebo calibration: HC1's empirical size on this series is 0.14 at nominal 0.05 (7/49; the oversizing catalogued as `natex` issue #51), 38 of 49 placebos exceed the observed |kink|, and the calibrated p is 0.80 on the full grid and 0.76 on the clean 37-cutoff subgrid. Third, curvature: degree-2 refits flip the R1 contrast *negative* (-0.77 at bandwidth 1.5 triangular; -0.90, $p = 0.015$, at bandwidth 2.0 uniform) — a local quadratic absorbs the "kink" entirely. Fourth, the nominally significant stragglers (uniform-with-donut cells at $p = 0.011 - 0.033$; seasonal Newey–West uniform cells at $p = 0.001 - 0.033$, the latter spending 15 parameters on 35 observations) all inherit the same edge-driven uniform estimate and are overridden by the calibration and the localization evidence.

Where the action actually is. Sub-window OLS slopes decompose the post-period: the pre-R1 window runs at $+0.280$ ln/yr; the first nine post-R1 months (February–October 2025) run at $+0.103$ — *shallower* than the pre-trend; November 2025 – June 2026 runs at $+1.19$ ln/yr. All steepening starts roughly ten months after the release. Year-over-year log growth is ≈ 0.30 both in the twelve months before R1 and in the last observed twelve months — the surge restores the trend as much as it breaks it. Blind LoRD3 localization agrees: at $k = 24$ the top discovery centers are November/December 2025 (scan $p = 0.02$), then February–April 2026; at $k = 36$ the same centers lead (scan $p = 0.01$); at $k = 12$ the late-2025/2026 cluster leads with the April/May 2020 COVID-era dip next; R1's month appears in no top-20 at any k. The dip-then-surge shape — November 2025 at 3,116 publications (fit residual -0.45, the series' anomaly), December at 5,643, June 2026 at 8,001 — is consistent with an NVD operational disruption followed by backlog catch-up (the October–November 2025 federal shutdown is the obvious candidate) on top of continuing CNA-program expansion [11]; that attribution is a hypothesis consistent with the blind localization, not a result verified in this run.

5 Caveats and Conclusion

Five caveats bound the claims. First, and principally: this is an NVD *published-date* series. Backlog clearance mechanically shifts counts late — the February 2024 NVD crisis and the 2025–26 catch-up are both in-sample — so publication-date slopes conflate discovery and processing; reserve-date robustness via the `cvelistV5` git history (`dateReserved` by CNA, with the Linux-kernel CNA split out) is mandatory before any attribution claim, and this run did not perform it. Second, the attribution of the November 2025 dip and December 2025 spike to the October–November 2025 federal shutdown plus NVD catch-up is a hypothesis consistent with the blind-scan localization, not verified in-run. Third, HC1 — and every nominal p-value here — is oversized on this smooth serially

Table 1: Sharp RKD in calendar time on ln monthly CVE publications. Headline specification: cutoff R1 ($t = 2025.055$), bandwidth 1.5 years, triangular kernel, degree 1, donut 0, HC1.

Cell	Kernel/spec	Kink	se	z	p
Panel A: headline cells (ln-points/yr; bandwidth 1.5, degree 1)					
R1 ($t = 2025.055$), headline	triangular	+0.038	0.128	0.30	0.764
R1, scout's cell	uniform	+0.184	0.104	1.77	0.077
R1, Newey–West (6 / 12 lags)	uniform	+0.184	—	—	0.129 / 0.077
o1 ($t = 2024.70$), contrast	triangular	−0.035	0.179	−0.20	0.844
Panel B: triage of the uniform estimate					
R1, uniform, bandwidth 1.0 / 2.0	uniform	+0.068 / +0.217	—	—	0.769 / 0.022
R1, degree 2, bw 1.5	triangular	−0.773	0.529	−1.46	0.144
R1, degree 2, bw 2.0	uniform	−0.900	0.369	−2.44	0.015
Largest placebo (Nov 2020)	triangular	+0.385	0.109	3.54	0.0004
Panel C: placebo calibration of the R1 cell (headline spec)					
Empirical size at nominal 0.05	49 cutoffs	0.14 (7/49; `natex` issue #51)			
Placebo-calibrated p, full / clean	$\|z\|$ rank	0.80 (49 cutoffs) / 0.76 (37 clean)			
Placebos exceeding observed |kink|	—	38 of 49			

Notes: kinks in ln-points per year; HC1 standard errors against standard normal critical values unless noted. The placebo grid re-estimates the headline specification at 49 monthly cutoffs (2020.5–2024.5); the clean subgrid keeps the 37 cutoffs whose windows end before R1. Sub-window OLS slopes (ln/yr): pre-R1 +0.280; February–October 2025 +0.103; November 2025 – June 2026 +1.19. Blind LoRD3 top centers: November/December 2025 (scan $p = 0.02$ at $k = 24$, 0.01 at $k = 36$), then February–April 2026 and April/May 2020; R1 in no top-20 at any k.

correlated aggregate (empirical placebo size 0.14 at nominal 0.05; `natex` issue #51): the placebo-calibrated inference is the inference of record, and the few uniform-with-donut and seasonal-HAC cells at nominal $p = 0.001$–0.033 inherit the same edge-driven uniform estimate and are overridden by calibration and localization. Fourth, calendar time is the running variable: any event co-timed with January 2025 is confounded by construction [7] — moot given the null at the cutoff, but binding for any future positive claim — and this is a single national series with no cross-sectional contrast. Fifth, the post window is truncated at June 2026 (17 post- versus 18 pre-months), and late placebo-cutoff windows unavoidably overlap the post-R1 region, which is why the clean 37-cutoff subgrid is reported alongside the full grid.

The conclusion is a calibrated null plus an identified artifact. At the DeepSeek-R1 release the ln CVE-publication trend shows no kink (+0.038, $p = 0.764$; calibrated $p = 0.80$) on a design with demonstrated power; the o1 contrast is equally null. The scout-flagged +0.18 is real arithmetic but wrong attribution: it is the uniform kernel handing the window edges full weight, and every localization tool — kernel/bandwidth pattern, sub-window slopes, curvature absorption, placebo calibration, and a seed-0 blind scan that never sees the cutoff — places the action at November/December 2025 through 2026, ten to seventeen months after the release, with the first nine post-R1 months *shallower* than the pre-trend. Attribution of the surge to AI-assisted vulnerability discovery [5, 4] is unsupported on this series. The result is promotable only as an artifact-identification and backlog case study — the dip-then-surge is consistent with NVD operational disruption and catch-up on top of CNA expansion [11] — and, per the collection's rules, any attribution-bearing follow-up must start from reserve dates, not publication dates.

Reproducibility. All estimates were produced with `natex` v0.2.0 [8]; estimation is deterministic (local weighted least squares), seed 0 was declared, and the only stochastic step anywhere in the pipeline is LoRD3's seeded scan calibration. The NVD monthly counts are fetched by a committed script in the run of record and are not committed to the repository. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates, the recomputed sub-window slopes, the placebo-calibration quantities, and the LoRD3 center ranks against the numbers of record before drawing.

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